"It Takes Two to Tango": Joint Location Choices and Local Shocks*

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Abstract

I examine how couples, who make decisions together and represent more than half of US households, shape the adjustment to local shocks and who bears them. Using new evidence from the PSID on how couples responded to the China Trade Shock, I build a quantitative spatial model in which couples choose jointly where to live and whether the wife works. In the model, the shock raises aggregate employment, but the gain would be 18% larger if every couple were split into two singles who make decisions independently. The smaller response comes from wives: a single woman decides whether to work based on her own wage, while a couple decides based on the wife's wage relative to her husband's, and the shock raises both spouses' wages. Yet if couples could not adjust whether the wife works in response to the shock, aggregate welfare gains would be at least 5% smaller. Couples also change who moves: they move when the husband's job is hit, but rarely when only the wife's is, so workers directly hit by the shock move less, but more people relocate overall. If everyone decided on their own, 13% fewer people would end up in a different state. Joint decisions also shift gains from wives to husbands and, through wages, from single men to single women.

Keywords: Spatial equilibrium, Local labor markets, Household labor supply
JEL Codes: F16, R13, J12

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I. Introduction

Couples, who represent more than half of US households (Anderson et al., 2024), respond to local shocks together. Unlike single individuals, couples must weigh two labor-market exposures against each other: a move that pays off for one spouse may not pay off for the other, making couples less likely to relocate in pursuit of better opportunities. At the same time, couples can adjust in a way that singles cannot: when one spouse loses a job or earnings, the other spouse, usually the wife, may increase her labor supply to offset the household's lost income, a response known as the *added-worker effect* (Lundberg, 1985; Attanasio et al., 2005).

In the presence of local shocks, couples therefore have two margins of adjustment: they can move, although they are less mobile than singles, and they can adjust labor supply within the household. How couples use these two margins shapes not only how the economy adjusts to a local shock, but also who bears its effects. Yet the spatial models used to study local shocks typically assume that each worker decides alone.

In this paper, I examine how joint decision-making shapes adjustment to local shocks and who benefits from them. To do so, I combine new reduced-form evidence on how couples respond to local shocks in the US with a dynamic spatial and trade model in which couples decide together where to live and whether the wife works. My main finding is that because couples make decisions together, a local shock to one spouse becomes a shock to the household, thereby changing both how the economy adjusts and who benefits.

The first part of the paper documents novel empirical facts on how the China Trade Shock affected employment, out-migration, and within-household responses. I use geocoded data from the 1991–2013 waves of the Panel Study of Income Dynamics (PSID), which tracks each household's composition, labor-market outcomes, and migration across commuting zones over time. The shock is the increase in import competition faced by US manufacturing following China's accession to the WTO in 2001 (Autor et al., 2013, 2021). I measure exposure at the individual level: an individual is exposed if they were employed in 1999 in an industry that faced rising competition from Chinese imports between 2000 and 2012. Unlike work that compares commuting zones with different exposure, I compare exposed and unexposed individuals within the same commuting zone, and because I observe each individual's household, I can distinguish couples in which the husband is exposed from couples in which the wife is. I estimate the shock's effects using a difference-in-differences design.

I establish three empirical facts that show couples adjust to the shock through two margins: migration and wives' labor supply. First, within the same commuting zone, exposed

individuals were 6.3 p.p. less likely to be employed during the shock and, on average, 4.7 p.p. more likely to leave their commuting zone than non-exposed individuals. Second, couples, who already migrate less than singles, move in response to the shock only when the husband is exposed: their out-migration then rises by 5 p.p. Third, wives' labor force participation also responds asymmetrically: it rises by 4.2 p.p. when their husbands are exposed and falls by 11 p.p. when they themselves are exposed, with no offsetting response in husbands' labor supply. These results suggest that how couples use each margin depends on which spouse is exposed.

To quantify what joint decision-making implies for the economy as a whole, I develop a dynamic spatial and trade equilibrium model in which the decision-making unit is a household, which can be a single man, a single woman, or a couple. The model builds on Caliendo et al. (2019): regions and sectors trade with one another, and each period, households choose where to live and which sectors to work in, subject to migration frictions and idiosyncratic preferences that let households switch sectors more readily than they move across locations, as in Rodríguez-Clare et al. (2026). I add three features. First, couples choose a location for the household and a sector for each spouse, weighting each spouse's well-being according to their bargaining power, so that any move must benefit the couple as a whole. Second, after choosing where to live, the couple decides whether the wife works in the market or at home, so her participation is a margin of adjustment alongside migration. Third, firms combine male and female labor using a CES technology with sector-specific gender intensities, so the same shock can affect the two spouses differently.

The model also captures the economies of scale from marriage. Part of each household's spending goes to a non-rival household public good, such as housing: a couple pays for it once, and both spouses enjoy it in full, whereas two singles living apart must each pay for their own. When the wife stays at home, her home production raises the amount of this good; the more productive she is at home, the more she raises it, and home productivity varies across couples. The couple weighs this gain against her wage relative to her husband's. Because a dollar of income goes further in a couple than for a single person, the gains of spouses and singles cannot be compared directly or summed to measure aggregate effects. So, following Chiappori and Meghir (2015), I construct a measure that converts each person's utility into the income a single person would need to reach it.

I bring the model to the data by calibrating the parameters that are new to my model and the dispersion of preferences across locations; everything else is standard. I calibrate the spouses' bargaining weights and the share of the budget couples spend on the household public good directly from household expenditure data. I pin down how strongly wives' participation responds to their wage relative to their husbands' with the third empirical

fact, the rise in participation among wives whose husbands were exposed. I also internally calibrate the lowest productivity wives can have at home so that the model matches married women's participation in 2000. Outside the couple's problem, I follow Rodríguez-Clare et al. (2026), and I calibrate the dispersion of preferences across locations via indirect inference, so that the model reproduces the population response to import competition between 2000 and 2007, as estimated by Autor et al. (2013): the more willing households are to move, the more a state's population falls when it is exposed to Chinese imports.

Using the calibrated model, I find that in the long run the China Trade Shock lowers the manufacturing share of employment by 0.29 p.p., but it does not push workers out of work. The losses in manufacturing fall mostly on men, and most of all on married men, partly because men were more concentrated in the manufacturing sectors the shock hits. Geographically, the losses fall on the states where manufacturing was initially concentrated, around the Great Lakes and across a Southern band. At the same time, cheaper Chinese goods raise real wages, so working becomes more attractive than staying out of work. The expanding service sectors absorb workers leaving manufacturing and draw in people who were not working, so the employment rate rises across all groups, by 0.11 p.p. in the aggregate.

In couples, the shock to one spouse becomes a shock to the household: unlike singles, who work more simply because real wages rise, couples respond as a unit. Wives increase their participation by 0.10 p.p., because the shock hits male-intensive manufacturing and raises wives' wages relative to their husbands'. Husbands, most of whom were already working, instead move out of manufacturing and fill a disproportionate share of the new jobs in other sectors. The couple's location follows the husband's job: in response to the shock, exposed husbands migrate about five times as much as exposed wives.

As real wages increase and more people work, every group is better off: the average individual gains the equivalent of a 0.18% increase in consumption. How much each group gains, however, depends on who people live with. Single women gain the most, 34% more than the average individual, because they work in expanding sectors and, unlike wives, are not tied to a husband's job or wage. Single men gain the least, and couples fall in between, much closer to single men, because the husband is usually the main earner, so the household's income largely depends on his wage.

Because couples make decisions together, they change both how the economy adjusts to the shock and who benefits from it. To show this, I compare the baseline economy with an all-singles economy, in which every couple is split into two singles, and with an economy in which I shut down wives' participation. Joint decisions make employment respond less to the shock: the aggregate employment gain is 18% larger in the all-singles economy, and the smaller response comes from wives. Since the shock also raises husbands' wages, a wife's

return to work increases by less than that of a single woman: married women's employment responds 21% less than single women's to the same increase in female wages. This smaller response is not a friction but the result of how couples optimize. The wife's participation also matters for the economy as a whole: every couple adjusts through it, not only those in which the husband is exposed, and shutting down this margin lowers aggregate welfare gains by at least 5%.

Joint decisions also change who moves. An exposed wife is tied to her husband's location: when only she is exposed, she would migrate almost four times as much if she were single, whereas an exposed husband migrates about as much as he would alone. Exposed workers therefore move less than they would without couples: in the all-singles economy, their migration response is 17% larger. Yet, even though couples are less mobile than singles, more people relocate overall because an unexposed wife follows her husband when he moves. In fact, without couples, 13% fewer people would end up in a different state. Most of this difference comes from wives. As singles, they would move only in response to their own wages, and because the shock raises women's wages everywhere, they would have little reason to pay the cost of moving.

Joint decisions also change who bears the shock: gains shift from wives to husbands and, through wages, from single men to single women. Within the couple, married women would gain more as singles, and married men less. A wife gains less than she would alone because her decision to work is tied to her husband's wage and her location to his job, so she cannot fully take advantage of the expanding sectors, and because the two spouses share a single household gain. When the wife is exposed to the shock, this tie costs both spouses: she adjusts less than she would alone, and her husband shares her loss, so both would gain more as singles. Singles are affected through wages: because wives' employment responds less to the shock, women's wages rise more; without couples, single women would gain 17% less, and single men 6% more. These results show that understanding the incidence of local shocks requires treating the household, and not only the worker, as the unit that decides where to live and work.

Related literature. This paper contributes to three strands of the literature. First, it contributes to the dynamic spatial general-equilibrium literature on trade and migration responses to local shocks (Redding and Rossi-Hansberg, 2017; Desmet et al., 2018; Monte et al., 2018; Allen and Donaldson, 2020, 2022; Donald et al., 2025; Desmet and Parro, 2025; Pellegrina and Sotelo, 2025). It is particularly closely related to Caliendo et al. (2019), which emphasizes the interaction between migration, productivity, and the spatial distribution of economic activity under the China Trade Shock, and to Rodríguez-Clare et al. (2026), who

extend Caliendo et al. (2019) to separate mobility across sectors within a location from mobility across locations. My contribution is to bring the household into this class of models, in which each worker usually decides alone: in my model, spouses choose together where to live and whether the wife works. As a result, women's employment responds less, and more people relocate, than in a model in which each worker decides alone, and gains shift from wives to husbands and from single men to single women.

Second, the paper contributes to the literature that studies how dual-earner households respond to shocks, highlighting the roles of risk sharing within couples (Attanasio et al., 2005; Pilossoph and Wee, 2021; Calvo et al., 2024) and co-location frictions (Guler et al., 2012; Flabbi and Mabli, 2018; Foerster and Ulbricht, 2023; Jayachandran et al., 2024; Yan, 2025).[1] This literature studies how couples respond to idiosyncratic shocks, such as one spouse's job loss, by reallocating labor within the household or by searching for jobs jointly across locations. A local shock is different: it hits many households in the same place at once, so couples adjust both by moving, though less than singles, and by reallocating labor within the household, and their responses change local wages and, with them, the options of every other household. My contribution is to study these two margins in general equilibrium, where they interact with wages and with other households' decisions. In general equilibrium, how couples use these margins also affects local wages and, with them, what singles gain.

Third, the paper contributes to the literature on the China Trade Shock and its effects on US local labor markets (Autor et al., 2013, 2014, 2016, 2019, 2021, 2025; Acemoglu et al., 2016; Pierce and Schott, 2016; Greenland et al., 2019; Caliendo et al., 2019; Faber et al., 2022; Borusyak et al., 2026). This literature has documented substantial heterogeneity in the local effects of rising competition from Chinese imports.[2] My contribution to this literature is twofold. First, using the PSID, I provide new empirical results on the effects of the shock in the US by measuring individual-level exposure and comparing individuals within the same commuting zone. Although my analysis uses a different dataset, sample definition, and empirical design, these results are consistent with previous evidence that increased import competition from China harmed more exposed workers and places, with persistent

[1] There is an extensive literature on family migration and household adjustment. Starting with Mincer (1978), this literature has emphasized that migration is often a joint decision and that such decisions can have unequal consequences within the household. Subsequent work has documented the role of "trailing wives" and the labor market costs of family migration for women (Costa and Kahn, 2000; Cooke et al., 2009; Benson, 2014; Blau and Kahn, 2017; Fadlon et al., 2020; Braun et al., 2021; Le Barbanchon et al., 2021; Venator, 2026, 2024). Although beyond the scope of this paper, the outcomes of the "trailing wives" under a trade shock remain an open research question.

[2] In particular, Autor et al. (2013) and subsequent papers document a significant decrease in manufacturing employment in US commuting zones with larger manufacturing shares, with long-lasting consequences that include increases in political polarization (Autor et al., 2020), child and adult poverty, and even mortality among young men (Autor et al., 2019), among several other topics.

employment effects on manufacturing workers, even though it improved aggregate welfare in the US economy (Autor et al., 2013, 2021; Caliendo et al., 2019). Second, to the best of my knowledge, this is the first paper to provide empirical evidence on within-household responses to the shock in the US.

The rest of the paper proceeds as follows. Section II presents the empirical analysis, detailing the approach and presenting the main empirical findings. Section III presents the theoretical model. Section IV brings the model to the data, and Section V presents the results of the quantitative exercise. Finally, Section VI concludes.

II. Empirical Analysis

This section presents the empirical analysis of how couples respond to the China Trade Shock and shows that their response depends on which spouse is exposed: couples move only when the husband is exposed, and wives enter the labor force when their husbands are. I first describe the data, define exposure at the individual and couple levels, and outline the empirical design. I then document three empirical facts concerning labor market outcomes, migration responses, and intra-household labor supply adjustments. Together, these findings motivate the theoretical framework that I develop in the next section.

II.1 Empirical Design

Data source. The primary data source for the empirical analysis is the Panel Study of Income Dynamics (PSID), which follows individuals and their households over time. Most studies of the China Trade Shock use commuting zones as the unit of observation and rely on repeated cross-sections, a design well suited to measuring effects on local labor markets. My research question instead requires me to distinguish how exposure varies within households and to track individuals across time and space. Because the PSID follows both spouses within the same household over time and across locations, it allows me to do both.

The PSID has three additional advantages that make it particularly well suited to this analysis. First, the restricted geocoded version records household addresses at the census-tract level in every survey wave.[3] This information allows me to follow individuals across local labor markets, compare individuals within the same market, and study migration between markets. Following the previous literature, I define local labor markets as US commuting zones (CZs).

[3]The PSID staff has previously disclosed all results; access to the restricted data is under Contract 921.

Second, the PSID allows me to construct marital status from household composition and each person's relationship to the household head in every period. I therefore do not have to rely exclusively on current legal marital status, which may understate the number of couples by excluding cohabitation or overstate it by including individuals who are separated but not divorced. Given the PSID's sampling structure, I restrict the sample to three marital-status categories:[4]

1. "Married": Individuals in their first union who are legally married or cohabiting with a different-sex partner;

2. "Divorced": Individuals who are legally divorced or separated, or whose cohabiting union ended;

3. "Single": Individuals whom I never observe as "married" to another individual.

I then pool single and divorced individuals into an unmarried category because neither group makes joint household decisions with a partner.[5]

Third, the PSID contains detailed information on labor market outcomes, allowing me to examine employment, unemployment, and labor force participation. For employed individuals, I observe average weekly hours, hourly wages, annual labor income, and several additional outcomes. I also observe total household income for couples.[6]

I use all PSID waves conducted in odd-numbered years between 1991 and 2013.[7] I restrict the sample to individuals aged 20–55 who lived in the mainland US and were present in the 1999 wave. The final sample contains 4,731 individuals and 38,634 observations across 12 waves.

Individuals exposed to the China Trade Shock. The China Trade Shock refers to the increase in competition from Chinese manufactured goods in the US after China joined the WTO in 2001 and continuing through 2012 (Autor et al., 2013, 2021). I identify exposed individuals in two steps. First, following Autor et al. (2021) (henceforth ADH), I define the growth in Chinese import competition faced by US manufacturing industry j between 2000

[4]I exclude remarried individuals from the sample. I classify an individual who separated after a cohabitation as "divorced", even if their legal marital status is single.

[5]Assuming that divorced individuals do not make joint decisions follows previous literature studying life after divorce (Weiss and Willis, 1993; Del Boca and Flinn, 1995; Flinn, 2000; Reynoso, 2024; Didier and Lopez, 2026).

[6]I convert all monetary variables to 2017 dollars using the Personal Consumption Expenditures Price Index (PCE) from the US Bureau of Economic Analysis.

[7]Because the PSID was conducted annually until 1997 and biennially afterward, I consider only odd years to construct a biennial sample before and after 1997, which avoids double-counting migration.

and 2012 as:

$$\Delta IC_j = 100 \times \frac{\Delta M_j^{\text{China,US}}}{Y_{j,2000} + M_{j,2000} - X_{j,2000}},$$

where $\Delta M_j^{\text{China,US}}$ is the change in US imports from China in industry j between 2000 and 2012, $Y_{j,2000}$ denotes industry j's shipments, and $M_{j,2000} - X_{j,2000}$ denotes its net imports in 2000. A positive value of ΔIC_j therefore indicates that US manufacturing industry j experienced increased competition from Chinese imports between 2000 and 2012.

To isolate the foreign supply-driven component of US imports from China, I follow ADH and measure industry-level growth in Chinese exports to eight other high-income countries. Growth in these exports reflects China's own productivity gains and falling trade costs, which raise Chinese exports everywhere, rather than changes in US demand.[8] I also use three-year-lagged values of Y_j, M_j, and X_j rather than 2000 values to avoid any endogeneity on the demand side. The resulting measure of import competition is

$$\widehat{\Delta IC_j} = 100 \times \frac{\Delta M_j^{\text{China,Other Countries}}}{Y_{j,1997} + M_{j,1997} - X_{j,1997}}.$$

A positive value of this measure indicates that industry j experienced increased competition from Chinese imports in other high-income countries. I therefore classify the corresponding US industry j as exposed to the China Trade Shock.

Second, I identify the industry in which each individual i was employed in 1999 and define the indicator Exposed_i as follows:

$$\text{Exposed}_i = \begin{cases} 1 & \text{if } i \text{ worked in 1999 in industry } j \text{ with } \widehat{\Delta IC_j} > 0 \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

For the analysis of within-household mechanisms, I further restrict the sample to couples. I divide couples into three groups: those in which neither spouse worked in an exposed industry in 1999, those in which the husband worked in an exposed industry, and those in which the wife worked in an exposed industry. I define couple h as exposed through the husband if

$$\text{Exposed Husband}_h = \begin{cases} 1 & \text{if he worked in 1999 in industry } j \text{ with } \widehat{\Delta IC_j} > 0 \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

[8] Autor et al. (2013) and their follow-up papers consider Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland. They select these countries based on the availability of data for the full sample period.

and as exposed through the wife if

$$\text{Exposed Wife}_h = \begin{cases} 1 & \text{if she worked in 1999 in industry } j \text{ with } \widehat{\Delta IC_j} > 0 \\ 0 & \text{otherwise.} \end{cases} \tag{3}$$

Summary statistics. Table 1 reports summary statistics for the pooled 1991–1999 sample. The first column describes the full sample; the second pools single and divorced individuals into an unmarried category; and the third and fourth report statistics for husbands and wives, respectively. Married individuals constitute approximately 60% of the sample, single individuals 24%, and divorced individuals the remainder.

Unmarried individuals are slightly younger and less likely to be college graduates or white than married individuals. The average individual is 37 years old; two-fifths of the sample are women, one-third are college graduates, and 86% are white. Half of the sample have children under 18, including nearly one-quarter of unmarried individuals. The youngest child in the household is, on average, 4 years old and tends to be younger among unmarried individuals, consistent with the younger average age of unmarried individuals.

Married and unmarried individuals live in similarly exposed CZs but differ sharply in how often they migrate. According to ADH's measure, one in ten individuals lives in a CZ in the top two deciles of exposure to Chinese import competition. Exposure does not vary meaningfully by marital status, suggesting little spatial sorting along this dimension. The average migration rate is 8%, with unmarried individuals migrating at nearly twice the rate of married individuals.

In 1999, 15% of the sample worked in an industry exposed to competition from Chinese imports, with substantial heterogeneity across groups. Exposure rates were 14% among unmarried individuals, 19% among husbands, and 10% among wives. At the household level, 20% of couples had at least one exposed spouse.

As noted by Autor et al. (2013, 2021), the period of the China Trade Shock also featured two related economic forces: increasing automation and broader globalization. Following Autor and Dorn (2013), I calculate routine-task intensity and offshorability indices for the occupations represented in the sample. Slightly less than one-quarter of observations correspond to individuals employed in routine-task-intensive occupations, while approximately one in ten workers is employed in an offshorable occupation. Offshorability varies substantially by gender, reflecting the occupational sorting of men and women. Because both forces also affected manufacturing workers, omitting them could lead to attributing their effects to the China Trade Shock, so I control for both occupational characteristics in the empirical analysis.

Table 1: Summary Statistics Before the China Trade Shock

	(1)	(2)	(3)	(4)
	All	Unmarried	Couples	
			Husbands	Wives
Panel A: Distribution by "Marital Status"				
Share of Single Individuals	0.24			
Share of Married Individuals	0.59			
Share of Divorced Individuals	0.16			
Panel B: Demographic Characteristics				
Age^{+}	36.86	34.69	38.92	37.19
Share of Female Individuals	0.42	0.55		
Share of College Graduates	0.33	0.29	0.37	0.32
Share of White Individuals	0.86	0.75	0.93	0.93
Share of Individuals with Kids	0.50	0.23	0.69	
Age of Youngest Kid^{+}	3.67	1.75	4.98	
Share in Highly Exposed CZs^{++}	0.11	0.12	0.11	
Migration Rate	0.08	0.11	0.06	
Panel C: Labor Market Outcomes				
Share of Exposed Individuals	0.15	0.14	0.19	0.10
Share in Routine Occupations	0.23	0.22	0.23	0.26
Share in Offshorable Occupations	0.12	0.14	0.08	0.16
Share Employed	0.91	0.89	0.97	0.85
Share Unemployed	0.03	0.06	0.02	0.02
Share Out of the Labor Force	0.05	0.05	0.01	0.13
Weekly Hours^{+}	40.80	39.89	45.34	33.60
Hourly Wage^{+}	19.54	16.98	23.70	16.45
Labor Income^{+}	42671.70	35953.36	57073.74	27617.31
Number of Observations	16,645	4,343	6,151	6,151

Notes. The unmarried category pools single and divorced individuals. Monetary values are expressed in 2017 USD using the PCE price index. $^{+}$: Indicates the sample mean. $^{++}$: Highly exposed CZs are those in the top 20% of exposure according to ADH.

Because the sample consists of prime-age individuals for whom I can identify their industry in 1999, employment and labor force participation rates are high, at 91% and 95%, but the composition of non-employment differs by marital status. Among non-employed individuals, unemployment is the most common status for unmarried individuals and husbands,

whereas non-participation in the labor force is most common among wives. This pattern is consistent with unmarried individuals relying more heavily on reemployment after job loss, whereas married individuals can adjust labor supply within the household.

The average individual works 40 hours per week, earns $19 per hour in 2017 USD, and receives approximately $42,000 in annual labor income. Husbands work the most hours, earn the highest hourly wages, and consequently receive the highest annual labor income; wives have the lowest values for all three outcomes. Married and unmarried individuals work similar hours on average, but married individuals earn approximately 20% higher hourly wages and therefore receive more labor income per capita. Although selection into marriage likely explains part of this gap, this pattern is also consistent with couples using intra-household labor supply adjustments to allocate resources more effectively.

Difference-in-differences design. To estimate the overall effect of exposure to the China Trade Shock, I estimate the following event-study specification for individual i in commuting zone c and year t:

$$\begin{aligned}\text{Outcome}_{i,c,t} =&\alpha \cdot \mathbb{1}[\text{Exposed}_i = 1] + \sum_{\tau \neq 1999} \beta_\tau \cdot \mathbb{1}[\text{Exposed}_i = 1] \cdot \mathbb{1}[t = \tau] \\ &+ X'\delta + \gamma_t + \gamma_c + \varepsilon_{i,c,t}.\end{aligned} \tag{4}$$

The control vector includes age and indicators for being female, holding a college degree, having children younger than 18, working in manufacturing in 1999, working in a routine occupation, and working in an offshorable occupation. The specification also includes commuting-zone and year fixed effects. I weight all regressions by PSID sampling weights and two-way cluster standard errors at the individual and industry levels.

Because the specification includes commuting-zone and year fixed effects, the coefficients of interest, β_τ, measure the difference in outcomes between exposed and non-exposed individuals in year τ, relative to the corresponding difference in 1999. Thus, the comparison is between individuals in the same local labor market and year who worked in different industries in 1999. I fix exposure at the 1999 industry level, before China joined the WTO, so that exposure does not reflect workers' later moves across industries. The identifying assumption is that, absent the shock, the outcomes of exposed and non-exposed individuals would have evolved in parallel, and the coefficients before 1999 test this assumption.

To summarize the average effect during the shock, I also estimate a difference-in-differences version of equation (4) that replaces the year-specific interactions with a single interaction:

$$\begin{aligned}\text{Outcome}_{i,c,t} =&\beta_1 \cdot \mathbb{1}[\text{Exposed}_i = 1] + \beta_2 \cdot \mathbb{1}[\text{Exposed}_i = 1] \cdot \mathbb{1}[t \geq 2001] \\ &+ X'\delta + \gamma_c + \gamma_t + \varepsilon_{i,c,t}.\end{aligned} \tag{5}$$

To examine joint household decision-making, I estimate the following specifications for spouse s in couple h, commuting zone c, and year t. First, I focus on whether the husband is exposed:

$$\begin{aligned}\text{Outcome}_{s,h,c,t} =& \beta_1 \cdot \mathbb{1}[\text{Exposed Husband}_h = 1] \\ &+ \beta_2 \cdot \mathbb{1}[\text{Exposed Husband}_h = 1] \cdot \mathbb{1}[t \geq 2001] \\ &+ X'\delta + \gamma_c + \gamma_t + \varepsilon_{s,h,c,t},\end{aligned} \tag{6}$$

and then on whether the wife is exposed:

$$\begin{aligned}\text{Outcome}_{s,h,c,t} =& \beta_1 \cdot \mathbb{1}[\text{Exposed Wife}_h = 1] + \beta_2 \cdot \mathbb{1}[\text{Exposed Wife}_h = 1] \cdot \mathbb{1}[t \geq 2001] \\ &+ X'\delta + \gamma_c + \gamma_t + \varepsilon_{s,h,c,t}.\end{aligned} \tag{7}$$

In these specifications, the coefficient of interest is β_2, which captures the post-2001 difference in outcomes between couples with an exposed spouse and those without, relative to the corresponding pre-2001 difference.[9] I estimate both specifications on the same sample of couples, so the comparison group in equation (6) includes couples in which only the wife is exposed, and vice versa. Both equations include the controls used in equation (4). For controls measured at the individual level, I include separate values for each spouse, such as the husband's and wife's ages.[10]

II.2 Empirical Facts

The three facts proceed from individual-level effects to within-household adjustment. The first establishes the labor market effects of the shock and examines whether exposed individuals respond by leaving their commuting zone. The second shows that couples migrate in response only when the husband is the exposed spouse. The third examines wives' labor supply responses, a margin of adjustment that couples can use alongside or instead of migration. In all three, the comparison is between exposed and non-exposed individuals facing the same local labor market, rather than between local labor markets with different levels of exposure.

[9] For this analysis, I drop 4% of couple-year observations across the full 1991–2013 panel in which both spouses were working in exposed industries in 1999, because I cannot measure whether one spouse's job was more important to the couple than the other's.

[10] I two-way cluster standard errors by the exposed spouse's industry and by household or individual, depending on how the outcome is measured. The industry dimension refers to the husband's industry in equation (6) and the wife's in equation (7). I cluster at the household level for outcomes measured at the household level, such as out-migration and household income, and at the individual level for outcomes measured for each spouse, such as employment and labor force participation.

Fact 1: Exposure to the China Trade Shock worsened labor market outcomes and increased out-migration.

Figure 1 presents estimates of equation (4) for employment, unemployment, labor force participation, weekly hours worked, hourly wages, and annual labor income. Table A.2 reports estimates from the corresponding difference-in-differences specification in equation (5), which summarizes the average effect during the China Trade Shock.

Panel A of Figure 1 shows that, during the first seven years of the shock, exposed individuals were less likely to be employed than non-exposed individuals. On average, exposure was associated with a 6.3 p.p. decline in the probability of employment. After 2009, the point estimates remain close to their 2007 level, but the confidence intervals widen, and the estimates are no longer statistically significant. Most of the decline in employment between 2001 and 2007 reflects individuals leaving the labor force (Panel C) rather than entering unemployment (Panel B). Between 2001 and 2013, exposed individuals were, on average, 3.9 p.p. less likely to participate in the labor force than non-exposed individuals.

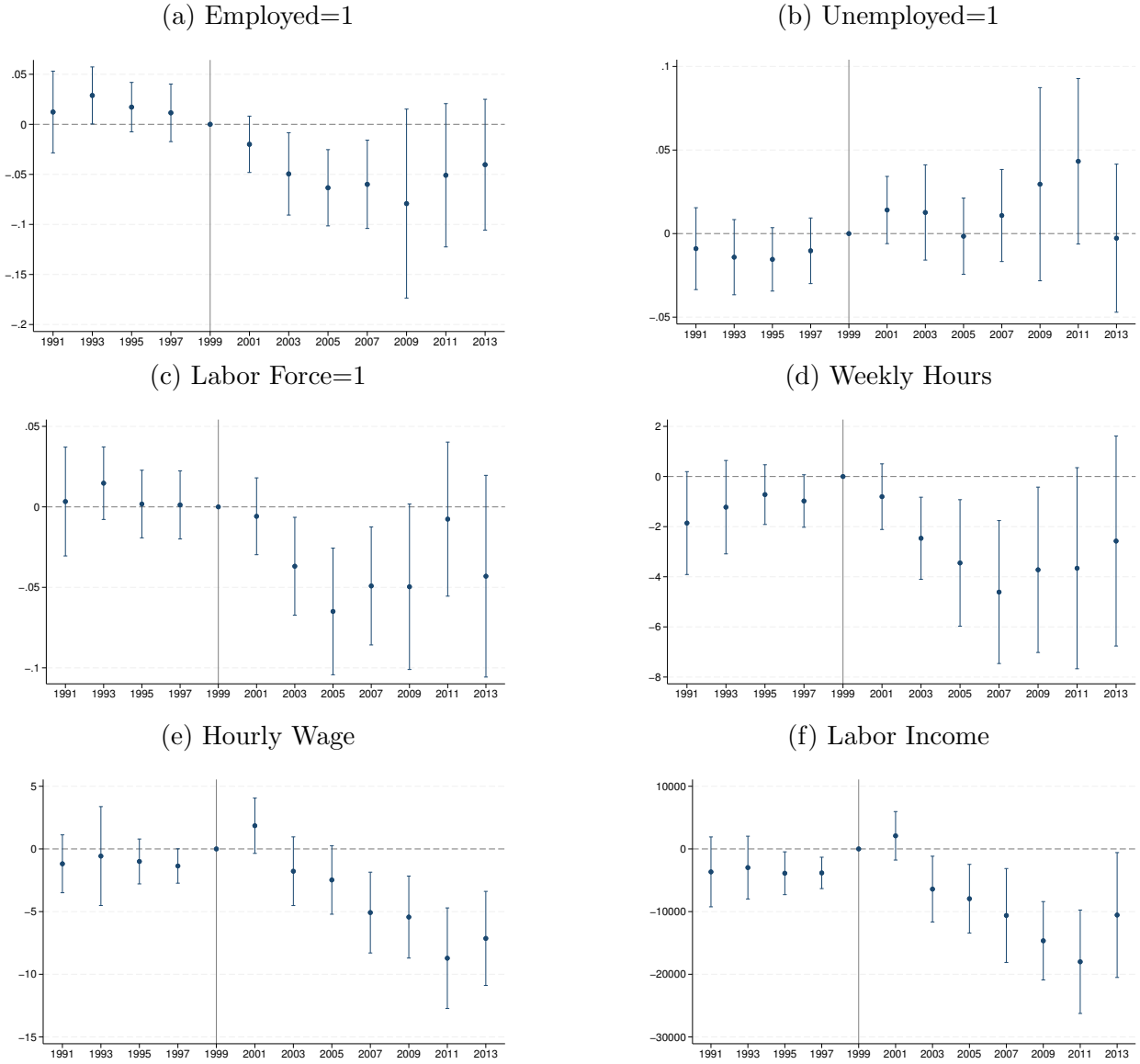
Weekly hours worked follow a pattern similar to employment (Panel D): individuals employed in exposed industries in 1999 worked 5% fewer hours per week during the shock than individuals employed in other industries that year. Hourly wages exhibit a different pattern (Panel E). Although the estimate is positive in 2001, the hourly wages of exposed individuals declined through the final year of the sample. On average, exposed individuals earned 10% lower hourly wages during the shock than non-exposed individuals. Combined with the reduction in hours worked, this decline resulted in an 11% lower annual labor income among exposed individuals (Panel F).

My employment estimates are similar in magnitude to ADH's when I use their continuous measure of import competition. To compare my estimates more directly with the previous literature, I replace the binary exposure measure in equation (1) with the continuous import-competition measure used by ADH (Table A.3).[11] The average decadalized growth in Chinese import competition implies a 0.5 p.p. average decline in the probability of employment among exposed individuals relative to non-exposed individuals between 2001 and 2013. For comparison, converting ADH's cumulative estimate for the employment-to-population ratio between 2001 and 2019 into an evenly distributed effect and retaining only odd-numbered years, as in the PSID sample, implies an average decline of 0.55 p.p. These findings are consistent with previous evidence on the adverse effects of the China Trade Shock in the US (Autor et al., 2013, 2014, 2016, 2021; Acemoglu et al., 2016; Pierce and Schott, 2016).

[11] One potential concern is that, given the small PSID sample, the particular set of CZs in my sample may drive the estimates. To address this concern, Appendix A.1 replicates ADH's findings using the same CZs included in my sample, ruling out this alternative explanation.

Despite using a different dataset, sample restrictions, and empirical approach, I similarly find a negative employment response among individuals working in exposed industries.

Figure 1: Exposure to the China Trade Shock and Labor Market Outcomes



Notes. Each panel reports estimates from equation (4), using the variable named in the panel title as the outcome. Monetary outcomes are expressed in 2017 USD using the PCE price index and include zero values. Robust standard errors are two-way clustered at the industry and individual levels. The bars indicate 95% confidence intervals.

Given the deterioration in labor market outcomes among individuals employed in industries exposed to Chinese import competition, particularly the decline in employment during the first years of the shock, I next examine whether exposed individuals moved to other CZs in search of better opportunities. For each individual i, I define out-migration in year t as

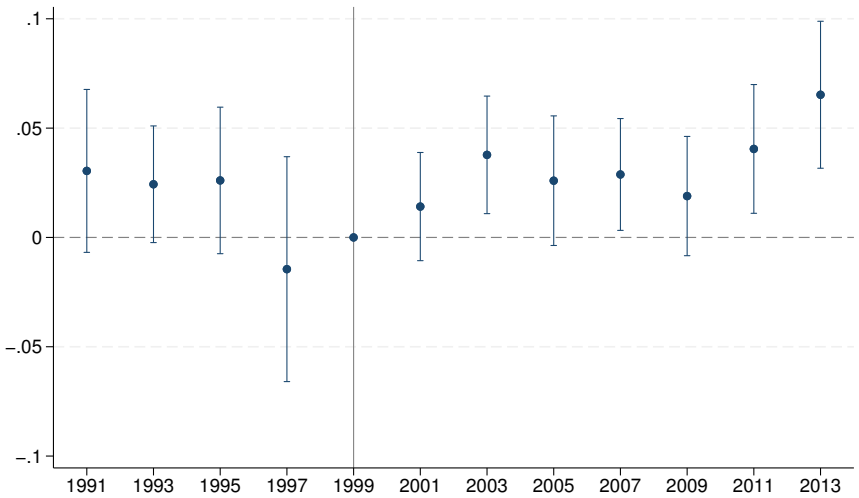
follows:

$$\text{Out-Migration}_{i,t} = \begin{cases} 1 & \text{if} \quad \text{CZ}_{i,t-2} \neq CZ_{i,t} \\ 0 & \text{if} \quad CZ_{i,t-2} = CZ_{i,t}. \end{cases} \tag{8}$$

During the shock, exposed individuals became increasingly likely to leave their CZ. Figure 2 presents estimates of equation (4) with out-migration as the outcome: the estimated difference increased by approximately 2 p.p. every two years after 2009, reaching 6 p.p. in 2013. Thus, most of the response emerges with a lag of seven to ten years, consistent with Greenland et al. (2019)'s evidence on the effects of the China Trade Shock on internal migration in the US. On average, exposed individuals were 4.7 p.p. more likely to out-migrate than non-exposed individuals within the same commuting zone, as reported in Table A.4.

The out-migration response is consistent with ADH's evidence of a statistically significant decline in the population aged 25 to 39 and a statistically insignificant decline in the overall population of more exposed commuting zones. Moreover, Borusyak et al. (2026) argue that shift-share measures of industry-level shocks may attenuate estimated migration responses when they abstract from workers' outside options, as in ADH. Accordingly, the individual-level out-migration response I estimate should be larger than ADH's aggregate estimate.

Figure 2: Exposure to the China Trade Shock and Out-Migration Responses



Notes. Out-migration is defined as equation (8). Robust standard errors are two-way clustered at the industry and individual levels. The bars indicate 95% confidence intervals.

Fact 2: Couples out-migrate only when the husband is exposed.

Couples leave their commuting zone when the husband is exposed to the shock, but not when the wife is. To examine how couples' migration responses differ from those of individuals, I estimate equations (6) and (7) using out-migration as the outcome at the household level.[12] Column 1 of Table 2 shows that couples in which the husband was exposed to the shock were 5 p.p. more likely to leave their commuting zone than couples in which the husband was not exposed.[13] By contrast, exposure of the wife produces no statistically significant migration response, and the point estimate in Column 2 is negative. Although this asymmetry may initially appear counterintuitive, it is consistent with evidence that husbands' employment plays a larger role in family migration decisions (Mincer, 1978; Costa and Kahn, 2000; Cooke et al., 2009; Benson, 2014; Blau and Kahn, 2017; Fadlon et al., 2020; Braun et al., 2021; Le Barbanchon et al., 2021).

I can rule out two alternative explanations for this asymmetry: changes in family composition and differences in migration responses among parents or women. First, the asymmetry could be driven by changes in family composition. If, for example, couples divorce when the wife is exposed but not when the husband is exposed, this would bias my findings. In Appendix A.2, I study the effect of the trade shock on the probabilities of marriage, divorce, and fertility and find no effect on any of them. Second, the asymmetry could reflect nothing more than different migration responses among parents or among exposed women. In Appendix A.3, I expand equation (4) to include a triple interaction for having children and for being a woman. In both cases, I cannot reject that parents or women responded in the same way as non-parents or men.

Fact 3: Wives' labor force participation falls when they are directly exposed and rises when husbands are exposed.

Couples have an additional margin of adjustment besides migration: one spouse's labor supply can respond to the other's exposure, as in the so-called "added-worker effect" (Lundberg, 1985), which refers to a spouse who specializes in home production entering the labor market when the other spouse loses their job, to offset the household's income

[12]The PSID records only the census tract of the household in which the interview takes place, which it always treats as the family's primary household. As a result, if only one spouse migrates, the PSID does not record the couple as migrating. Instead, it records the couple as "married, spouse absent," and the respondent reports for both spouses. According to PSID staff, this case is very unlikely. My estimates are therefore a lower bound on couples' migration responses, as they only account for couples who migrate together.

[13]The negative coefficient in the first row of Table 2 reflects the relatively favorable employment conditions in those manufacturing industries before the shock, which made couples less likely to migrate if the husband was employed there.

Table 2: Exposure to the China Trade Shock and Couples' Out-Migration Responses

	(1)	(2)
	Out-Migration=1	
Exposed Husband=1	-0.121***	
	(0.043)	
(Exposed Husband=1)x(Post=1)	0.050**	
	(0.022)	
Exposed Wife=1		0.064
		(0.074)
(Exposed Wife=1)x(Post=1)		-0.010
		(0.028)
Observations	13,695	13,695
R^2	0.10	0.09
Baseline Mean	0.05	0.05

Notes. Out-migration is measured at the household level and defined as equation (8). Robust standard errors are two-way clustered by household and the husband's industry in Column 1, and by household and the wife's industry in Column 2. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

loss. I therefore study labor market responses to the shock, distinguishing between direct exposure, through a spouse's own 1999 industry, and indirect exposure, through the other spouse's. When wives are directly exposed to the shock, they are 9.9 p.p. less likely to be employed and 11 p.p. less likely to be in the labor force than non-exposed wives (Panel A of Table 3). However, when wives are indirectly exposed because their husbands worked in an exposed industry, they are 4.2 p.p. more likely to be in the labor force relative to the wives of non-exposed husbands (Panel B of Table 3). This provides suggestive evidence of the added-worker effect among wives indirectly exposed to the China Trade Shock through their husbands.

Husbands, by contrast, respond to their wives' exposure through their wages rather than through an added-worker effect. When husbands are indirectly exposed to the shock, they have the same probability of employment as non-exposed husbands, while earning a lower hourly wage (Table A.5). This pattern suggests that indirectly exposed husbands have a lower reservation wage. When the wife's earnings fall, the household values the husband's

job more, so he accepts a lower wage rather than risk a longer spell without work.

Table 3: Wife's Exposure to the China Trade Shock and Labor Market Outcomes

	Wife Employed=1	Wife Labor Force=1
Exposed Wife=1	0.041	0.016
	(0.099)	(0.110)
(Exposed Wife=1)x(Post=1)	-0.099**	-0.110**
	(0.045)	(0.048)
Exposed Husband=1	0.027	0.027
	(0.077)	(0.068)
(Exposed Husband=1)x(Post=1)	0.041***	0.042***
	(0.013)	(0.013)
Observations	14,083	14,083
Baseline Mean	0.77	0.78

Notes. Robust standard errors are two-way clustered at the wife's industry and individual levels in Panel A and at the husband's industry and individual levels in Panel B. * for p<.10, ** for p<0.05, *** for p<0.01.

The absence of an added-worker effect among husbands contrasts with the findings of Parey et al. (2025) for England and Wales. They show that when English husbands are indirectly exposed to the trade shock, older husbands increase their labor force participation and move into self-employment. They find a positive, though not statistically significant, response among indirectly exposed wives. The difference in the results for husbands could be driven by the fact that 93% of husbands in their sample are in the labor force, compared with 99% in my sample (Table 1). As a result, husbands in my sample face a ceiling effect that hinders identification of an added-worker effect for husbands.

The household income of exposed couples remained close to that of non-exposed couples during the shock, regardless of which spouse was exposed. To see whether these asymmetric responses help couples absorb the shock, I estimate equations (6) and (7) using a measure of household income from the PSID. When husbands are exposed, the coefficient is statistically insignificant, suggesting that the combination of out-migration and the wife's labor force participation helps stabilize household income during the shock (Table A.6).

When the wife is exposed, household income is stable because the loss is small, not because the couple offsets it by migrating or working more. These couples do not migrate, the wife's employment falls, and the husband's employment does not change (Tables 3 and

A.5). Before the shock, however, wives earned roughly half as much labor income as husbands (Table 1), so the loss of the wife's earnings is small relative to total household income. The point estimate in Column 2 of Table A.6 is negative, consistent with this smaller loss, but imprecisely estimated. The smaller loss also helps explain why these couples do not move: the income at stake is lower than it would be when the husband, the household's main earner, is exposed to the shock. Household income is also an incomplete measure of welfare, because it does not capture changes in home production, leisure, or the costs of moving. I quantify the welfare effects of both margins with the model developed in the next section.

The empirical evidence points to three central patterns that, together, suggest that couples adjust to the shock both by moving and by adjusting labor supply within the household, and that the use of each margin depends on which spouse is exposed. First, exposure to the China Trade Shock worsened labor market outcomes for directly exposed individuals, who responded by leaving their commuting zones. Second, among couples, this migration response was concentrated in households where the husband was exposed. Third, I find evidence of an added-worker effect during the trade shock: wives entered the labor market when their husbands were exposed to the shock, whereas wives directly exposed to the shock left the labor force.

The empirical evidence compares exposed with non-exposed individuals, so it cannot show the general equilibrium effects of couples' adjustment: what each margin does for the economy as a whole and who bears the shock, including individuals who were not exposed. In the next section, I develop a framework in which households jointly decide on location and labor supply, within a quantitative spatial and trade model.

III. A Model of Trade and Migration with Couples

This section develops a dynamic spatial and trade model in which households are the decision-making unit. The model extends the framework of Caliendo et al. (2019) (henceforth CDP), and I use it to rationalize the three empirical facts documented in Section II and to quantify how couples change the way the economy adjusts to the China Trade Shock and who bears it. The model's mechanisms map directly onto these facts, and I point out each link as the corresponding part of the model is introduced.

The economy consists of L locations, indexed by i as origins and j as destinations, populated by three household types, indexed by h: single men ($h = s_m$), single women ($h = s_f$), and couples ($h = c$). In each location, a population mass N is born at $t = 0$, comprising $N/2$ men and $N/2$ women. A fraction $\bar{s}_i$ of individuals are single, while the

remaining fraction, $1-\bar{s}_i$, belong to different-sex couples.[14] Time is discrete and indexed by $t \in \{0, 1, 2, \ldots\}$.

All households are forward-looking and respond to local conditions by choosing their place of residence and sector of employment. Couples make these choices jointly, choosing one location and a sector for each spouse. After making these choices, a couple draws a home-production productivity shock β, which, together with the spouses' relative wage, determines whether the wife participates in market work.

The central feature of the model is the interaction between migration and married women's labor force participation. Because men and women are exposed to different sectors, a sectoral productivity shock affects the two spouses differently, and a couple can respond by moving, by changing the wife's participation, or both. Together, these two decisions shape how the economy adjusts to sectoral productivity shocks, through where people live and who works, and how the effects of these shocks differ across households and locations.

III.1 Environment

1) Technology

In each period t, every region i contains K productive sectors, indexed by $k, n \in \{1, \ldots, K\}$. Let $A_{i,k,t} > 0$ denote Hicks-neutral productivity in sector k of region i. Let $\psi_{k,m}, \psi_{k,f} > 0$ denote sector-specific gender-intensity parameters that are common across locations and satisfy $\psi_{k,m} + \psi_{k,f} = 1$.

A representative intermediate-good firm in sector k of origin i combines male labor, $M_{i,k,t}$, and female labor, $F_{i,k,t}$, into an effective labor aggregate using a CES production function with elasticity of substitution $\sigma > 1$:

$$\mathcal{L}_{i,k,t} = \left(\psi_{k,m} M_{i,k,t}^{(\sigma-1)/\sigma} + \psi_{k,f} F_{i,k,t}^{(\sigma-1)/\sigma}\right)^{\sigma/(\sigma-1)}. \tag{9}$$

Output is then given by $Y_{i,k,t} = A_{i,k,t}\mathcal{L}_{i,k,t}$.

Under perfect competition and constant returns to scale, each sectoral intermediate good is sold at its unit cost, and shipping it from origin-sector pair ik to destination-sector pair jn entails iceberg trade costs $\tau_{ik,jn} \geq 1$. The delivered price paid by sector n in region j is therefore $p_{ik,jn,t} = p_{i,k,t}\,\tau_{ik,jn}$, where $p_{i,k,t} = \bar{w}_{i,k,t}/A_{i,k,t}$ is the producer price and $\bar{w}_{i,k,t}$ is the CES wage index. The CES wage index is the unit cost of effective labor given the average

[14] The model holds household composition fixed: marriage, divorce, fertility, and population growth are excluded, so couples remain together throughout the transition. Figure A.4 provides supporting evidence for this assumption.

wages paid to men and women in sector k of location i, $w^M_{i,k,t}$ and $w^F_{i,k,t}$:

$$\bar{w}_{i,k,t} = \left(\psi^{\sigma}_{k,m}\left(w^M_{i,k,t}\right)^{1-\sigma} + \psi^{\sigma}_{k,f}\left(w^F_{i,k,t}\right)^{1-\sigma}\right)^{\frac{1}{1-\sigma}}. \tag{10}$$

The first-order conditions of the firm's cost-minimization problem yield optimal demands for male and female labor as functions of the effective labor aggregate and relative wages:

$$M_{i,k,t} = \psi^{\sigma}_{k,m}\left(\frac{w^M_{i,k,t}}{\bar{w}_{i,k,t}}\right)^{-\sigma}\mathcal{L}_{i,k,t}, \tag{11}$$

$$F_{i,k,t} = \psi^{\sigma}_{k,f}\left(\frac{w^F_{i,k,t}}{\bar{w}_{i,k,t}}\right)^{-\sigma}\mathcal{L}_{i,k,t}. \tag{12}$$

Inverting the labor demand functions in equations (11) and (12) yields the equilibrium gender-specific factor prices within each region-sector pair:

$$w^M_{i,k,t} = \bar{w}_{i,k,t}\psi_{k,m}\left(\frac{\mathcal{L}_{i,k,t}}{M_{i,k,t}}\right)^{1/\sigma}, \qquad w^F_{i,k,t} = \bar{w}_{i,k,t}\psi_{k,f}\left(\frac{\mathcal{L}_{i,k,t}}{F_{i,k,t}}\right)^{1/\sigma}. \tag{13}$$

Because men and women are distributed differently across sectors, a change in $A_{i,k,t}$ changes men's and women's average wages differently. For a given level of employment, it moves men's and women's wages within a region-sector cell in the same proportion, via the wage index $\bar{w}_{i,k,t}$, so the difference arises from where men and women work. As a result, the two spouses in a couple, who may work in different sectors, can experience very different changes in wages in response to the same change in $A_{i,k,t}$.

In each destination region j and sector n, a representative sectoral final-good firm combines goods from all origin-sector pairs into a sector-n composite, $Q_{j,n,t}$, using a CES aggregator with elasticity of substitution $\eta_n > 1$. The sectoral price index and the share of sector n's expenditure in region j spent on goods from sector k in region i are

$$P_{j,n,t} = \left(\sum_{i=1}^{L}\sum_{k=1}^{K} p^{1-\eta_n}_{ik,jn,t}\right)^{1/(1-\eta_n)}, \qquad \pi_{ik,jn,t} = \left(\frac{p_{ik,jn,t}}{P_{j,n,t}}\right)^{1-\eta_n}. \tag{14}$$

A representative regional final-good firm then combines the sectoral composites into the regional consumption good, $Q_{j,t}$, using a Cobb-Douglas aggregator with expenditure shares $\iota_k > 0$ and $\sum_{k=1}^{K}\iota_k = 1$.[15] The regional price index and the expenditure on sector n are

[15]The aggregator is $Q_{j,t} = \prod_{k=1}^{K}(Q_{j,k,t}/\iota_k)^{\iota_k}$, where dividing each composite by its share ι_k normalizes the aggregator so that the regional price index takes the simple form in equation (15).

therefore

$$P_{j,t} = \prod_{k=1}^{K} P_{j,k,t}^{\iota_k}, \qquad P_{j,n,t}Q_{j,n,t} = \iota_n P_{j,t} Q_{j,t}. \tag{15}$$

Each location j also contains a non-employment sector, indexed by 0, that is available to single men, single women, and married men.[16] The non-employment sector produces no output. An individual who chooses sector 0 in region j receives an exogenous nominal benefit b_j, which is constant over time:

$$w_{j,0,t}^{M} = w_{j,0,t}^{F} = b_j.$$

2) Preferences

Location and sector choices. The household's mobility problem is dynamic: each household $h \in \{s_m, s_f, c\}$ is forward-looking and discounts the future with discount factor $\delta \in (0,1)$. At the start of every period, households begin in a particular location-sector position and must decide whether to switch sectors and whether to stay in the same location or move to a different one. Every household occupies a single location, so spouses always live together, but the sector component differs by household type: for single individuals, it is one sector in $\{0, 1, \ldots, K\}$, while for couples, it is the pair formed by the husband's and wife's sectors. The husband's current and destination sectors satisfy $k_m, n_m \in \{0, 1, \ldots, K\}$, while the wife's current and destination market sectors satisfy $k_f, n_f \in \{1, \ldots, K\}$.

For convenience, I collapse a household's location-sector position into a single *bundle*. I write the origin (current) bundle as $\boldsymbol{o} \equiv (i, k_m, k_f)$, the destination (next-period) bundle as $\boldsymbol{d} \equiv (j, n_m, n_f)$, and a generic bundle summed over as $\boldsymbol{z} \equiv (\ell, q_m, q_f)$. For single men, a bundle is (i, k_m); for single women, (i, k_f); and for couples, (i, k_m, k_f). The same symbols $\boldsymbol{o}, \boldsymbol{d}, \boldsymbol{z}$ denote the bundle for every household type, and I unbundle them into components only where a formula refers to a particular location or sector.

Let $\mathcal{S}_h$ denote the set of feasible bundles for household type h, and define

$$\mathcal{S}_h(j) \equiv \{\boldsymbol{d} \in \mathcal{S}_h : \text{ the location of } \boldsymbol{d} \text{ is } j\}$$

as the set of feasible bundles available to type h that are located in j. For singles, $\mathcal{S}_h(j)$ contains the $K+1$ sectors available in j; for couples, it contains the $(K+1) \times K$ sector pairs available in j.

To decide their bundle for the next period, each household h in current bundle $\boldsymbol{o}$ draws

[16] Married women do not choose sector 0; instead, couples choose a productive sector for the wife and then decide whether she works in the market or engages in home production, as explained below.

two idiosyncratic preference shocks: a *location* shock $\varepsilon^{h,L}_{\boldsymbol{o}j,t}$ for each destination location j, and a *sector* shock $\varepsilon^{h,S}_{\boldsymbol{od},t}$ for each destination sector bundle within that location. The two enter additively, so the total preference shock attached to destination bundle $\boldsymbol{d}$, located in j, is

$$\varepsilon^{h}_{\boldsymbol{od},t} = \varepsilon^{h,L}_{\boldsymbol{o}j,t} + \varepsilon^{h,S}_{\boldsymbol{od},t}. \tag{16}$$

Following Rodríguez-Clare et al. (2026), I assume that the vector $\varepsilon^{h}_{\boldsymbol{o},t} \equiv (\varepsilon^{h}_{\boldsymbol{od},t})_{\boldsymbol{d}\in\mathcal{S}_h}$ is i.i.d. over time and households and has a nested Gumbel joint distribution,

$$F\left(\varepsilon^{h}_{\boldsymbol{o},t}\right) = \exp\left(-\sum_{j=1}^{L}\left(\sum_{\boldsymbol{d}\in\mathcal{S}_h(j)} \exp\left(-\varepsilon^{h}_{\boldsymbol{od},t}/\nu\right)\right)^{\nu/\theta}\right), \tag{17}$$

with $0 < \nu \leq \theta$.[17] The parameter ν governs the dispersion of preferences across sector bundles within a location, so $1/\nu$ determines substitution across sectors. Similarly, θ governs preference dispersion across locations, so $1/\theta$ determines substitution across locations.[18] The restriction $\nu \leq \theta$ implies that sector choices within a location respond at least as strongly to differences in continuation values as location choices do. Thus, households substitute at least as much across sectors within a location as they do across locations.[19]

A household moving from origin bundle $\boldsymbol{o}$ to destination bundle $\boldsymbol{d}$ incurs a mobility cost $\mu_{\boldsymbol{od}} \geq 0$, with $\mu_{\boldsymbol{oo}} = 0$. The lifetime utility of a household of type h occupying bundle $\boldsymbol{o}$ at time t is

$$v^{h}_{\boldsymbol{o},t} = \log(W^{h}_{\boldsymbol{o},t}) + \max_{\boldsymbol{d}\in\mathcal{S}_h}\left\{\delta\mathbb{E}\left(v^{h}_{\boldsymbol{d},t+1}\right) - \mu_{\boldsymbol{od}} + \varepsilon^{h}_{\boldsymbol{od},t}\right\},$$

where $W^{h}_{\boldsymbol{o},t}$ denotes the period-t indirect utility of household type h in current bundle $\boldsymbol{o}$. For single men and single women, the indirect utility corresponds to $W^{s_m}_{\boldsymbol{o},t} = w^{M}_{i,k,t}/P_{i,t}$ and $W^{s_f}_{\boldsymbol{o},t} = w^{F}_{i,k,t}/P_{i,t}$, respectively.[20] For couples, $W^{c}_{\boldsymbol{o},t}$ also depends on the wife's market-work decision, as I describe below.

Let $V^{h}_{\boldsymbol{o},t} \equiv \mathbb{E}(v^{h}_{\boldsymbol{o},t})$. Then, using the properties of the nested Gumbel distribution (Ap-

[17] Rodríguez-Clare et al. (2026) extend Caliendo et al. (2019) by allowing the dispersion of preferences that governs mobility across sectors within a region and across regions to differ, adding a margin that better matches the empirical evidence in Autor et al. (2013). In this setting, the extension distinguishes between mobility across sectors and mobility across regions, with the latter being the relevant margin for couples.

[18] Neither ν nor θ is itself an elasticity: both are scale parameters of the preference shocks. Their inverses are semi-elasticities of choice shares with respect to continuation values, which are measured in log-utility units. See Appendix B.1 for more details.

[19] When $\nu = \theta$, equation (17) reduces to the product of independent Gumbel distributions with scale θ.

[20] This is the same as CDP's indirect utility, but allowing for gender-specific wages.

pendix B.1), I write the value function as

$$V^h_{\boldsymbol{o},t} = \log(W^h_{\boldsymbol{o},t}) + \theta \log\left(\sum_{\ell=1}^{L}\left(\sum_{\boldsymbol{z}\in\mathcal{S}_h(\ell)} \exp\left[\delta V^h_{\boldsymbol{z},t+1} - \mu_{\boldsymbol{oz}}\right]^{1/\nu}\right)^{\nu/\theta}\right). \tag{18}$$

I decompose the choice probability into a location choice and a conditional sector-bundle choice. Intuitively, a household first compares locations based on the value of the sector bundles each offers, and then chooses a bundle within the chosen location. The value of the bundles that location j offers is summarized by

$$\Omega^h_{\boldsymbol{o}j,t} \equiv \sum_{\boldsymbol{z}\in\mathcal{S}_h(j)} \exp\left[\delta V^h_{\boldsymbol{z},t+1} - \mu_{\boldsymbol{oz}}\right]^{1/\nu}.$$

The share of type-h households in bundle $\boldsymbol{o}$ that choose location j is

$$\lambda^h_{\boldsymbol{o}j,t} = \frac{\left(\Omega^h_{\boldsymbol{o}j,t}\right)^{\nu/\theta}}{\sum_{\ell=1}^{L}\left(\Omega^h_{\boldsymbol{o}\ell,t}\right)^{\nu/\theta}}. \tag{19}$$

Conditional on choosing location j, the share of households h that selects destination bundle $\boldsymbol{d}\in\mathcal{S}_h(j)$ is

$$\lambda^h_{\boldsymbol{od}|j,t} = \frac{\exp\left[\delta V^h_{\boldsymbol{d},t+1} - \mu_{\boldsymbol{od}}\right]^{1/\nu}}{\Omega^h_{\boldsymbol{o}j,t}}. \tag{20}$$

The unconditional share of type-h households in bundle $\boldsymbol{o}$ that choose destination bundle $\boldsymbol{d}\in\mathcal{S}_h(j)$ is therefore

$$\lambda^h_{\boldsymbol{od},t} = \lambda^h_{\boldsymbol{od}|j,t}\,\lambda^h_{\boldsymbol{o}j,t}. \tag{21}$$

Thus, $1/\theta$ governs the response of location shares to each location's inclusive value, $\nu \log \Omega^h_{\boldsymbol{o}j,t}$, whereas $1/\nu$ governs the response of within-location shares to the value of each sector bundle.

In the model, exposure to a sectoral productivity shock worsens labor market outcomes and increases out-migration, consistent with Fact 1 in Section II. Changes in $A_{i,k,t}$ transmitted through trade lower wages in exposed region-sector cells, which in turn lowers the value of those bundles. Based on the choice probabilities in equations (19) and (20), households then leave these bundles and move to other sectors and locations.

Couples' consumption and wives' participation choices. After choosing its bundle, a couple jointly decides how to spend household income and whether the wife works in the market or stays at home, based on the spouses' relative wage and a home-production productivity draw. Each couple draws an independently and identically distributed home-production pro-

ductivity shock $\beta \sim \text{Pareto}(\chi, \kappa)$, with support $\beta \geq \chi > 0$ and shape parameter $\kappa > 0$.[21] The couple then jointly chooses each spouse's private consumption, $c_{\boldsymbol{d},f,t}$ and $c_{\boldsymbol{d},m,t}$, expenditure on the household public good, $q_{\boldsymbol{d},t}$, and whether the wife stays at home, $e_{\boldsymbol{d},t} \in \{0,1\}$, with $e_{\boldsymbol{d},t} = 1$ if she does.[22] When the wife stays at home, the productivity shock increases the effective level of the household public good, $Q_{\boldsymbol{d},t} = \beta^{e_{\boldsymbol{d},t}} q_{\boldsymbol{d},t}$. Home production therefore acts as a multiplier on the public good the household buys.

The public good $Q_{\boldsymbol{d},t}$ captures the economies of scale that couples obtain from living together. A good is public within the household when it is *non-rival*: one spouse's consumption does not reduce what is available to the other, so both spouses consume the full quantity the household buys, like housing or heating (Browning et al., 2014). Two singles who live apart must each pay for their own housing, whereas a couple pays for it once and both spouses enjoy it. Because both spouses value the public good, pooling income always makes a couple potentially better off than the two spouses living apart: the couple can replicate the private consumption each spouse would have as a single, buy the larger of the two amounts of the public good the spouses would have bought alone, and still have income left over. The public good then generates economies of scale from marriage because a given standard of living costs less per person in a couple than for an individual.

Conditional on choosing bundle $\boldsymbol{d}$, the couple solves

$$\begin{aligned} \max_{c_{\boldsymbol{d},f,t}, c_{\boldsymbol{d},m,t}, q_{\boldsymbol{d},t}, e_{\boldsymbol{d},t} \in \{0,1\}} \quad & \left(Q_{\boldsymbol{d},t}^{\gamma} c_{\boldsymbol{d},f,t}^{1-\gamma}\right)^{\alpha} \left(Q_{\boldsymbol{d},t}^{\gamma} c_{\boldsymbol{d},m,t}^{1-\gamma}\right)^{1-\alpha} \\ \text{s.t.} \quad & P_{j,t}\left(c_{\boldsymbol{d},f,t} + c_{\boldsymbol{d},m,t} + q_{\boldsymbol{d},t}\right) = \underbrace{w_{j,n_m,t}^{M} + w_{j,n_f,t}^{F}(1 - e_{\boldsymbol{d},t})}_{I(e)}, \\ & Q_{\boldsymbol{d},t} = \beta^{e_{\boldsymbol{d},t}} q_{\boldsymbol{d},t}, \end{aligned} \tag{22}$$

where $\alpha \in (0,1)$ is the wife's bargaining weight and $\gamma \in (0,1)$ is the expenditure share on the public good.[23] The parameter γ governs the size of the economies of scale from marriage: the larger the share of income spent on the public good, the larger the gain from sharing it. Home production adds to this gain, because when the wife stays at home, her home productivity β raises the effective quantity of the public good, $Q_{\boldsymbol{d},t}$, that both spouses enjoy.[24]

[21] The home-production productivity shock captures the non-monetary value generated when the wife engages in home production (Becker, 1965).

[22] The model abstracts from the intensive margin of the wife's labor supply: the wife either participates in market work or does not. This restriction appears empirically innocuous because, alongside the participation responses documented in Fact 3, wives' hours exhibit no statistically significant response to either spouse's exposure.

[23] The model holds α fixed, thereby ruling out negotiation of intra-household bargaining power.

[24] Section IV.3 and Appendix D.3 describe the calibration process for α, the public-good expenditure share γ, and the scale and shape of the home-production shock, χ and κ, respectively.

Following Chiappori and Mazzocco (2017), I solve the couple's problem sequentially and obtain the optimal allocations for private and public consumption, conditional on $\boldsymbol{d}$ and $e_{\boldsymbol{d},t}$,

$$\begin{aligned} c^*_{\boldsymbol{d},f,t}(e) &= \alpha(1-\gamma)\frac{I(e)}{P_{j,t}}, \\ c^*_{\boldsymbol{d},m,t}(e) &= (1-\alpha)(1-\gamma)\frac{I(e)}{P_{j,t}}, \\ q^*_{\boldsymbol{d},t}(e) &= \gamma\frac{I(e)}{P_{j,t}} \quad \Longrightarrow \quad Q^*_{\boldsymbol{d},t}(e) = \beta^e\gamma\frac{I(e)}{P_{j,t}} \end{aligned} \tag{23}$$

The wife works in the market when the gain in household income from her earnings outweighs the gain in the household public good from her home production. To find the decision rule for $e_{\boldsymbol{d},t}$, define the relative wage r faced by a couple in $\boldsymbol{d}$ as

$$r_{\boldsymbol{d},t} \equiv \frac{w^F_{j,n_f,t}}{w^M_{j,n_m,t}}.$$

The couple's indirect utility conditional on choosing $\boldsymbol{d}$ and drawing β in period t is

$$\mathcal{U}_{\boldsymbol{d},t}(\beta) = \mathcal{A}\frac{w^M_{j,n_m,t}}{P_{j,t}} \begin{cases} (1+r_{\boldsymbol{d},t}) & \text{if} \quad e_{\boldsymbol{d},t}=0 \text{ (wife works in the market)} \\ \beta^\gamma & \text{if} \quad e_{\boldsymbol{d},t}=1 \text{ (wife works at home)}, \end{cases}$$

where $\mathcal{A} \equiv \alpha^{\alpha(1-\gamma)}(1-\alpha)^{(1-\alpha)(1-\gamma)}\gamma^\gamma(1-\gamma)^{1-\gamma}$ collects constants. Hence, the wife works in the market if and only if

$$\begin{aligned} \mathcal{U}_{\boldsymbol{d},t}(e_{\boldsymbol{d},t}=1) &< \mathcal{U}_{\boldsymbol{d},t}(e_{\boldsymbol{d},t}=0) \\ \Longleftrightarrow \quad \beta &< (1+r_{\boldsymbol{d},t})^{1/\gamma}, \end{aligned} \tag{24}$$

which implies that the value of β such that the couple is indifferent between the wife working at home and at the market is $\beta^*_{\boldsymbol{d},t} = (1+r_{\boldsymbol{d},t})^{1/\gamma}$. Working raises household income by the factor $1+r_{\boldsymbol{d},t}$, whereas staying at home raises the effective public good by the factor β, which enters utility with weight γ. The wife therefore stays at home only when her home productivity is high relative to her earnings. Thus, the probability that the wife works in the market conditional on bundle $\boldsymbol{d}$, denoted by $\rho_{\boldsymbol{d},t}$, is[25]

$$\rho_{\boldsymbol{d},t} \equiv \Pr(e_{\boldsymbol{d},t}=0) = 1 - \left(\frac{\chi}{\beta^*_{\boldsymbol{d},t}}\right)^\kappa = 1 - \chi^\kappa(1+r_{\boldsymbol{d},t})^{-\kappa/\gamma}. \tag{25}$$

[25]To avoid corner solutions in the couple's problem, I require $0 < \rho_{\boldsymbol{d},t} < 1$. The upper bound, $\rho_{\boldsymbol{d},t} < 1$, holds for any $\chi > 0$, whereas the lower bound, $\rho_{\boldsymbol{d},t} > 0$, requires χ to lie below the market-work threshold, i.e., $\chi < (1+r_{\boldsymbol{d},t})^{1/\gamma}$. Moreover, because the home-production productivity shock enters the couple's problem as $Q_{\boldsymbol{d},t} = \beta^{e_{\boldsymbol{d},t}}q_{\boldsymbol{d},t}$, the model requires $\chi \geq 1$, ensuring that staying at home never reduces the effective level of the public good. For this condition to hold in every location-sector-time cell, the relevant restriction is $1 \leq \chi < \min_{\boldsymbol{d},t}(1+r_{\boldsymbol{d},t})^{1/\gamma}$.

Given the wife's labor-supply rule in equation (24), the ex ante expected utility from choosing $\boldsymbol{d}$ at time t is[26]

$$W^c_{\boldsymbol{d},t} \equiv \mathbb{E}[\mathcal{U}_{\boldsymbol{d},t}(\beta)] = \mathcal{A}\frac{w^M_{j,n_m,t}}{P_{j,t}}\left[(1+r_{\boldsymbol{d},t})F_\beta(\beta^*_{\boldsymbol{d},t}) + \int_{\beta^*_{\boldsymbol{d},t}}^{\infty}\beta^\gamma f_\beta(\beta)\,d\beta\right]$$
$$= \mathcal{A}\frac{w^M_{j,n_m,t}}{P_{j,t}} \cdot \underbrace{\chi^\gamma\left(1-\rho_{\boldsymbol{d},t}\right)^{-\gamma/\kappa}\frac{\kappa/\gamma-\rho_{\boldsymbol{d},t}}{\kappa/\gamma-1}}_{B_{\boldsymbol{d},t}}.$$

Conditional on choosing $\boldsymbol{d}$, the couple's payoff therefore combines the husband's sector-specific wage with the expected gain from the wife's endogenous labor-supply decision, summarized by $B_{\boldsymbol{d},t}$. Because the couple picks the better of the two options once β is drawn, $B_{\boldsymbol{d},t}$ has a simple interpretation:

$$B_{\boldsymbol{d},t} = \mathbb{E}\left[\max\left\{1+r_{\boldsymbol{d},t},\ \beta^\gamma\right\}\right].$$

It is the expected value, per unit of the husband's wage, of the better of two options: market work, which adds the wife's relative wage to household income, and home production, which scales up the public good. It therefore rises with $r_{\boldsymbol{d},t}$.

Indirect utility. The indirect utility relevant for household h's dynamic problem in equation (18) is[27]

$$W^h_{\boldsymbol{o},t} = \begin{cases} \frac{w^M_{i,k,t}}{P_{i,t}} & \text{if} \quad h = s_m \\ \frac{w^F_{i,k,t}}{P_{i,t}} & \text{if} \quad h = s_f \\ \frac{\mathcal{A}\cdot B_{\boldsymbol{o},t}\cdot w^M_{i,k_m,t}}{P_{i,t}} & \text{if} \quad h = c. \end{cases} \tag{26}$$

Here, $P_{i,t}$ denotes the regional aggregate consumption price index defined in equation (15).

The problem with equation (26) is that the units are not comparable across households. For couples, \$1 of market expenditure can generate more than \$1 in consumption-equivalent value due to economies of scale within marriage, as explained above. Thus, a 1% gain for a spouse and a 1% gain for a single person are percentages of different bases. So, following Chiappori and Meghir (2015), I construct a Money Metric Welfare Index (MMWI) for singles and couples to obtain a comparable money-metric measure of welfare changes for the quantitative analysis.

[26] For $W^c_{\boldsymbol{d},t}$ to be finite, I require $\kappa > \gamma$.

[27] Singles have the same preferences over private consumption and the household good as spouses, $\gamma\log Q + (1-\gamma)\log c$, but do not share Q with a spouse. Their indirect utility is therefore $\log(w/P)$ up to the constant $\gamma\log\gamma + (1-\gamma)\log(1-\gamma)$, which cancels from all welfare changes. This formulation assumes that welfare is comparable across marital statuses; the money-metric index below makes this comparison explicit.

The MMWI converts each spouse's utility into the income a single person would need to achieve that utility.[28] The key difference between a spouse and a single is the price each pays for the public good. A single pays the full market price for the public good, so the MMWI of a single is simply the real wage, w/P. A spouse pays only part of it, a share equal to their bargaining weight, because the other spouse pays the rest and both enjoy all of it. When the wife stays at home, her home production further lowers this price. Let α_g denote the bargaining weight of the spouse of gender g, so that $\alpha_f = \alpha$ and $\alpha_m = 1 - \alpha$. I start from the resources each spouse commands: a share α_g of household income $I(e)$. A wife who stays at home earns no wage, but she is still allocated resources equivalent to αw^M, a share of her husband's wage. I then scale these resources by how much cheaper the public good is for the spouse than for a single individual, weighted by the importance of the public good in consumption, γ. The MMWI of spouse g, denoted by m_g, is therefore

$$m_g = \alpha_g^{1-\gamma} \beta^{e\gamma} I(e). \tag{27}$$

The two spouses' MMWIs add up to more than the household's income, which is how the economies of scale from marriage appear in money terms. Summing across spouses,

$$m_f + m_m = \left(\alpha^{1-\gamma} + (1-\alpha)^{1-\gamma}\right) \beta^{e\gamma} I(e),$$

where $\alpha^{1-\gamma} + (1-\alpha)^{1-\gamma}$ represents the economies of scale within marriage in money terms. Thus, the spouses would jointly require more income than the household possesses if each had to purchase the public good separately. With the MMWI, I measure the indirect utility of couples in the same units as wages and prices, so I can compare it with that of singles.[29]

III.2 Analytical Results for Couples

The couple's problem described above contains two mechanisms that generate Facts 2 and 3, holding wages and prices fixed: the couple's payoff responds asymmetrically to the two spouses' wages, and the probability that the wife works in the market depends on the spouses' relative wages.

The two spouses' wages enter the couple's location choice asymmetrically, which provides the mechanism underlying Fact 2 (couples out-migrate only when the husband is exposed): a decline in the husband's wage lowers the couple's payoff in a location by a larger percentage than the same percentage decline in the wife's wage, as long as she earns less than he does. The husband's wage scales the couple's entire payoff, whereas the wife's wage matters only

[28] Appendix B.2 derives the index step by step.

[29] Because couples choose their bundle before β is realized, I weight welfare changes in Section V by the ex ante index, $\bar{m}_g$, developed in Appendix B.2.

when she works, and even then it is only part of household income. By the envelope theorem, the elasticities of the couple's payoff with respect to each spouse's wage are

$$\varepsilon^{F}_{\boldsymbol{d},t} \equiv \frac{\partial \log W^{c}_{\boldsymbol{d},t}}{\partial \log w^{F}_{j,n_f,t}} = \frac{\rho_{\boldsymbol{d},t}\, r_{\boldsymbol{d},t}}{B_{\boldsymbol{d},t}} \leq \frac{\rho_{\boldsymbol{d},t}\, r_{\boldsymbol{d},t}}{1 + r_{\boldsymbol{d},t}},$$
$$\varepsilon^{M}_{\boldsymbol{d},t} \equiv \frac{\partial \log W^{c}_{\boldsymbol{d},t}}{\partial \log w^{M}_{j,n_m,t}} = 1 - \varepsilon^{F}_{\boldsymbol{d},t}.$$

When $r_{\boldsymbol{d},t} < 1$, $\varepsilon^{F}_{\boldsymbol{d},t} < \varepsilon^{M}_{\boldsymbol{d},t}$: the value of the couple's current location falls by more when the husband's sector is hit, so couples respond to his exposure by moving far more than to hers.

The probability that the wife works in the market in equation (25) depends only on the relative wage r, the parameters of the Pareto distribution, χ and κ, and the public-good expenditure share, γ. I show next how ρ depends on each of them. These results also guide the calibration of χ, κ, and γ in Section IV.

1. *Participation rises with* r:

$$\frac{\partial \rho}{\partial r} = \frac{\kappa}{\gamma} \cdot \frac{1-\rho}{1+r} > 0. \tag{28}$$

A higher wage of the wife relative to her husband's raises the opportunity cost of staying home. Thus, the threshold β^* rises, and some draws that used to justify staying home no longer do, so more wives work.

The response of participation to the relative wage generates the adjustment margin in Fact 3 (wives' participation falls when they are directly exposed and rises when their husbands are), with the direction observed in the data. A decline in the husband's sectoral wage raises $r_{\boldsymbol{d},t}$, increases the threshold $\beta^{*}_{\boldsymbol{d},t}$, and induces more married women to enter market work, whereas a decline in the wife's own sectoral wage lowers $r_{\boldsymbol{d},t}$ and has the opposite effect.

2. *Participation falls with the scale parameter,* χ*, and rises with the shape parameter,* κ:

$$\frac{\partial \rho}{\partial \chi} = -\frac{\kappa}{\chi}(1-\rho) < 0, \qquad \frac{\partial \rho}{\partial \kappa} = -\frac{(1-\rho)\log(1-\rho)}{\kappa} > 0. \tag{29}$$

Raising χ scales the whole distribution of home productivity upward while leaving the threshold unchanged, so every couple gains more when she stays at home. As a result, more couples draw above β^*, so participation falls. Because χ is a scale parameter, the effect is proportional: the elasticity of $1-\rho$ with respect to χ is κ. By contrast, a larger shape parameter concentrates the draws near χ and thins the right tail. Staying home takes an exceptionally good draw, one above β^*, and those draws become rarer, so more wives work.

3. *Participation falls with the public-good share, γ:*

$$\frac{\partial \rho}{\partial \gamma} = -\frac{\kappa(1-\rho)}{\gamma^2}\log(1+r) < 0. \tag{30}$$

A higher public-good share raises the utility from any given β, since β^γ grows with γ when $\beta > 1$. The threshold β^* therefore falls: a smaller home-productivity draw now matches the income gain from working, so more couples choose home production. The effect is proportional to $\log(1+r)$ because $\log \beta^* = \log(1+r)/\gamma$, so a change in γ moves the threshold more when the income gain from working is larger.

III.3 Equilibrium

The economic geography at time t, denoted by Γ_t, consists of the time-varying exogenous sectoral productivities and the time-invariant shares of single individuals born in each region, migration costs, iceberg trade costs, and non-employment benefits: $\Gamma_t = \{A_{i,k,t}, \bar{s}_i, \mu_{\boldsymbol{od}}, \tau_{ik,jn}, b_j\}_{i,j=1,\,k,n=1}^{L,K}$. I collect the remaining time-invariant parameters in the vector $\bar{\Gamma} = \{N, L, K, \delta, \chi, \kappa, \alpha, \gamma, \sigma, \nu, \theta, \{\psi_{k,m}, \psi_{k,f}, \eta_k, \iota_k\}_{k=1}^K\}$.

Let $N_{\boldsymbol{d},t}^{s_m}$ and $N_{\boldsymbol{d},t}^{s_f}$ denote, respectively, the masses of single men and single women living in location j and working in sector $n \in \{0, \ldots, K\}$ at time t. Let $N_{\boldsymbol{d},t}^{c}$ denote the mass of couples living in location j in which the husband is assigned to sector $n_m \in \{0, \ldots, K\}$ and the wife is assigned to productive market sector $n_f \in \{1, \ldots, K\}$. The supplies of male and female labor to productive sectors $n = 1, \ldots, K$ are therefore

$$M_{j,n,t} = N_{j,n,t}^{s_m} + \sum_{n_f=1}^{K} N_{j,n,n_f,t}^{c}, \qquad F_{j,n,t} = N_{j,n,t}^{s_f} + \sum_{n_m=0}^{K} \rho_{j,n_m,n,t} N_{j,n_m,n,t}^{c}. \tag{31}$$

Male labor supply in sector n sums over the wife's sector, whereas female labor supply in sector n sums over the husband's sector and weights each cell by the wife's probability of participation. Thus, female labor supply responds to wages through both mobility and participation, whereas male labor supply responds only through mobility.

Let $N_t = \left\{\{N_{\boldsymbol{d},t}^{s_m}\},\ \{N_{\boldsymbol{d},t}^{s_f}\},\ \{N_{\boldsymbol{d},t}^{c}\}\right\}$ denote the complete distribution of household masses across locations and sectoral bundles at time t. Because the problems of intermediate-good firms, final-good firms, and married women's labor supply are static, I define the *temporary equilibrium* as follows.

Definition 1. *Given $(N_t, \Gamma_t, \bar{\Gamma})$, a **temporary equilibrium** is a vector of productive-sector wages $\{\bar{w}_{j,n,t}, w_{j,n,t}^M, w_{j,n,t}^F\}$, sectoral prices $\{p_{j,n,t}, P_{j,n,t}\}$, regional price indices $\{P_{j,t}\}$, and allocations $\{\rho_{\boldsymbol{d},t}, M_{j,n,t}, F_{j,n,t}, Y_{j,n,t}, Q_{j,n,t}, Q_{j,t}, \pi_{ik,jn,t}\}$ such that, for every productive region-sector pair jn and period t:*

(i) Intermediate-good firms maximize profits subject to the CES production technology, and wages satisfy (10) and (13);

(ii) Final-good firms maximize profits subject to sectoral CES and regional Cobb-Douglas aggregation. Sectoral price indices satisfy (14), and regional price indices satisfy (15);

(iii) Couples optimally choose the wife's labor supply, and participation probabilities satisfy (25) for every sector pair (n_m, n_f);

(iv) Labor markets clear: $p_{j,n,t}Y_{j,n,t} = w^M_{j,n,t}M_{j,n,t} + w^F_{j,n,t}F_{j,n,t}$;

(v) Goods markets clear under balanced trade: $p_{i,k,t}Y_{i,k,t} = \sum_{j=1}^{L}\sum_{n=1}^{K}\pi_{ik,jn,t}\,\iota_n P_{j,t}Q_{j,t}$.

Given the path of exogenous fundamentals and an initial distribution of households across bundles, denoted by $N_0 = \left\{\{N^{s_m}_{\boldsymbol{o},0}\},\ \{N^{s_f}_{\boldsymbol{o},0}\},\ \{N^{c}_{\boldsymbol{o},0}\}\right\}$, I define the model's *sequential competitive equilibrium*. Using the two-stage choice probabilities in equations (19) and (20), the population of each household type evolves according to

$$N^h_{\boldsymbol{d},t+1} = \sum_{\boldsymbol{o}\in\mathcal{S}_h} \lambda^h_{\boldsymbol{od}|j,t}\,\lambda^h_{\boldsymbol{o}j,t}\,N^h_{\boldsymbol{o},t}, \qquad \boldsymbol{d}\in\mathcal{S}_h(j), \quad \forall h. \tag{32}$$

Definition 2. *Given $(N_0, \Gamma_t, \bar{\Gamma})$, a **sequential competitive equilibrium** is a sequence of productive-sector wages $\{\bar{w}_{j,n,t}, w^M_{j,n,t}, w^F_{j,n,t}\}_{t=0}^{\infty}$, prices $\{p_{j,n,t}, P_{j,n,t}, P_{j,t}\}_{t=0}^{\infty}$, values $\{V^h_{\boldsymbol{o},t}\}_{t=0}^{\infty}$, and allocations $\{\rho_{\boldsymbol{d},t}, \pi_{ik,jn,t}, \lambda^h_{\boldsymbol{o}j,t}, \lambda^h_{\boldsymbol{od}|j,t}, N^h_{\boldsymbol{o},t}, M_{j,n,t}, F_{j,n,t}, Y_{j,n,t}, Q_{j,n,t}, Q_{j,t}\}_{t=0}^{\infty}$ for every household type h, such that, in each period t:*

(i) The temporary equilibrium holds;

(ii) Couples and single individuals solve their respective migration problems: value functions satisfy (18), while location and conditional sectoral choice probabilities satisfy (19) and (20);

(iii) The population of each household type evolves according to (32);

(iv) Productive-sector male and female labor supplies satisfy (31).

III.4 Welfare Analysis of a Sectoral Productivity Shock

To quantify welfare, I combine equation (18) with $\mu_{\boldsymbol{oo}} = 0$ to express the expected lifetime utility of household type h in bundle $\boldsymbol{o}$ at time t as

$$V^h_{\boldsymbol{o},t} = \log(W^h_{\boldsymbol{o},t}) + \delta V^h_{\boldsymbol{o},t+1} - \theta\log\left(\lambda^h_{\boldsymbol{o}i,t}\right) - \nu\log\left(\lambda^h_{\boldsymbol{oo}|i,t}\right)$$

$$= \sum_{s=t}^{\infty} \delta^{s-t} \log \left(\frac{W^h_{\boldsymbol{o},s}}{\left(\lambda^h_{\boldsymbol{o}i,s}\right)^{\theta} \left(\lambda^h_{\boldsymbol{oo}|i,s}\right)^{\nu}} \right). \tag{33}$$

Here, $\lambda^h_{\boldsymbol{o}i,t}$ denotes the probability of remaining in the origin location, and $\lambda^h_{\boldsymbol{oo}|i,t}$ denotes the probability of remaining in the origin sector bundle, conditional on remaining in the origin location. Appendix B.1 derives this recursion.

To measure the welfare effect of a sectoral productivity shock, I compare two economies that differ only in the path of sectoral productivities from $t = 1$ on. I call the economy with the new path, $\{A'_{j,n,t}\}_{t\geq 1}$, the *with-shock economy*, and the economy with the original path, $\{A_{j,n,t}\}_{t\geq 1}$, the *no-shock economy*. The change is unanticipated at $t = 0$, and households fully learn the new productivity path at $t = 1$, as in CDP. Let x denote the value of a given variable in the no-shock economy and x' the value of the same variable in the with-shock economy. Given that the changes in productivities happen from $t \geq 1$, $x_{t=0} = x'_{t=0}$.

Let $\dot{x}_{t+1} \equiv x_{t+1}/x_t$, and $\hat{x}_{t+1} = \dot{x}'_{t+1}/\dot{x}_{t+1}$ denote the ratio of changes over time in the with-shock and no-shock equilibria.[30] Using this notation, I compute expected lifetime utility in the no-shock economy, $V^h_{\boldsymbol{o},0}$, and in the with-shock economy, $V'^h_{\boldsymbol{o},0}$. I define the consumption compensating variation for household type h in bundle $\boldsymbol{o}$ at time $t = 0$, denoted by $\zeta^h_{\boldsymbol{o}}$, implicitly by

$$V'^h_{\boldsymbol{o},0} = V^h_{\boldsymbol{o},0} + \sum_{s=0}^{\infty} \delta^s \log(\zeta^h_{\boldsymbol{o}}) \iff \log(\zeta^h_{\boldsymbol{o}}) = (1-\delta)(V'^h_{\boldsymbol{o},0} - V^h_{\boldsymbol{o},0}).$$

Because the temporary equilibrium and migration shares are identical across the two economies at $t = 0$, the welfare change in consumption-equivalent units is

$$Z^h_{\boldsymbol{o}} \equiv \log(\zeta^h_{\boldsymbol{o}}) = \sum_{s=1}^{\infty} \delta^s \log \left(\frac{\hat{W}^h_{\boldsymbol{o},s}}{\left(\hat{\lambda}^h_{\boldsymbol{o}i,s}\right)^{\theta} \left(\hat{\lambda}^h_{\boldsymbol{oo}|i,s}\right)^{\nu}} \right), \tag{34}$$

where

$$\hat{W}^h_{\boldsymbol{o},s} = \begin{cases} \hat{w}^M_{i,k_m,s}/\hat{P}_{i,s} & \text{if} \quad \boldsymbol{o} = (i,k_m),\ h = s_m, \\ \hat{w}^F_{i,k_f,s}/\hat{P}_{i,s} & \text{if} \quad \boldsymbol{o} = (i,k_f),\ h = s_f, \\ \hat{B}_{\boldsymbol{o},s}\hat{w}^M_{i,k_m,s}/\hat{P}_{i,s} & \text{if} \quad \boldsymbol{o} = (i,k_m,k_f),\ h = c. \end{cases}$$

Equation (34) decomposes the welfare change into changes in real income, which for couples include the change in the value of the wife's participation option, $\hat{B}_{\boldsymbol{o},s}$, and changes in the option value of moving, captured by the staying probabilities. The staying probabilities

[30] See Appendix C for the full derivation of the model in Dynamic Hat Algebra.

matter because a lower probability of staying indicates that a household's options elsewhere have improved relative to its current bundle, raising its welfare even if it ultimately stays. Conditional on the observed staying shares, a larger θ increases the welfare weight assigned to changes in the probability of remaining in the origin location. Similarly, a larger ν increases the weight assigned to changes in the probability of remaining in the origin sector bundle conditional on staying in that location. Thus, declines in these probabilities enter the welfare measure with weights θ and ν, respectively, where $\theta \geq \nu$.

I aggregate welfare changes across initial location-sector bundles to obtain the aggregate welfare effect of a sectoral productivity shock:[31]

$$Z = \underbrace{\sum_{i=1}^{L}\sum_{k=0}^{K} N_{\boldsymbol{o},0}^{s_m} Z_{\boldsymbol{o}}^{s_m}}_{Z^{sm}} + \underbrace{\sum_{i=1}^{L}\sum_{k=0}^{K} N_{\boldsymbol{o},0}^{s_f} Z_{\boldsymbol{o}}^{s_f}}_{Z^{sf}} + \underbrace{\sum_{i=1}^{L}\sum_{k_m=0}^{K}\sum_{k_f=1}^{K} N_{\boldsymbol{o},0}^{c} Z_{\boldsymbol{o}}^{c}}_{Z^{c}}. \tag{35}$$

This welfare measure lets me compare, in consumption-equivalent units, how a sectoral productivity shock affects each household type across initial locations and sectors. In particular, I can compare the gains of singles and couples, of couples in which only one spouse works in an exposed sector, and of people living in different locations. The next section brings the model to the data, and Section V then uses this measure to show how the welfare effects of the China Trade Shock differ across household types and locations, and how couples change who bears the shock.

IV. Bringing the Model to the Data

I bring the model to the data by measuring most inputs directly in the data, taking a few parameters from the literature, and calibrating only the parameters that govern migration across states and the couple's problem. I first describe the quantitative environment, distinguishing parameters drawn from the literature from inputs observed directly in the data. I then present the dispersion of preferences across sectors and locations, the calibration of the couple-specific parameters, and the method for quantifying the China Trade Shock.

[31] In Section V, I aggregate these welfare changes, weighting each person by their money metric before the shock, so that the aggregate adds up gains in money terms rather than percentages on different bases (Appendix B.2).

IV.1 Quantitative Environment

The economy comprises 50 US states, 37 foreign trade partners, and 22 productive sectors, plus a non-employment sector.[32] Each model period corresponds to one quarter, and I solve the model for 200 quarters beginning in 2000Q1. The quantitative exercise and the empirical analysis in Section II cover different windows: the empirical analysis follows ADH's 2001–2012 window (Autor et al., 2021), whereas the quantitative exercise uses the 2000–2007 window (Autor et al., 2013), because I calibrate both the shock and the dispersion of preferences across locations to the estimates in that paper.

Parameters from previous literature. I set the discount factor to $\delta = 0.99$, the standard quarterly value, take the sectoral trade elasticities $\{\eta_n\}$ from Caliendo and Parro (2015), and take the within-sector elasticity of substitution between male and female labor, σ, from Acemoglu et al. (2004) rather than estimating it directly. Estimating σ myself would be difficult. The CES technology in Section III implies:

$$\log\left(\frac{w^M_{i,k,t}}{w^F_{i,k,t}}\right) = \log\left(\frac{\psi_{k,m}}{\psi_{k,f}}\right) - \frac{1}{\sigma}\log\left(\frac{M_{i,k,t}}{F_{i,k,t}}\right).$$

Identifying σ requires exogenous variation in the relative supply of male and female labor, because relative supply also responds to relative wages; without such variation, the estimated coefficient would be biased. Acemoglu et al. (2004) estimate an analogous relationship (their equation (11)) by treating the mobilization of men during World War II as an exogenous shift in women's relative labor supply. Accordingly, I set σ to 1.4, the lower bound of their estimated range.[33]

Directly observed inputs. I measure the trade and production side of the economy over 2000–2007 using international and US data. Bilateral trade shares $\{\pi_{ik,jn,t}\}$ and region-sector output and expenditure come from the World Input-Output Database (WIOD) for countries, and from the 2002 Commodity Flow Survey (CFS) and the Bureau of Economic Analysis (BEA) for US states. Final-good expenditure shares $\{\iota_n\}$ come from the BEA input-output

[32] The foreign trade partners are China, 35 other countries in the World Input-Output Database, and a constructed Rest of the World. The productive sectors are 12 manufacturing sectors, wholesale and retail trade, construction, and 8 service sectors. This geography and sectoral partition follow CDP.

[33] This conservative choice reflects the fact that the cohorts examined in their study were almost entirely retired by 2000Q1, when the quantitative model begins. Since then, as more women entered the labor force, they specialized in occupations where they had a comparative advantage, so men and women sort differently across occupations and are less substitutable within a sector (Ngai and Petrongolo, 2017; Cortes et al., 2023; Bhalotra et al., 2025).

tables.[34] Panel A of Table 4 summarizes these inputs and their sources.

I construct household-side inputs from US microdata. I measure the initial distribution of households across states and sector pairs, $\{N^{s,m}_{\boldsymbol{o},0}, N^{s,f}_{\boldsymbol{o},0}, N^{c}_{\boldsymbol{o},0}\}$, and the state-level share of singles, $\bar{s}_i$, using the 5% IPUMS Census 2000. I measure male and female employment, wages, and wives' labor force participation using the 2000 Census and the American Community Survey (ACS) for 2001–2007. I recover the sector-specific gender-intensity parameters $\{\psi_{k,m}, \psi_{k,f}\}$ from the hours worked by gender in each sector. I measure regional non-employment income, b_j, using state unemployment-insurance benefits from the US Department of Labor and hold it fixed at its 2000 value. Panel B of Table 4 summarizes these inputs and their sources.

Because I solve the model using Dynamic Hat Algebra, I do not need to estimate mobility costs; instead, I measure the initial unconditional transition probability $\lambda^{h}_{\boldsymbol{od},0}$ in equation (21) as the product of an interstate migration probability from the ACS, which gives the location share, and a sectoral transition probability from the Current Population Survey (CPS), which gives the within-location share. Because the CPS follows addresses and drops respondents who move, it records no sectoral transitions for interstate movers, so for movers, I impute the sectoral transition probability using the destination state's sectoral transition matrix:[35]

$$\lambda^{h}_{(i,k)(j,n),0} = \underbrace{f^{h}_{ij}}_{\text{ACS}} \times \underbrace{P^{h}_{j}(n \mid k)}_{\text{CPS}}.$$

Finally, I assign each non-employed wife in the data to her potential sector. The problem is that the model assigns every wife to a productive sector, but the data do not reveal the potential sector of a non-employed wife. If I were to assign non-employed wives to productive sectors in proportion to the sectoral distribution of employed wives, I would create a selection problem: sectors employing larger shares of wives would also receive larger shares of non-employed wives, even though wives assigned to those sectors are precisely the least likely to be out of the labor force. To recover each non-employed wife's potential sector, I use the model to derive a selection correction.[36] The implied mass of non-employed married women

[34] The quantitative exercise uses CDP's full production block rather than the simplified version presented in Section III. Intermediate-good firms use both labor and intermediate inputs from other sectors, and total factor productivity combines the Hicks-neutral component introduced in Section III with a variety-specific component. Implementing the household extension therefore requires replacing the labor input demanded by intermediate-good firms with the effective labor aggregate in equation (9).

[35] This construction presents two challenges: the surveys differ in frequency, and CPS cells become sparse once split by household type. First, the ACS is annual, whereas the CPS is monthly. I address this mismatch following CDP. Second, to address the sparse cells, I temporarily aggregate the 12 manufacturing sectors into a single sector, and do the same for the service sectors. I then disaggregate the resulting transition probabilities using national gender-specific sorting patterns. Appendix D.2 documents each step in the construction of the migration matrices.

[36] See Appendix D.1 for details.

[Proposed: Chinese productivity and trade elasticity rows removed]

Table 4: Observed Quantitative Inputs

Object	Description	Source
Panel A: Trade and Production Inputs		
$\pi_{ik,jn,t}$	Trade shares between countries	WIOD, 2000–2007
	Trade shares among US states	2002 CFS and WIOD
	Trade shares, US states and other countries	WIOD and BEA state employment
$p_{i,k,t}Y_{i,k,t}$	Output and expenditure by region × sector	BEA (US states), WIOD (other countries)
—	Value-added shares and input-output coefficients	BEA (US states), WIOD (other countries)
ι_n	Final-good expenditure shares	BEA input-output tables
Panel B: Household Inputs		
$\bar{s}_i$	State-level share of singles	IPUMS Census 2000
N_i	State-level population	IPUMS Census 2000 and ACS 2001–2007
$N^{s,m}_{\boldsymbol{o},0}, N^{s,f}_{\boldsymbol{o},0}$	Initial populations of single men and women by state × sector	IPUMS Census/ACS
$N^{c}_{\boldsymbol{o},0}$	Initial population of couples by state × husband-wife sector pair	IPUMS Census/ACS
$M_{\boldsymbol{o},t}, F_{\boldsymbol{o},t}$	Male and female employment by state × sector	IPUMS Census/ACS
$w^{M}_{i,k,t}, w^{F}_{i,k,t}$	Male and female wages by state × sector	IPUMS Census/ACS
$\psi_{k,m}, \psi_{k,f}$	Sector-specific gender intensities	Hours worked by gender, IPUMS Census/ACS
b_j	State-level non-employment income	US Department of Labor (unemployment insurance)
$\lambda^{h}_{\boldsymbol{od},0}$	Initial transition probabilities	ACS (states) and CPS (sectors)

Notes. WIOD: World Input-Output Database; CFS: Commodity Flow Survey; BEA: Bureau of Economic Analysis; CPS: Current Population Survey. In Panel B, the sample includes individuals aged 25–65 across 50 states, excluding widows. Non-employment refers to being out of the labor force rather than being unemployed. Sectors are classified according to a 22-sector partition of NAICS industries.

in state j and sector n is

$$NonF^c_{j,n,t} = \underbrace{F^c_{j,n,t}}_{\text{observed}} \cdot \underbrace{\frac{1-\rho_{j,n,t}}{\rho_{j,n,t}}}_{\text{the model's odds of non-employment}},$$

where $\rho_{j,n,t}$ is the participation rate of married women in state j and sector n implied by the model (equation (25)). Scaling observed employment by $1/\rho_{j,n,t}$ recovers all wives attached to each sector, employed or not, so sectors in which wives rarely work receive more non-employed wives. This corrects for selection and assigns non-employed wives to their potential productive sectors.

IV.2 Dispersion of Preferences across Sectors and Locations

The model features ν, which governs the dispersion of preferences across sectors within a location (equation (20)), and θ, which governs their dispersion across locations (equation (19)). The larger a dispersion parameter, the more choices are driven by idiosyncratic preferences and the less they respond to differences in continuation values.

I set ν to 5.34, the quarterly value estimated by CDP, because their estimate comes only from workers who switch sectors without changing location, which is precisely the choice that ν governs in equation (20).[37] The matched panel therefore records sectoral transitions only among individuals who remain in the same location and contains no cross-location transitions. The only difference from CDP's setting is that in my model, continuation values V^h are household-type-specific, reflecting the economies of scale associated with marriage. I use the same value for all household types, so the model does not impose differential rates of sectoral substitution across household types.

I calibrate θ by indirect inference to match the population response to import competition in Autor et al. (2013), following Rodríguez-Clare et al. (2026). This moment disciplines θ because, given ν, θ determines how much a state's population responds when the value of living there changes. Autor et al. (2013) regress the change in the log of the working-age population of a commuting zone on its exposure to Chinese imports and obtain a coefficient of -0.050. I estimate the same regression using model-generated state-level data.[38] The

[37] Following Artuc et al. (2010), CDP estimate their dispersion parameter using sectoral transitions among men in the March CPS from 1975 to 2000. As noted above, the CPS drops respondents who change residence.

[38] The dependent variable is the percentage change in state population between 2000 and 2007, scaled to a decade as in Autor et al. (2013). The regressor is the state's exposure to Chinese imports over 2000–2007, predicted using the Autor et al. (2013) instrument and rescaled to the mean of their commuting-zone measure so that the model and empirical coefficients have the same units. I restrict the model sample to the 48 states used in the Autor et al. (2013) estimation, excluding Alaska and Hawaii. Using first-stage fitted exposure makes the ordinary least squares coefficient in the model the counterpart of the two-stage least squares coefficient in the data.

calibrated value, $\theta = 10.67$, generates a coefficient of -0.051, compared with the target of -0.050. Because $\nu < \theta$, households substitute more across sectors within a location than across locations.

IV.3 Calibrating the Couple's Problem

The couple's decisions about private consumption, public consumption, and whether the wife works in the market or at home depend on the wife's weight in the household objective α, the public-good expenditure share γ, and the lower bound and shape of the home-production shock, χ and κ. Because I solve the couple's problem sequentially, following Chiappori and Mazzocco (2017), I also identify them sequentially, each from a distinct piece of variation: α and γ from household expenditure shares, the slope κ/γ from the China Trade Shock, and χ from the level of married women's participation. Table 5 summarizes the resulting values; Appendix D.3 describes the construction in detail and discusses their values in comparison with the previous literature.

I calibrate α and γ to match household expenditure shares. Because the couple's utility function in equation (22) is Cobb–Douglas, α corresponds to the wife's share of total private consumption, while γ corresponds to the share of public-good expenditure in total household expenditure. I measure both expenditure shares using the 2000–2007 Consumer Expenditure Interview Survey. For γ, I use the share of total expenditure devoted to market inputs into home production.[39] The wife's weight α, in contrast, cannot be directly calculated from expenditures because the observed assignable private good, adult apparel, represents only a subset of private consumption. I therefore recover the spouses' resource shares using the method of Browning et al. (2013), which relies on expenditures by *singles*, and set α to the wife's share of the spouses' combined resources. The resulting values are $\alpha = 0.47$ and $\gamma = 0.49$.

I calibrate κ and χ using the wife's participation rule in equation (25), which is log-linear in the relative wage. Expressing the equation in terms of the non-participation probability and taking logs yields

$$\log\big(1 - \rho(r)\big) = \underbrace{\kappa \log \chi}_{\text{intercept}} - \underbrace{(\kappa/\gamma)}_{\text{slope}} \log(1 + r). \tag{36}$$

Direct estimation of equation (36) is complicated by endogeneity and sample selection. First, the relative wage r faced by a couple is an equilibrium outcome. Second, I observe relative wage changes only for couples in which the wife participates in the market, creating a trun-

[39] The home-production goods comprise shelter, utilities, food at home, and market substitutes for home time, such as domestic services and appliance repair.

cation problem.

To address the endogeneity of r, I use the husband's exposure to the China Trade Shock in the PSID (Section II) as the source of exogenous variation in r.[40] When the husband is exposed, the China Trade Shock directly affects his wage but not his wife's wage, influencing her participation decision through the relative wage r, as required by the exclusion restriction underlying equation (36). Equation (36) is linear, so I identify its slope using the following Wald ratio, measuring both changes within the same empirical experiment:

$$\frac{\kappa}{\gamma} = -\frac{\Delta \log(1-\rho)}{\Delta \log(1+r)}.$$

The labor force participation response of wives of exposed husbands in Panel B of Table 3 implies $\Delta \log(1-\rho) = -0.212$, while the change in relative wages of spouses when the husband is exposed (Table D.1) implies $\Delta \log(1+r) = 0.048$.[41] Together, these estimates imply[42]

$$\frac{\kappa}{\gamma} = -\frac{-0.212}{0.048} = 4.44 \quad \Longrightarrow \quad \kappa = \gamma \times \frac{\kappa}{\gamma} = 2.17.$$

Conditional on κ/γ, I internally calibrate χ to match the *level* of married women's labor force participation at $t = 0$, so that the model's mass-weighted average participation rate equals its empirical counterpart in 2000Q1, 0.621. The resulting value is $\chi = 1.92$.[43]

The calibrated parameters imply that married women's labor force participation responds roughly one-for-one to the female wage, holding the husband's wage fixed. Differentiating ρ with respect to the female wage and evaluating the resulting expression at $\rho_{t=0} = 0.621$ and $r_{t=0} = 0.607$ yields

$$e \;=\; \frac{\partial \log \rho}{\partial \log w_f} \;=\; \frac{\kappa}{\gamma} \cdot \frac{1-\rho_{t=0}}{\rho_{t=0}} \cdot \frac{r_{t=0}}{1+r_{t=0}} \;=\; 1.02.$$

Thus, a 1% increase in the female wage raises married women's labor force participation by 1.02%. This elasticity lies within the range for married women surveyed by Blundell and MaCurdy (1999).

[40] Because I simulate this same shock in the quantitative exercise, the model's participation response is pinned down by how wives actually responded to it, rather than by unrelated variation in wages.

[41] Wives of exposed husbands increase their labor force participation by 4.2 p.p. from a baseline of 0.78. The husband's exposure raises the relative wage r by 12.2 log points, which corresponds to a 4.8 log-point increase in $1+r$. Because I observe wages only for employed individuals, I evaluate this change at $r_{t=0} = 0.607$, the relative wage of all couples, rather than at the mean among couples in which the wife works (Appendix 2).

[42] The resulting calibration satisfies the model's theoretical restriction, $\kappa > \gamma$.

[43] This value of χ satisfies the support restriction $\chi \geq 1$ and yields an interior participation margin in every cell. The calibration uses two participation rates for different purposes: the slope comes from the PSID experiment, whose baseline participation rate is 0.78, whereas χ matches the participation rate of the model population, 0.621.

Table 5: Calibrated Couple's Parameters

Object	Name	Value
α	Wife's weight in the household objective	0.47
γ	Public-good expenditure share	0.49
κ/γ	Slope of the participation rule	4.44
κ	Pareto shape parameter for home productivity	2.17
χ	Pareto lower bound for home productivity	1.92

Notes. See Appendix D for details.

IV.4 Quantifying the China Trade Shock

I represent the China Trade Shock, as in CDP, as an increase in total factor productivity in *Chinese* manufacturing sectors and calibrate the path of Chinese sectoral productivity so that the resulting growth in US imports from China reproduces the first stage in Autor et al. (2013), and I hold Chinese productivity constant after 2007.[44] The model therefore defines the shock at the industry level, consistent with the reduced-form evidence in Section II. I quantify the shock by comparing the two economies defined for productivity shocks in Section III.4. The *with-shock* economy includes the calibrated growth in Chinese productivity, whereas the *no-shock* economy eliminates this growth.

All results reported below are differences between the with-shock and no-shock economies. Accordingly, a positive employment effect indicates that the China Trade Shock increased employment relative to a world without the shock, while a positive welfare effect indicates that the relevant household type was better off as a result.

V. Couples and the Adjustment to the China Trade Shock

Couples change how the economy adjusts to the China Trade Shock and who bears it. In this section, I quantify how the China Trade Shock affected each household type and how couples matter for this effect. First, I examine the effects of the China Trade Shock on manufacturing employment, total employment, population changes, participation responses, and welfare. Second, to understand the mechanisms behind these effects and how couples matter for them, I compare the baseline economy with an all-singles economy in which every

[44] The shock is a foreign supply shock that is transmitted through the trade block as a demand shock to US producers: cheaper Chinese imports reduce both the prices and quantities at which US manufacturers sell in domestic and third-country markets.

couple is split into two singles, and with economies in which I shut down migration, partially shut down wives' participation response, or both.

V.1 Quantitative Effects of the China Trade Shock

Through the lens of the model, the China Trade Shock moves workers out of manufacturing, draws more people into work, and raises welfare across all groups, but households absorb it very differently depending on who they live with and where they live. I first show how large the shock is and how it reallocates employment, then how it changes welfare, and finally how both vary across states.

The China Trade Shock permanently shrinks manufacturing. The manufacturing share of employment declines over the first 15 years (Figure 3a) and never recovers, ending 0.29 p.p. lower by $t = 199$ (Panel A of Table 6).[45] The losses fall mostly on men. Women's manufacturing decline is about half that of men's and varies little by marital status, whereas married men's decline is 44% larger than single men's. Part of the gap between married and single men arises because more married men worked in manufacturing in 2000Q1, but not all of it: even accounting for that, married men's decline is still 22% larger.

Rather than pushing workers into non-employment, the shock raises the aggregate employment rate by 0.11 p.p. (Panel A of Table 6). Every group works more, especially single women (Figure 3b). The shock raises real wages: cheaper Chinese manufactured goods lower the US price index, and because the non-employment benefit is fixed, working becomes more attractive than staying out of work. Non-manufacturing sectors expand, absorbing workers leaving manufacturing and drawing in people who were not working. Services, the sector that grows the most, gain 18% more jobs per capita than manufacturing loses (Figure 3c).

Higher real wages are enough to explain why singles work more, but couples adjust differently, with each spouse using a different margin. Wives enter the labor force through the added-worker effect: the shock hits male-intensive manufacturing and lowers husbands' wages relative to their wives', so more couples choose to have the wife work (equation (28)). Husbands, most of whom were already working, instead move out of manufacturing into other sectors. By $t = 199$, married men fill 53% of the non-manufacturing jobs created by the shock, although they held only 33% of them in 2000Q1.

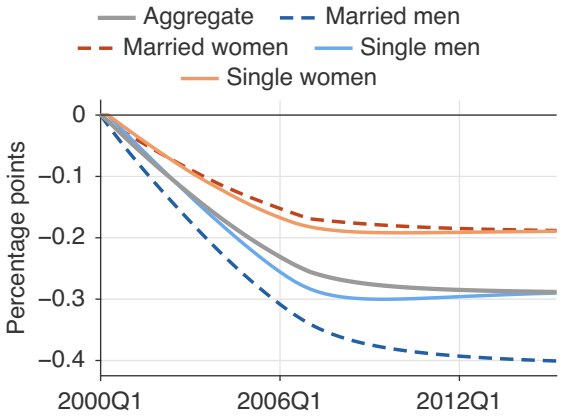
These adjustments leave every group better off. The average individual in the US gains 0.18% in consumption-equivalent terms, because real wages rise and more people work.[46]

[45] Measured as manufacturing jobs per capita, the metric in CDP, the decline is similar in my model and in CDP (-0.21 vs -0.23 p.p.; Table E.1).

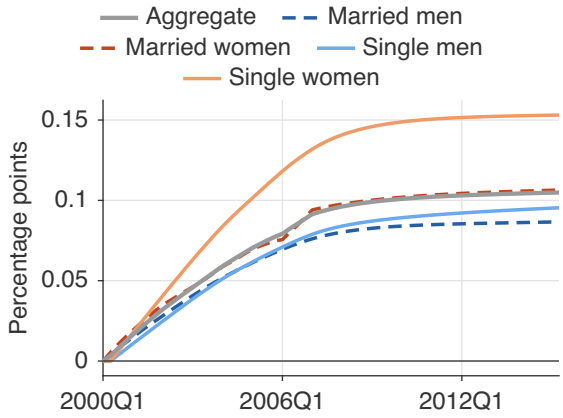
[46] One potential concern would be that considering couples changes the trade equilibrium. In Figure E.5, I compare the welfare gains for 35 countries, excluding China, and the rest of the world, in my model and in CDP. The two series correlate at 0.997, and the same 30 of 36 regions gain in both. Welfare in the rest

Figure 3: Employment Effects of the China Trade Shock in the Short Run

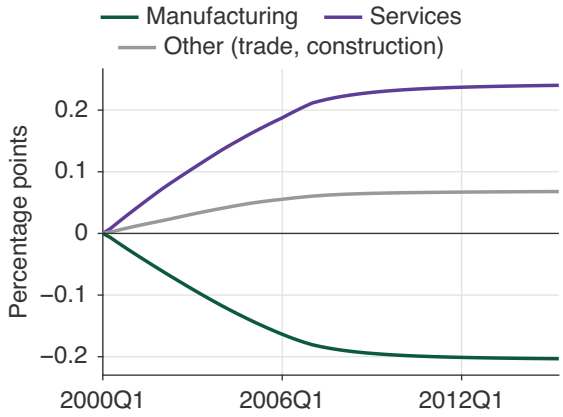
(a) Manufacturing share of employment



(b) Employment rate



(c) Employment per capita, by sector



Notes. Responses to the shock (with-shock minus no-shock economy) over the first 62 quarters (2000Q1–2015Q2). Panel (a) is the manufacturing share of employment, in p.p. of employment; panel (b) is the employment rate, in p.p. of population; panel (c) is employment per capita by sector, in p.p. of population, where manufacturing covers the 12 manufacturing sectors and other covers wholesale and retail trade and construction.

These gains, however, differ across household types. Single women gain the most, 34% more than the average individual, and single men the least (Panel A of Table 6). Single women benefit because they work in expanding sectors, rarely work in the 12 manufacturing sectors hit by the shock, and can move on their own. Couples fall in between single men and single women, but much closer to single men, because the husband's wage drives the household's gain, while the wife's contribution depends on whether she works.

For the same reason, the husband's exposure to the shock matters more for a couple's

of the world is also the same in every economy I study below (Table E.1).

gain than the wife's (Panel B of Table 6). When his wage falls, the household's whole income falls with it. When hers falls, the couple can fall back on her home production, which absorbs part of the loss. Couples in which neither starts in manufacturing gain the most, about as much as the average individual. When only the wife is exposed, the couple's gain falls by 10%; when only the husband is, it falls by more than twice as much, 22%. Couples in which both spouses are exposed gain 30% less.

Who is exposed also shapes how couples adjust: couples move when the husband is exposed, but not when the wife is (Panel B of Table 6). Exposed married women are the least likely of all workers to move (Panel A), because the couple's location follows the husband's job, not hers. Instead, wives adjust by working: their participation rises in every type of couple, slightly more when the husband is exposed.

These national averages hide large differences across states, because manufacturing was far from evenly spread. In 2000Q1, it was concentrated around the Great Lakes, in Indiana, Michigan, Wisconsin, and Ohio, and across a Southern band from Arkansas to the Carolinas, and it was much less present in the Mountain West and Florida (Figure 4a). These are the states the shock hits hardest, and people there respond in two ways: some leave, and wives go to work. The larger a state's manufacturing share in 2000Q1, the more people it loses (Figure 4b), and the more married women's participation rises (Figure 4c), so wives work more in the states losing people.

Welfare gains, however, track where people leave much more closely than where manufacturing was. The four states that lose the most people, Louisiana, Kentucky, Indiana, and Mississippi, are also the four that gain the least (Figures E.6a and E.6c), and only one of them, Indiana, is among the Great Lakes states with the highest manufacturing shares. The link to population is tight because people leave places whose value falls the most, so population losses reveal where the shock hurts. The geography of gains is nearly identical across household types (Figure E.7), but where a couple lives matters more for its gains than it does for singles: across states, couples' gains vary 35% more than single men's and about twice as much as single women's. A couple must live in one place, so it cannot take one job in one state and the other in another, and its gains depend more on where it starts.

V.2 Why Couples Matter for How the Economy Adjusts

The previous subsection shows what the China Trade Shock does through the lens of the model; to learn why, and how couples matter for it, I turn to counterfactual economies. In the baseline economy, couples' joint decisions, migration, and wives' participation all respond to the shock simultaneously, so the baseline economy alone cannot distinguish among them.

Table 6: Long-Run Effects of the China Trade Shock

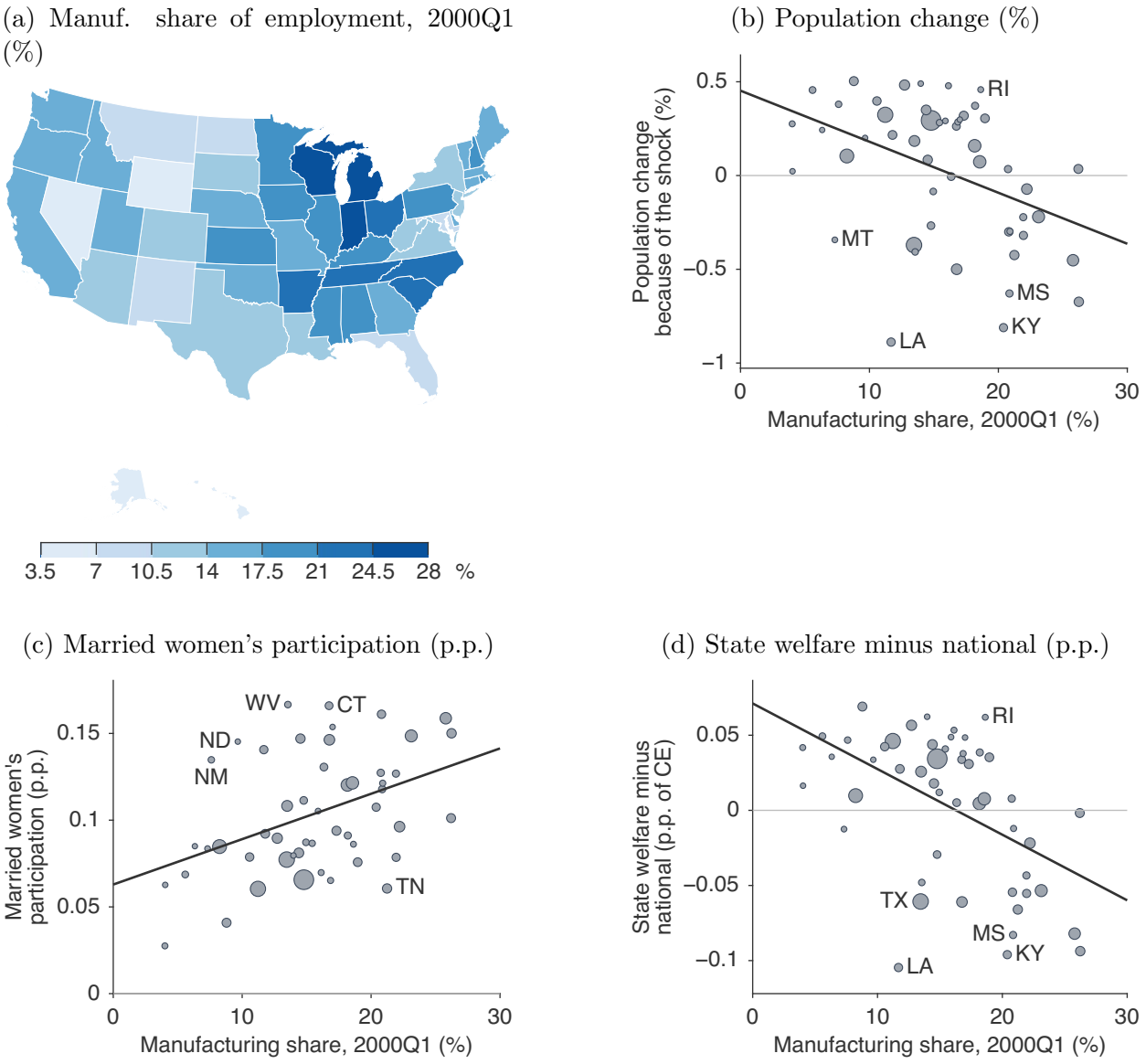
	Share (%)	Manuf. share of employment (p.p.)	Employment rate (p.p.)	Married women's participation (p.p.)	Migration response (p.p.)	Welfare (% CE)	Welfare relative (%)
Panel A. By household type							
Aggregate	100.0	−0.293	0.112		+0.0166	0.175	—
Single men	11.7	−0.287	0.106		+0.0206	0.157	−10.1
Single women	15.0	−0.188	0.158		+0.0115	0.234	+33.9
Married men	36.7	−0.412	0.096		+0.0259	0.170	−2.9
Married women	36.7	−0.192	0.111	0.101	+0.0053	0.170	−2.9
Panel B. Couples by exposure at $t = 1$							
All couples	100.0			0.101	−0.0066	0.170	−5.5
Neither	76.6			0.100	−0.0109	0.180	—
Wife only	6.6			0.101	−0.0062	0.161	−10.3
Husband only	13.7			0.105	+0.0141	0.140	−22.1
Both in manuf.	3.1			0.103	+0.0078	0.125	−30.4

Notes. Responses at $t = 199$ (with-shock minus no-shock economy), except shares (2000Q1) and welfare. The manufacturing share is in p.p. of employment, and the employment rate is in p.p. of population. The participation response is the change in married women's quarterly participation rate, in p.p.; a non-working wife is tagged by her potential sector. In Panel A, the migration response is the difference between the shares of manufacturing and non-manufacturing workers who have changed state at least once by $t = 199$, in p.p., counting each person once; individuals are tagged by their sector at $t = 1$, married men by the husband's sector, and married women by the wife's (potential) sector. In Panel B, couples are tagged by both spouses' sectors at $t = 1$, and the migration response is the change in the share of couples in the cell that have changed state at least once by $t = 199$, in p.p. Welfare is the money-metric consumption-equivalent gain, weighting each person by their money metric at 2000Q1. Welfare relative is $100 \times (G_g/G - 1)$ relative to the aggregate in Panel A and to couples in which neither spouse starts in manufacturing in Panel B. Blank cells are not defined for the group.

I therefore compare the baseline economy with economies that remove these features one at a time: a counterfactual world in which everyone is single, which I call the all-singles economy, shows what joint decisions do, and economies in which I shut down migration or wives' participation show what each margin does, both for couples and for everyone else.

These comparisons point to four results. First, all couples adjust through wives' participation, and without it, the economy as a whole gains less. Second, women's employment responds less to the shock than it would in the absence of couples, because whether a wife works depends on her wage relative to her husband's. Third, couples move together, which changes who relocates and where the gains go. Fourth, couples change who bears the shock.

Figure 4: The China Trade Shock across States, by Initial Manufacturing Share



Notes. Panel (a) is the manufacturing share of the state's employment in 2000Q1. In panels (b) to (d), each dot is a state, sized by its population in 2000Q1. The horizontal axis is the state's manufacturing share of employment in 2000Q1, in %. All outcomes are long-run responses ($t = 199$) in the baseline economy, defined as: the change in the state's population as a percentage of its no-shock level, which sums to zero across states because the national population is fixed; the change in married women's participation rate, in p.p., averaged over the couples living in the state; and the state's money-metric welfare gain minus the national gain (0.175), in p.p., for the people living in the state in 2000Q1. Lines are OLS fits; correlations, unweighted across the 50 states, are −0.41, +0.43, and −0.50. Labeled states are the five with the largest residuals.

Construction of the counterfactual economies. All counterfactual economies face the same shock, so they differ only in how they absorb it.[47] The all-singles economy shows how the same people would respond to the shock if each of them chose on their own. To build it, I split every couple into two singles, reconstruct all inputs from the data, and recalibrate the location dispersion parameter θ to match the same moment as in the baseline economy. With ν unchanged, the recalibrated θ is 5% lower than in the baseline economy.[48] I compare the same people across the two economies: a married woman in the baseline economy is, in the all-singles economy, a single woman who chooses her location and sector on her own.

To see what each of the couple's two margins does, I shut them down one at a time and together. In the first one, I shut down migration by setting $1/\theta \to 0$, so that location choices no longer respond to differences in continuation values, while switching sectors within a state is still allowed. In the second one, I partially shut down wives' participation response by setting $\kappa/\gamma \to 1$, keeping their participation rate in 2000Q1 unchanged. Because the model requires $\kappa > \gamma$, the shutdown is only partial: wives' participation response to the shock is 59% smaller than in the baseline economy, not zero.[49] In the third one, I combine both changes. I also shut down migration in the all-singles economy, so that the economy without migration has a counterpart without couples. Each exercise is a full re-solve of the model with the corresponding parameter changed, while all other parameters (except χ whenever I partially shut down participation) are held at their baseline values (Table 7).

I report the main responses to the shock for every economy in Table 8 and the welfare gains in Table 9. Tables E.3 and E.4 report results for every group. These tables report levels, so each economy can be compared with any other. Unless stated otherwise, every percentage change in the text is relative to the baseline economy.

[47] Manufacturing jobs per capita fall within 4% of the baseline in every economy (Table E.1). Welfare in the rest of the world barely moves. The mean gain of the other 36 regions differs by less than 2% across economies, and their gains correlate with those in CDP at 0.997.

[48] The target is a short-run moment, the population response to the shock over 2000–2007. Over that window, at the baseline economy's θ, the all-singles economy responds 3.6% less to the shock, so matching the same moment requires a slightly lower θ. I return to the point below when I compare how many people move in each economy.

[49] The participation response depends on κ/γ, as equation (28) shows. Holding the participation rate fixed, the response scales with κ/γ, so lowering κ/γ shrinks the response proportionally. In the baseline economy, $\kappa/\gamma = 4.44$, and the model requires $\kappa > \gamma$, so the response can be made at most 4.44 times smaller. For computational reasons, I set $\kappa = 0.65$, which shrinks it by a factor of 3.39. Once wages adjust in general equilibrium, the participation response is 59% smaller. To keep married women's participation rate in 2000Q1 unchanged, I also recalibrate χ.

Table 7: Economies Compared in the Quantitative Exercise

Economy	Couples	Migration	Participation	Change relative to the baseline economy
Baseline	✓	✓	✓	—
All-singles	—	✓	—	Couples split into two singles; θ recalibrated
All-singles, no migration	—	—	—	All-singles economy with $1/\theta \to 0$
No migration	✓	—	✓	$1/\theta \to 0$
No participation	✓	✓	partially	$\kappa/\gamma \to 1$
No migration & no participation	✓	—	partially	Both changes

Notes. A check mark indicates that the margin responds to the shock as in the baseline economy; a dash indicates that it is shut down or does not exist; "partially" indicates that it is partially shut down. I shut down migration by setting $\theta = 10^6$. Because the model requires $\kappa > \gamma$, I dampen participation by setting $\kappa = 0.65$ and recalibrate χ to keep married women's participation rate in 2000Q1 unchanged.

Result 1: All couples adjust through wives' participation, and without it, the economy as a whole gains less.

As Section V.1 shows, wives' participation rises in every type of couple, because a wife's labor supply responds not only to her own exposure to the shock but also to her husband's. In the short run, wives respond as in the reduced-form evidence in Table 3: wives whose husbands are exposed enter the market at the start of the shock, whereas wives who are exposed themselves leave it (Figure E.3). Wives in unexposed couples enter gradually, because the shock raises women's relative wage everywhere: by $t = 199$, women's relative wage rises in all 50 states, and so does married women's participation. By $t = 199$, the response is nearly the same in every type of couple: it is largest when only the husband is exposed, but only 5% larger than when neither spouse is (Panel B of Table 6).

Without this margin, the economy as a whole gains less. When I partially shut down participation, the employment gain is 24% smaller (Table 8), aggregate welfare gains are 5% lower (Tables 9 and E.2), the average couple gains 7% less, and welfare falls in all 50 states (Figure E.4). Because wives' participation response is only 59% smaller, these losses understate the value of this margin.

Every type of couple loses when I partially shut down participation. The losses are largest when only the husband is exposed (-10%; Table E.2): these wives would have entered the market the most, and now enter much less. Couples in which only the wife is exposed lose

Table 8: Responses to the China Trade Shock across Economies

	Baseline	All-singles	All-singles, no mig.	No migration	No participation	No mig. & no part.
Panel A. Employment						
Manuf. share (p.p.)	−0.293	−0.291	−0.286	−0.287	−0.285	−0.279
Employment rate (p.p.)	0.112	0.132	0.119	0.099	0.085	0.073
Female emp. rate (p.p.)	0.124	0.170	0.161	0.117	0.080	0.075
Panel B. Migration						
Exposed (p.p.)	+0.0166	+0.0194	—	—	+0.0170	—
Share living in a different state because of the shock (%)						
Aggregate	0.150	0.130	—	—	0.148	—
Married women	0.159	0.116	—	—	0.158	—
Panel C. Women's labor supply						
Wives' particip. (p.p.)	0.101			0.097	0.041	0.040
Elasticity	0.452	0.682	0.644	0.426	0.276	0.258

Notes. Responses at $t = 199$, in levels. Panel A is in p.p. of employment (manufacturing share) and of population (employment rates). In Panel B, the migration response of the exposed is the difference between the shares of manufacturing and non-manufacturing workers who have changed state at least once by $t = 199$, in p.p., counting each person once. In Panel C, married women's participation is the change in their quarterly participation rate (in p.p.), and the female labor-supply elasticity is the log female employment response to the log female real-wage response. In all-singles economies, married women are the same people as singles. A dash marks a margin that is shut down; blank cells are not defined in that economy.

too (−5%), because the same margin also lets her leave the market and fall back on home production. Unexposed couples also gain 7% less, because their wives would also enter as women's relative wages rise.[50]

Using this margin of adjustment is not free for couples: wives who enter the market give up home production. If I value a non-working wife's time as a single's, i.e., at the non-employment benefit b_j, and keep every decision unchanged, the aggregate welfare gain in the baseline economy would be 11% higher (Table 9), because a wife who enters the market would then give up only b_j instead of her home production.[51]

[50] When I partially shut down participation, couples also move differently, although only slightly: couples with an exposed husband move relatively more, and all other couples move less (Panel D of Table E.3).

[51] See Appendix E.3 for more details.

Table 9: Welfare Effects of the China Trade Shock across Economies

	Baseline economy		Counterfactual economies				
	Home prod.	Valued as singles	All-singles	All-singles, no mig.	No mig.	No part.	No mig. & no part.
Panel A. Welfare gain (% CE)							
Aggregate	0.175	0.194	0.178	0.180	0.176	0.167	0.169
Single men	0.157	0.157	0.166	0.170	0.162	0.153	0.158
Single women	0.234	0.234	0.194	0.195	0.236	0.255	0.257
Married men	0.170	0.193	0.164	0.167	0.171	0.157	0.159
Married women	0.170	0.193	0.193	0.194	0.171	0.157	0.159
Panel B. Spatial							
CV of state welfare	0.278		0.204	0.327	0.448	0.289	0.462

Notes. Money-metric consumption-equivalent gains (%), summed over $t = 2, \ldots, 199$. Each row represents the same people across all economies, and each economy weights each person by their baseline money metric in 2000Q1. In the economies with couples, married men and women carry the couple's welfare; in the all-singles economies, they are the same people as singles. The first two columns are the baseline economy: the first values a non-working wife's time as home production, as in the model, and the second values it as a single, with the non-employment benefit b_j, with every decision unchanged. In Panel B, CV is the population-weighted coefficient of variation of state welfare gains.

Result 2: With couples, women's employment responds less to the shock.

Even though every couple adjusts through the wife's participation, women's employment responds less to the shock than it would in the absence of couples, because whether a wife works depends on her wage relative to her husband's, rather than on her own wage alone. The baseline economy already shows this. Married and single women's wages respond to the shock by the same amount, yet married women's labor supply is 21% less elastic than that of single women. By $t = 199$, a husband's wage rises by 61% as much as his wife's, so her wage relative to his rises by less than half as much as her own.

Without couples, every woman responds to her own wage, and women's employment responds much more. In the all-singles economy, female labor supply is 51% more elastic than in the baseline economy, and women's employment rises 37% more (Panel C and Panel A of Table 8). As a result, the all-singles economy generates 18% larger aggregate employment gains, even though men's employment rises slightly less.[52]

[52] Women account for 35% of the gain in non-manufacturing employment in the baseline economy and 41% in the all-singles economy, mostly in other services and in accommodation and food.

Result 3: Couples move together, which changes who relocates and where the gains go.

Because spouses relocate together, exposed workers in couples are less likely to relocate. As Section V.1 shows, couples move when the husband is exposed, but not when the wife is (Panel B of Table 6), as in the second empirical fact. Exposed wives are tied to their husbands' location: when only the wife is exposed, her counterfactual single would migrate almost four times as much as she does in the couple, whereas an exposed husband migrates about as much in the couple as his counterfactual single. As a result, the migration response of exposed workers is 17% larger in the all-singles economy (Panel B of Table 8).

Couples also lead more people to relocate, because wives follow their husbands. By $t = 199$, 13% fewer people end up in a different state in the all-singles economy than in the baseline economy (Panel B of Table 8).[53] Most of the difference comes from wives: 27% fewer of their counterfactual singles end up in a different state. A couple's location responds to the husband's exposure, whereas wives' counterfactual singles respond only to their own wages. Because the shock raises women's wages, they have little incentive to move elsewhere and pay the mobility cost.

Who moves also shapes where the shock's gains go. When I shut down migration, welfare becomes 61% more unequal across states, even though aggregate gains are only 1% higher (Panel B of Table 9). Migration barely changes the aggregate gain, because workers who cannot move can still switch sectors within their state. It does, however, shift gains across states. Without migration, residents of the states that people leave in the baseline economy are worse off, because they can no longer move out. Residents of the states people move to are better off because no new arrivals push down their wages.

For residents of the states that people leave, migration matters more when people live in couples. Without migration, the welfare of residents of the states people leave falls 42% more in the baseline economy than in the all-singles economy, because more of them would have left. Couples with an exposed husband lose more from shutting down migration than either spouse would as a single. When only the husband is exposed, the couple loses 3.4% of its gain, whereas his counterfactual single would lose 2.8% and hers only 0.3% (Panel B of Table 10).[54]

[53] This is also why the all-singles economy needs a lower θ to match the population response over 2000–2007: without unexposed spouses following their partners, the same θ moves fewer people. Because a lower θ makes singles more responsive to the shock, the recalibration, if anything, enlarges the 17% gap in the migration response of the exposed and shrinks the 13% gap here.

[54] Appendix E.2 maps the change in welfare in each state and details the participation response in this counterfactual.

Table 10: Couples and the Same Spouses as Singles

	Couple (baseline)	Husband's counterfactual single	Wife's counterfactual single	Couple, no participation
Panel A. Welfare gain (% CE), with migration				
All couples	0.170	0.164	0.193	
Neither	0.180	0.172	0.195	
Wife only	0.161	0.168	0.179	
Husband only	0.140	0.134	0.190	
Both in manuf.	0.125	0.135	0.177	
Panel B. Change in welfare from shutting down migration (% of each economy's own gain)				
All couples	+0.7	+1.7	+0.3	+1.1
Neither	+1.6	+2.7	+0.6	+2.0
Wife only	+0.7	+1.4	+0.1	+1.0
Husband only	−3.4	−2.8	−0.3	−3.4
Both in manuf.	−2.7	−2.1	−0.5	−2.5

Notes. Couples are tagged by both spouses' sectors at $t = 1$; a non-working wife is tagged by her potential sector. The couple's welfare is based on the baseline economy; the husband's and the wife's counterfactual singles are the two former spouses in the all-singles economy. Panel A: money-metric consumption-equivalent gains (%), weighting both economies by the baseline money metric at 2000Q1. Panel B: change in welfare from shutting down migration, as a percentage of each economy's own gain: no migration relative to the baseline economy for the couple, all-singles economy without migration relative to the all-singles economy for the spouses as singles, and no migration & no participation relative to no participation for the couple with participation partially shut down.

Result 4: Couples change who bears the shock.

Being in a couple shifts gains from the wife to the husband. As singles, married women would gain 14% more and married men 3% less (Table 9). A wife gains less than she would alone for two reasons: first, whether she works depends on her husband's wage, so she cannot fully take advantage of expanding sectors; second, the spouses share a common household gain.[55] The exception is when the wife is exposed: then both spouses would gain more as singles. A couple in which only the wife is exposed gains 4% less than the husband would alone and 10% less than she would (Panel A of Table 10). In the couple, she is tied to her husband's location and his wage, so she adjusts less than she would alone, and he shares her

[55] Couples barely change how much the economy gains: aggregate gains are 2% larger in the all-singles economy. When only the husband is exposed, the couple gains 5% more than his counterfactual single and 26% less than her counterfactual single (Panel A of Table 10).

loss.[56]

Couples also change what singles gain because, in general equilibrium, the amount of work married women supply affects the wages singles earn. In the all-singles economy, more women work, so women's wages rise less: single women gain 17% less and single men 6% more than in the baseline economy.[57] When I partially shut down participation, the opposite happens: fewer married women work, women's wages rise more, and single women are the only group better off: they gain 9% more, while single men gain 2% less.[58]

With couples, welfare is more unequal across states, as Section V.1 suggests: without couples, welfare gains would be 27% less dispersed (Panel B of Table 9), even though the map of gains barely changes (Figure E.8). The difference is much larger for wives than for husbands. As singles, married women's welfare would be 48% less dispersed across states and married men's 13% less, because living as a couple ties a wife's welfare to where the couple lives.

Taken together, the counterfactuals show that couples change how the economy adjusts to the China Trade Shock and who bears it. Couples adjust through wives' participation, without which the economy as a whole gains less, and because a wife's decision depends on her husband's wage, women's employment responds less. Couples also move together, which changes who relocates and where the gains go: more people relocate, and migration matters more for the states that people leave. Finally, couples shift gains from wives to husbands and from single men to single women.

VI. Conclusion

I show that couples change how the economy adjusts to a local shock and who bears it, because most adults decide where to live and work together with a spouse. Unlike singles, couples must weigh two labor-market exposures against each other, which makes them less likely to move. At the same time, they have a margin of adjustment that singles lack: when one spouse's earnings fall, the couple can decide that the other spouse works more.

Using individual-level exposure to the China Trade Shock in the PSID, I show that how

[56] When both spouses are exposed, the couple gains 7% less than the husband's and 29% less than the wife's counterfactual single. An exposed wife adjusts less on every margin than she would alone: as a single, she would switch to other sectors 24% more, raise her participation by 68%, and migrate almost four times as much.

[57] In the all-singles economy, the female wage response is 11% smaller, the male wage response is 9% larger, and the relative wage rises by 39% less (Panel E of Table E.3).

[58] The female wage response is 9% larger and the male wage response 3% smaller than in the baseline economy; because men and women are complements in production, fewer working women also lower men's wages.

couples respond to a local shock depends on which spouse is hit. Comparing individuals within the same commuting zone, I find that exposed workers experienced worse labor-market outcomes and were more likely to leave. Among couples, however, out-migration rose only when the husband was exposed, and wives entered the labor force when their husbands were exposed but left it when they themselves were.

To quantify what these joint decisions imply for the economy as a whole, I develop a dynamic spatial and trade model in which the decision-making unit is the household: a single man, a single woman, or a couple. Regions and sectors trade with one another, and each period, households choose where to live and which sectors to work in (Caliendo et al., 2019). The model's novelty is that couples decide together: they choose one location for the household and a sector for each spouse, so any move must benefit the couple as a whole, and then decide whether the wife works in the market or at home. The model also captures the economies of scale from marriage: a couple pays once for a household good, such as housing, that both spouses enjoy in full, and the wife's home production raises it when she stays at home.

Through the lens of the model, the China Trade Shock is, overall, a boom for US workers. Cheaper Chinese goods raise real wages, so even though the manufacturing share of employment falls by 0.29 p.p., the employment rate rises in every group, and the average individual gains the equivalent of a 0.18% increase in consumption. Couples adjust differently from singles: because the shock lowers husbands' wages relative to their wives', more couples choose to have the wife work, while husbands move out of manufacturing into other sectors, and the couple's location follows the husband's job. Single women gain the most, 34% more than the average individual; single men the least; and couples fall in between, much closer to single men, because the husband's wage drives the household's gain. Across states, the shock hits hardest where manufacturing was concentrated, and people there respond in two ways: some leave, and wives enter the labor force.

Couples change how the economy adjusts to the shock: all of them increase the wife's labor supply, yet women's employment responds less than it would without couples. Wives work more in every couple in the long run, not only in those in which the husband is exposed, because the shock raises women's wages relative to men's everywhere. Without this margin, the economy as a whole would gain at least 5% less. Yet a couple decides whether the wife works based on her wage relative to her husband's, and the shock raises his wage too, so without couples, the employment gain from the shock would be 18% larger. Couples also move together, following the husband's job, so exposed workers move less than they would without couples, but more people relocate overall.

Couples also change who bears the shock. Within the couple, gains shift from the wife

to the husband, because her decision to work is tied to his wage and her location to his job; when the wife is exposed, both spouses would gain more as singles, because she adjusts less than she would alone and he shares her loss. Across households, gains shift from single men to single women because wives' employment responds less to the shock, so women's wages rise more, benefiting the women who decide alone.

These results imply that aggregate gains, and even their geography, give an incomplete picture of who benefits from a local shock. Understanding how the economy adjusts to local shocks, and who bears them, requires treating the household, and not only the worker, as the unit that decides where to live and work. This matters for policy. Trade adjustment assistance, for instance, is designed around individual workers exposed to trade shocks. Yet an exposed wife's options depend on her husband's job, and the shock also reaches single workers who were never employed in an exposed industry, because how much married women work changes what everyone else earns.

The mechanisms behind these results are not specific to the China Trade Shock. In any local shock, a shock to one spouse becomes a shock to the household, because couples decide together where to live and whether the wife works. What is specific to this shock is who gains: because the China Trade Shock hits manufacturing, a male-intensive sector, it exposes husbands more than wives and draws wives into the labor market. A shock to a female-intensive sector could reverse this pattern, although its welfare consequences need not be a mirror image; the model provides a framework to study such shocks, which I leave for future research.

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Appendix

"It Takes Two to Tango": Joint Location Choices and Local Shocks

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A. Robustness Exercises

A.1 The China Trade Shock across Sample Commuting Zones

Given the PSID sample restrictions and the size of the PSID, I include 375 commuting zones out of the 722 that ADH use in their analysis. My results could therefore be driven by which zones are included in the PSID rather than by differential exposure to the China Trade Shock. To assess this concern, I replicate ADH's main results using the 375 CZs represented in the sample.[59][60]

Table A.1 first compares the CZs in the sample with the full set used by ADH. Because the restricted sample is a subset of the full sample, the two groups are not independent, and a conventional difference-in-means test is therefore ill-defined. The final column instead reports the standardized difference, defined as the difference in means between the restricted and full samples divided by the square root of the average of their variances. This measure is expressed in standard-deviation units, so it is comparable across variables measured on different scales. Following Imbens and Rubin (2015), absolute standardized differences below 0.25 indicate comparability. All standardized differences in Table A.1 are 0.13 or smaller, including the difference for the change in import penetration. By this criterion, the CZs in the PSID sample are comparable to the full ADH sample with respect to both the control variables and the China Trade Shock itself.

Next, I estimate ADH's Equation (2) for the CZs in the sample:

$$\Delta Y_{i,t+h} = \alpha_t + \beta_{1,h}\Delta IP_i^{China,US} + X'_{i,t}\beta_2 + \varepsilon_{i,t+h},$$

where $\Delta Y_{i,t+h}$ denotes the change in outcome Y for CZ i between $t = 2000$ and year $t + h$. As in ADH, $\Delta IP_i^{China,US}$ is instrumented with $\Delta IP_i^{China,Others}$, following the identification logic described in Section II. The analysis considers ADH's three main outcome categories: employment, income, and population headcount.

Estimated on the CZs in the PSID sample, ADH's specification yields responses that closely track ADH's full-sample estimates, except for the changes in population of younger age groups. Figure A.1 replicates ADH's employment results for the manufacturing employment-to-population ratio in panel a, the non-manufacturing employment-to-population ratio in panel b, and the total wage-and-salary employment-to-population ratio in panel c. Across all three outcomes, the estimates based on the full set of CZs closely align with those based only on the CZs in the PSID sample.

[59] Because the analysis uses the geocoded restricted version of the PSID, the identities of the commuting zones in the sample cannot be disclosed, nor can a map be provided at any geographic level below the state.

[60] This exercise uses the replication package provided by David Dorn on his website.

Table A.1: Comparing CZs in the Sample with All CZs

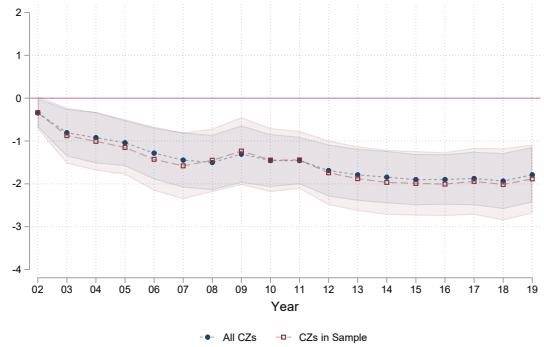
	Full		Restricted			Standardized
	Mean	SD	Mean	SD	Diff	Difference
Control variables (in 2000)						
Manufacturing emp./total emp.	16.18	7.46	15.68	6.91	-0.49	-0.07
Female emp./total emp.	64.38	5.51	64.54	5.21	0.16	0.03
Routine occupation emp./total emp.	31.90	2.37	32.20	2.16	0.30	0.13
Offshorability index	0.00	0.51	0.06	0.48	0.07	0.13
College educated/total pop.	53.61	7.47	54.41	6.88	0.80	0.11
Foreign born/total pop.	14.76	12.81	15.74	12.89	0.97	0.08
Non-white population/total pop.	18.18	10.95	18.81	10.36	0.63	0.06
Population 65+/total pop.	12.37	2.91	12.15	2.81	-0.21	-0.07
Population 40-64/total pop.	30.10	1.87	30.02	1.79	-0.08	-0.04
Population 0-17/total pop.	25.63	2.22	25.65	2.14	0.02	0.01
Δ log population 1970-1990/2	12.31	12.25	12.77	12.35	0.47	0.04
Change in CZ import penetration (2000 to 2012)						
Imports by US	0.89	0.59	0.88	0.55	-0.01	-0.02
Imports by other high-inc. countries	1.22	0.69	1.21	0.64	-0.01	-0.02
Number of Commuting Zones	722		375			

he income responses in the sample CZs also closely track ADH's full-sample estimates. Figure A.2 reports the corresponding results for income: personal income per capita in panel a, government transfers per capita in panel b, and total labor compensation per worker in panel c. The only visible divergence is a somewhat larger increase in government transfers among the sample CZs, although the coefficients are not statistically distinguishable.

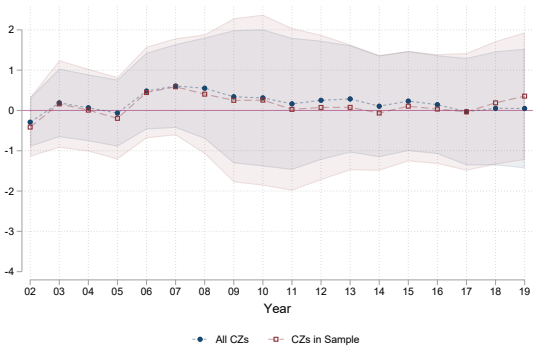
Finally, population headcounts, which ADH use to measure migration responses, are the only outcomes for which the two sets of estimates diverge, and only among younger age groups (Figure A.3). The estimates are similar for the total population and for individuals ages 40–64. However, the population decline among individuals ages 25–39 is larger in the sample CZs, and the estimates indicate a statistically significant decline among individuals ages 18–24 that is absent from ADH's full-sample results. The difference among individuals aged 25–39 is relevant for my sample, whose average age is 37: it suggests that young adults left the sample CZs somewhat more than they left the average CZ. My main estimates, however, compare exposed and non-exposed individuals, so a population decline common to both groups within a CZ does not, by itself, drive the estimated differences.

Figure A.1: China Trade Shock Impact on Employment

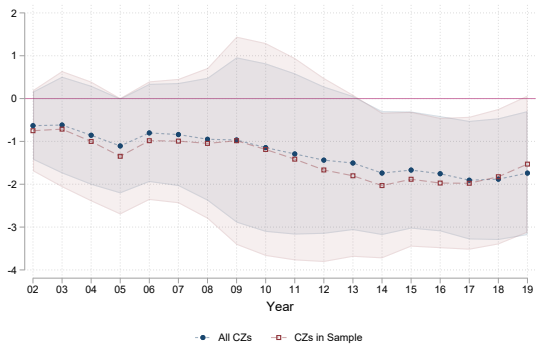
(a) Manufacturing Emp./Working-age pop.



(b) Non-manufacturing Emp./Working-age pop.



(c) Total Wage and Salary Emp./Working-age pop.



Notes. In each panel, I plot the original ADH result for that outcome variable in blue and, in red, the result when I consider only the CZs in the sample. A CZ is considered in the sample if I can estimate its fixed effect when estimating equation (5).

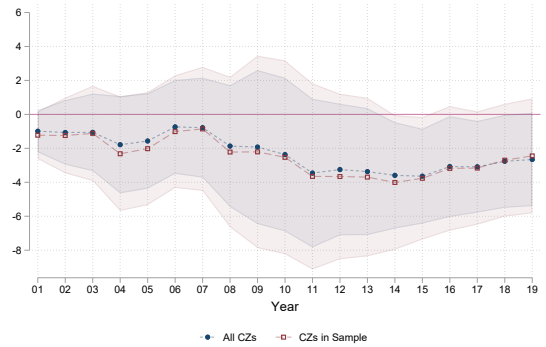
Taken together, the two exercises indicate that, although the PSID sample restrictions reduce geographic coverage, they do not appear to select local labor markets in which the China Trade Shock had systematically different effects. The CZs represented in the PSID are comparable to the full ADH sample in terms of observable characteristics, including exposure to the China Trade Shock, and estimating ADH's specification on these CZs produces similar coefficients.

A.2 The China Trade Shock and Household Composition

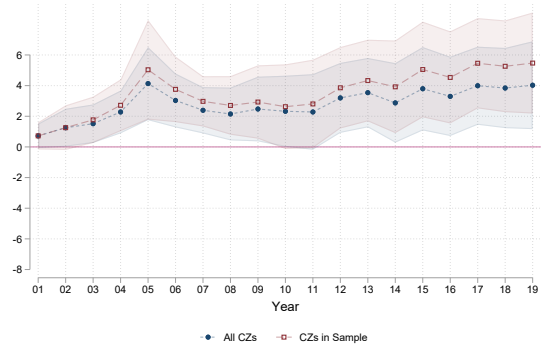
One potential concern is that the China Trade Shock alters household structure. Such changes could bias the results through two channels: the shock may affect marriage, divorce, or fertility decisions that subsequently shape out-migration responses, and couples who re-

Figure A.2: China Trade Shock Impact on Log Income

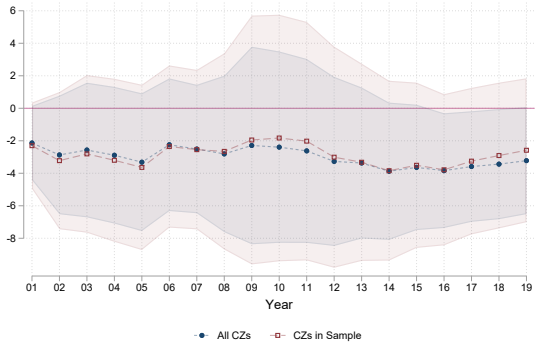
(a) Log(Personal Income per capita)



(b) Log(Government Transfers per capita)



(c) Log(Total Labor Compensation per worker)



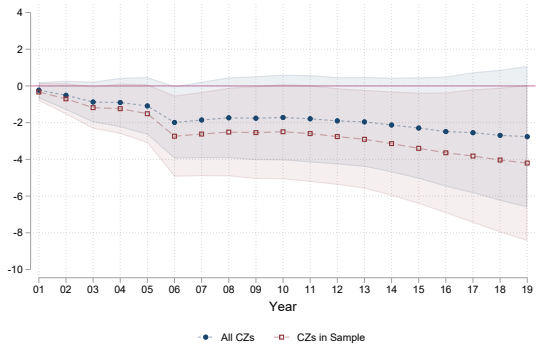
Notes. In each panel, I plot the original ADH result for that outcome variable in blue and, in red, the result when I consider only the CZs in the sample. A CZ is considered in the sample if I can estimate its fixed effect when estimating equation (5).

main together after the shock may differ systematically from those who divorce. To assess this concern, I use the full sample of individuals and estimate equation (4) for the probabilities of marrying, divorcing, and having a new child. Figure A.4 shows that, between 2001 and 2013, exposed individuals were no more likely than non-exposed individuals to marry, divorce, or have a child.

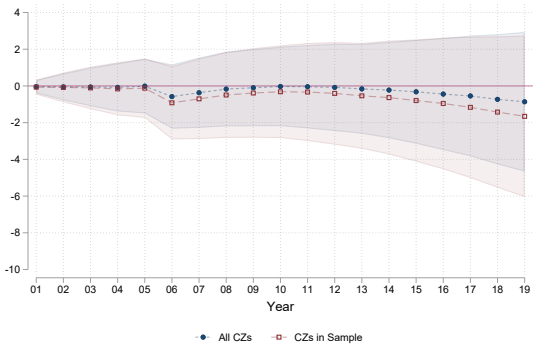
At first glance, this result may appear inconsistent with Autor et al. (2019), who find that a one-unit trade shock reduces the share of previously married women by 0.21 p.p. and increases the share of never-married women by 1.2 p.p.. The apparent discrepancy reflects differences in how marital status is measured. In the ACS, marital status records legal status and does not separately identify cohabiting women. Consequently, women who transition from single to cohabiting are classified as married in my data, whereas they are classified as single in Autor et al. (2019). In addition, because the ACS is a repeated cross-section, a

Figure A.3: China Trade Shock Impact on Log Population Headcounts

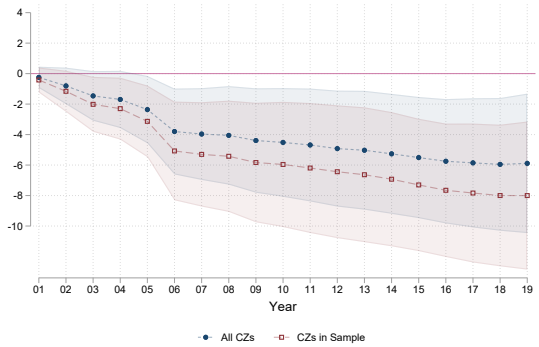
(a) Log(Population, 18–64)



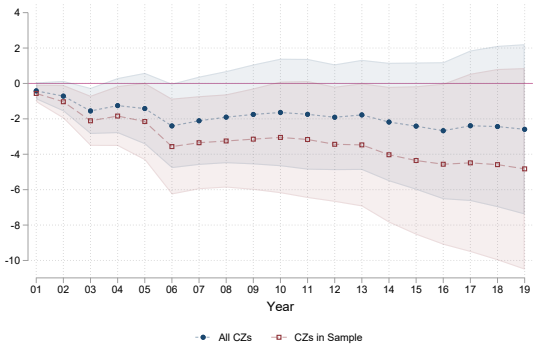
(b) Log(Population, 40–64)



(c) Log(Population, 25–39)



(d) Log(Population, 18–24)



Notes. In each panel, I plot the original ADH result for that outcome variable in blue and, in red, the result when I consider only the CZs in the sample. A CZ is considered in the sample if I can estimate its fixed effect when estimating equation (5).

woman's prior marital status cannot be recovered once she is observed as married. Women who transition from divorced to married are therefore classified simply as married in their analysis, whereas they are excluded from my sample.

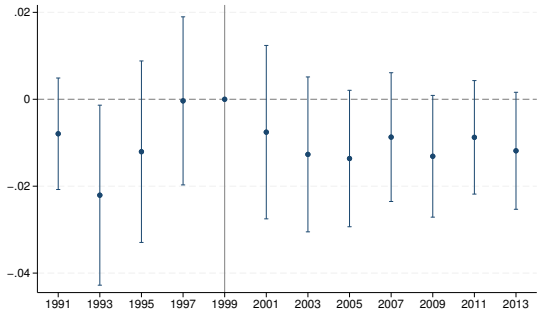
I also find no robust effect of trade shocks on fertility, in contrast to Autor et al. (2019). One likely explanation is that I allow both men and women to be exposed, whereas they focus exclusively on women. My estimates therefore capture a broader set of household shocks and adjustment margins.

A.3 Other Migration Drivers

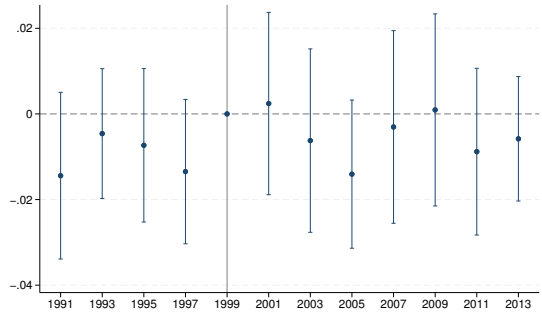
The asymmetric migration response within couples could instead reflect two other drivers: the presence of children or gender differences in responses to the shock. First, the behavioral differences I attribute to joint decision-making may reflect the presence of children rather than couple status itself. To address this concern, I extend equation (4) to a triple-difference

Figure A.4: Exposure to the China Trade Shock and Family Outcomes

(a) Marriage=1



(b) Divorce=1



(c) Fertility=1

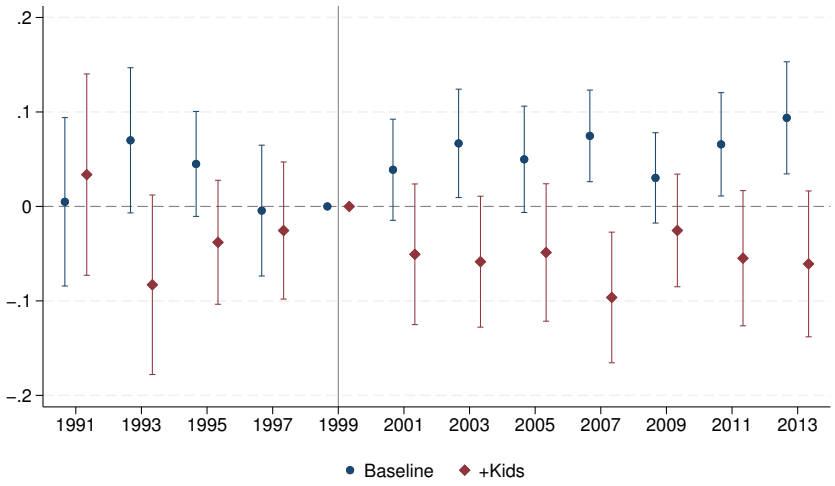


Notes. Each panel plots the estimates of equation (4) for the outcome named in its title. Robust standard errors are two-way clustered at the industry and individual levels. Confidence intervals are constructed at the 95% level. In the equivalent static difference-in-differences model, the coefficient of interest is not statistically significant for any of the three outcomes.

specification that accounts for whether an individual has a child younger than 18. If individuals with children respond differently from those without children, the coefficient on the triple interaction should be statistically different from zero. Figure A.5 shows that this coefficient is not statistically significant in any year except 2007. Thus, exposure does not generate systematically different out-migration responses among individuals with and without children. Individuals with children nevertheless have a baseline out-migration probability that is 4.2 p.p. lower than that of individuals without children. Therefore, the two groups differ in their intercepts, even though their responses to exposure are similar.

Second, the estimated differences could reflect gender-specific responses to the shock. To address this concern, I extend equation (4) to a triple-difference specification that accounts for whether the individual is a woman. If women and men respond differently to the shock, the coefficient on the triple interaction should be statistically different from zero. Figure A.6 shows that this coefficient is not statistically significant in any year during the shock. Thus, exposed women do not exhibit a systematically different out-migration response from

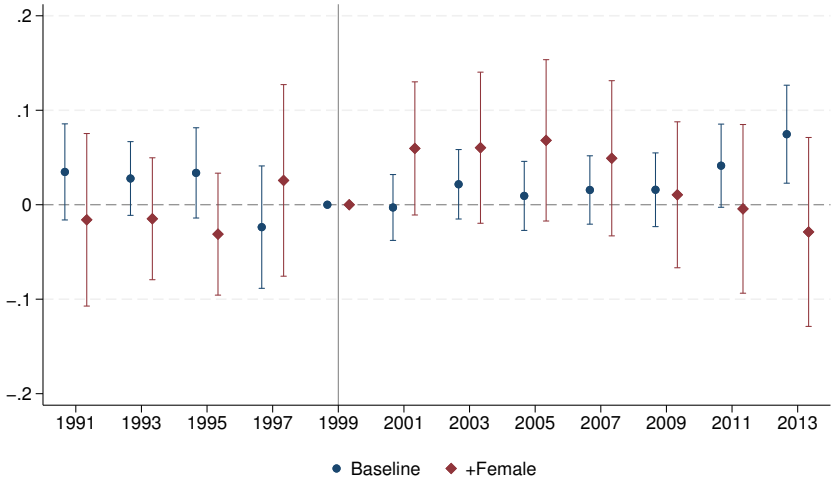
Figure A.5: Exposure to the China Trade Shock and Out-Migration Responses, by Parental Status



Notes. Robust standard errors are two-way clustered at the industry and individual levels. Confidence intervals are constructed at the 95% level. In blue, I plot the coefficient on the interaction between exposure and year. In red, I plot the coefficient on the triple interaction among exposure, year, and an indicator for having a child younger than 18.

exposed men.

Figure A.6: Exposure to the China Trade Shock and Out-Migration Responses, by Gender



Notes. Robust standard errors are two-way clustered at the industry and individual levels. Confidence intervals are constructed at the 95% level. In blue, I plot the coefficient on the interaction between exposure and year. In red, I plot the coefficient on the triple interaction among exposure, year, and an indicator for being a woman.

Taken together, these results indicate that the asymmetric migration response among

couples, depending on which partner is exposed, is driven neither by the presence of children in the household nor by general gender differences in responses to the shock.

A.4 Additional Tables

Table A.2: Exposure to the China Trade Shock and Labor Market Outcomes

	(1) Employed=1	(2) Unemployed=1	(3) Lab Force=1
Exposed=1	-0.007	-0.002	-0.008
	(0.019)	(0.008)	(0.017)
(Exposed=1)x(Post=1)	-0.063***	0.023**	-0.039***
	(0.017)	(0.010)	(0.012)
Baseline Mean	0.90	0.04	0.94
	(1) Weekly Hours	(2) Hourly Wage	(3) Labor Income
Exposed=1	-0.454	-0.677	1,978.003
	(1.644)	(4.011)	(8,565.422)
(Exposed=1)x(Post=1)	-2.058**	-2.581**	-5595.397**
	(0.889)	(1.160)	(2,273.988)
Baseline Mean	40.62	24.26	50,236.60
Observations	38,634	38,634	38,634

Notes. Robust standard errors are two-way clustered at the industry and individual levels. * for p<0.10, ** for p<0.05, *** for p<0.01.

Table A.3: Binary versus Continuous Exposure to the China Trade Shock

	(1) Employed=1	(2) Unemployed=1	(3) Labor Force=1
Exposed=1	-0.007	-0.002	-0.008
	(0.019)	(0.008)	(0.017)
(Exposed=1)x(Post=1)	-0.063***	0.023**	-0.039***
	(0.017)	(0.010)	(0.012)
Shock	0.003*	-0.000	0.003**
	(0.002)	(0.001)	(0.001)
(Shock)x(Post=1)	-0.007***	0.003***	-0.003***
	(0.002)	(0.001)	(0.001)
Observations	38,634	38,634	38,634
R2	0.08	0.05	0.06
Baseline Mean	0.90	0.04	0.94

Notes. Robust standard errors are two-way clustered at the industry and individual levels. * for p<0.10, ** for p<0.05, *** for p<0.01.

Table A.4: Exposure to the China Trade Shock and Out-Migration Responses

	(1) Out-Migration=1
Exposed=1	-0.064
	(0.042)
(Exposed=1)x(Post=1)	0.047**
	(0.021)
Observations	37,443
R^2	0.09
Baseline Mean	0.07

Notes. Robust standard errors are two-way clustered at the industry and individual levels. $*$ for p<0.10, $**$ for p<0.05, $***$ for p<0.01.

Table A.5: Wife's Exposure to the China Trade Shock and Husband's Labor Market Outcomes

	(1) Husband Employed=1	(2) Husband's Hourly Wage
Exposed Wife=1	-0.021	-1.527
	(0.020)	(1.355)
(Exposed Wife=1)x(Post=1)	-0.030	-9.349***
	(0.026)	(1.557)
Observations	14,083	14,083
R^2	0.10	0.33
Baseline Mean	0.95	30.42

Notes. All monetary values are in 2017 USD, using the PCE index, and include zeros. Robust standard errors are two-way clustered at the wife's industry and individual levels. * for p<0.10, ** for p<0.05, *** for p<0.01.

Table A.6: Couple's Exposure to the China Trade Shock and Household Income Responses

	(1)	(2)
	Household Income	
Exposed Husband=1	7,357.215	
	(11692.540)	
(Exposed Husband=1)x(Post=1)	3,930.551	
	(6,330.419)	
Exposed Wife=1		529.105
		(12003.901)
(Exposed Wife=1)x(Post=1)		-2348.701
		(4,316.531)
Observations	14,083	14,083
R^2	0.35	0.35
Baseline Mean	120226.29	120600.92

Notes. All monetary values are in 2017 USD, using the PCE index, and include zeros. Robust standard errors are two-way clustered at the husband's industry and household levels in Column 1 and the wife's industry and household levels in Column 2. * for p<0.10, ** for p<0.05, *** for p<0.01.

B. Model Derivations

This appendix derives the choice probabilities and value function under the nested Gumbel shock and then develops the money-metric measure used to compare welfare across singles and couples.

B.1 Choice Probabilities and Expected Values

The derivation follows Appendix B.2 of Rodríguez-Clare et al. (2026). Fix h, $\boldsymbol{o}$, and t, and suppress these indices whenever doing so creates no ambiguity. Define $x_{\boldsymbol{d}} \equiv \delta V_{\boldsymbol{d},t+1} - \mu_{\boldsymbol{od}}$. The household chooses $\boldsymbol{d}$ if and only if $x_{\boldsymbol{d}} + \varepsilon_{\boldsymbol{d}} \geq x_{\boldsymbol{z}} + \varepsilon_{\boldsymbol{z}}$ for every $\boldsymbol{z}$.

Choice probabilities. Let $G_{\boldsymbol{d}}(\cdot)$ denote the partial derivative of F in equation (17) with respect to its $\boldsymbol{d}$-th argument. For any $\boldsymbol{d} \in \mathcal{S}_h(j)$,

$$G_{\boldsymbol{d}}(\varepsilon) = \frac{1}{\theta}\left(\sum_{\boldsymbol{z}\in\mathcal{S}_h(j)} e^{-\varepsilon_{\boldsymbol{z}}/\nu}\right)^{\nu/\theta-1} e^{-\varepsilon_{\boldsymbol{d}}/\nu}\, F(\varepsilon).$$

The choice probability for $\boldsymbol{d}$ is $\int G_{\boldsymbol{d}}(\varepsilon_{\boldsymbol{d}}, \{\varepsilon_{\boldsymbol{d}} + x_{\boldsymbol{d}} - x_{\boldsymbol{z}}\}_{\boldsymbol{z}\neq\boldsymbol{d}})\, d\varepsilon_{\boldsymbol{d}}$. Evaluating each remaining argument at $\varepsilon_{\boldsymbol{z}} = \varepsilon_{\boldsymbol{d}} + x_{\boldsymbol{d}} - x_{\boldsymbol{z}}$ yields $e^{-\varepsilon_{\boldsymbol{z}}/\nu} = e^{-\varepsilon_{\boldsymbol{d}}/\nu} e^{(x_{\boldsymbol{z}} - x_{\boldsymbol{d}})/\nu}$ and therefore

$$\sum_{\boldsymbol{z}\in\mathcal{S}_h(\ell)} e^{-\varepsilon_{\boldsymbol{z}}/\nu} = e^{-\varepsilon_{\boldsymbol{d}}/\nu} e^{-x_{\boldsymbol{d}}/\nu}\, \Omega_\ell, \qquad \Omega_\ell \equiv \sum_{\boldsymbol{z}\in\mathcal{S}_h(\ell)} e^{x_{\boldsymbol{z}}/\nu},$$

so Ω_ℓ is shorthand for the inclusive value $\Omega^h_{\boldsymbol{o}\ell,t}$, where $\Omega^h_{\boldsymbol{o}j,t} \equiv \sum_{\boldsymbol{z}\in\mathcal{S}_h(j)} \exp\left[\delta V^h_{\boldsymbol{z},t+1} - \mu_{\boldsymbol{oz}}\right]^{1/\nu}$. Substituting, and writing $T \equiv e^{-x_{\boldsymbol{d}}/\theta} \sum_\ell \Omega_\ell^{\nu/\theta}$,

$$\Pr(\boldsymbol{d}) = \frac{e^{x_{\boldsymbol{d}}/\nu}}{\Omega_j} \cdot \frac{\Omega_j^{\nu/\theta}}{\sum_\ell \Omega_\ell^{\nu/\theta}} \int_{-\infty}^{\infty} \frac{1}{\theta}\, T\, e^{-\varepsilon_{\boldsymbol{d}}/\theta} \exp\left(-T e^{-\varepsilon_{\boldsymbol{d}}/\theta}\right) d\varepsilon_{\boldsymbol{d}}.$$

The integrand is the density of a univariate Gumbel distribution with scale θ, so the integral equals one. The first fraction corresponds to

$$\lambda^h_{\boldsymbol{od}|j,t} = \frac{\exp\left[\delta V^h_{\boldsymbol{d},t+1} - \mu_{\boldsymbol{od}}\right]^{1/\nu}}{\Omega^h_{\boldsymbol{o}j,t}}, \tag{B.1}$$

while the second corresponds to

$$\lambda^h_{\boldsymbol{o}j,t} = \frac{\left(\Omega^h_{\boldsymbol{o}j,t}\right)^{\nu/\theta}}{\sum_{\ell=1}^{L}\left(\Omega^h_{\boldsymbol{o}\ell,t}\right)^{\nu/\theta}}. \tag{B.2}$$

Interpretation for ν and θ. ν and θ are scale parameters of the preference shocks. Their inverses are semi-elasticities of choice shares with respect to continuation values, which are measured in log-utility units.

For two bundles $\boldsymbol{d}$ and $\boldsymbol{d}'$ in the same location j, equation (B.1) implies

$$\frac{\partial \log\left(\lambda^h_{\boldsymbol{od}|j,t}/\lambda^h_{\boldsymbol{od}'|j,t}\right)}{\partial\left(\delta V^h_{\boldsymbol{d},t+1}\right)} = \frac{1}{\nu}$$

while quation (B.2) implies

$$\frac{\partial \log\left(\lambda^h_{\boldsymbol{o}j,t}/\lambda^h_{\boldsymbol{o}j',t}\right)}{\partial\left(\nu \log \Omega^h_{\boldsymbol{o}j,t}\right)} = \frac{1}{\theta}.$$

The implied response to wages therefore depends on the discount factor δ and on how persistent the wage change is. Only in a static version of the model, populated with singles, $1/\nu$ and $1/\theta$ are wage elasticities: of a sector's within-location share with respect to its real wage, and of a location's share with respect to a proportional change in all of its real wages, for choices with small shares (see Rodríguez-Clare et al., 2026)

Expected value. The joint probability that $x_{\boldsymbol{d}} + \varepsilon_{\boldsymbol{d}} \leq a$ and that $\boldsymbol{d}$ is chosen is obtained by integrating the same expression up to $a - x_{\boldsymbol{d}}$. This yields $\lambda^h_{\boldsymbol{od},t} \exp\left(-\tilde{T} e^{-a/\theta}\right)$, where $\tilde{T} \equiv \sum_\ell \Omega^{\nu/\theta}_\ell$. Conditional on $\boldsymbol{d}$ being chosen, the maximized payoff therefore follows a Gumbel distribution with location $\theta \log \tilde{T}$ and scale θ, implying a mean of $\theta \log \tilde{T} + \theta\, \bar{e}$, where $\bar{e}$ is the Euler–Mascheroni constant. Because this expression does not depend on $\boldsymbol{d}$,

$$\mathbb{E}\left[\max_{\boldsymbol{d}} \{x_{\boldsymbol{d}} + \varepsilon_{\boldsymbol{d}}\}\right] = \theta \log\left(\sum_{\ell=1}^{L} \Omega^{\nu/\theta}_{\boldsymbol{o}\ell,t}\right) + \theta\, \bar{e},$$

which, after dropping the constant, delivers equation (18).

Welfare recursion. From equation (19) evaluated at the origin location i, $\sum_\ell \Omega^{\nu/\theta}_{\boldsymbol{o}\ell,t} = (\lambda^h_{\boldsymbol{o}i,t})^{-1} \Omega^{\nu/\theta}_{\boldsymbol{o}i,t}$. From equation (20) evaluated at the origin bundle and $\mu_{\boldsymbol{oo}} = 0$, $\Omega_{\boldsymbol{o}i,t} = (\lambda^h_{\boldsymbol{oo}|i,t})^{-1} \exp(\delta V^h_{\boldsymbol{o},t+1})^{1/\nu}$. Combining,

$$\theta \log\left(\sum_\ell \Omega^{\nu/\theta}_{\boldsymbol{o}\ell,t}\right) = \delta V^h_{\boldsymbol{o},t+1} - \theta \log\left(\lambda^h_{\boldsymbol{o}i,t}\right) - \nu \log\left(\lambda^h_{\boldsymbol{oo}|i,t}\right),$$

which, substituted into equation (18) and iterated forward, gives equation (33).

B.2 The Money-Metric Welfare Index

I follow Chiappori and Meghir (2015) to construct a Money-Metric Welfare Index (MMWI), defined as the income each spouse would require to attain their utility level if they faced the full price of public consumption. For notational simplicity, I suppress the indices $\boldsymbol{d}$ and t.

Let $u_g \equiv \gamma \log Q + (1-\gamma)\log c_g$, for $g \in \{f, m\}$, denote the log utility of spouse g. Substituting equation (23) into u_g, I obtain

$$\begin{aligned} u_f &= \log \frac{I(e)}{P} + e\gamma \log \beta + K + (1-\gamma)\log \alpha, \\ u_m &= \log \frac{I(e)}{P} + e\gamma \log \beta + K + (1-\gamma)\log(1-\alpha), \end{aligned}$$

where $K \equiv \gamma \log \gamma + (1-\gamma)\log(1-\gamma)$. In levels, $U_g = \mathcal{A}_g \beta^{e\gamma} I(e)/P$ with $\mathcal{A}_f = e^K \alpha^{1-\gamma}$ and $\mathcal{A}_m = e^K (1-\alpha)^{1-\gamma}$. Ex ante utility corresponds to

$$\mathbb{E}[U_g] = (\mathcal{A}_g / \mathcal{A})\, \mathbb{E}[\mathcal{U}]. \tag{B.3}$$

Thus, each spouse's expected utility is a fixed multiple of the couple's expected utility. The ratio of private consumption is $c_f^*/c_m^* = \alpha/(1-\alpha)$ and is independent of wages, β, and e. Because the model contains no distribution factors, an increase in the wife's wage raises household income without changing her share of it.

1. Conditional sharing rule (CSR). The CSR allocates between the spouses the income remaining after expenditure on the public good:

$$\tilde{\varsigma}_f = P c_f^* = \alpha(1-\gamma) I, \qquad \tilde{\varsigma}_m = P c_m^* = (1-\alpha)(1-\gamma) I, \qquad \tilde{\varsigma}_f + \tilde{\varsigma}_m = (1-\gamma) I.$$

The CSR therefore excludes both the γI spent on the public good and the increase in public-good production generated by home production.

2. Lindahl prices. One unit of Q costs the couple $P_Q \equiv P\beta^{-e}$. Thus, when the wife remains at home, home production reduces the effective price of the public good. Each spouse's personalized price equals their marginal willingness to pay for Q, expressed in monetary units:

$$P_g = P \frac{\partial u_g / \partial Q}{\partial u_g / \partial c_g} = \frac{\gamma}{1-\gamma} \cdot \frac{P c_g^*}{Q^*} \quad \Longrightarrow \quad P_f = \alpha P_Q, \qquad P_m = (1-\alpha) P_Q.$$

These personalized prices satisfy the Samuelson condition $P_f + P_m = P_Q$. Because both spouses value Q identically, each bears a share of its cost equal to their bargaining weight.

3. Generalized sharing rule (GSR). The GSR adds each spouse's valuation of public expenditure at their personalized price to their private expenditure:

$$\varsigma_f^* = Pc_f^* + P_f Q^* = \alpha(1-\gamma)I + \alpha\gamma I = \alpha I, \qquad \varsigma_m^* = (1-\alpha)I.$$

A wife who remains at home ($e = 1$, $I = w^M$) has no labor earnings but is allocated resources equivalent to αw^M. The home-production shock does not directly affect the GSR; instead, it operates through the Lindahl price.

4. Money-Metric Welfare Index (MMWI). The MMWI of spouse g, denoted by m_g, is the income that spouse g would need to attain utility u_g when facing a reference price vector rather than personalized prices. An individual with preferences $Q^\gamma c^{1-\gamma}$ and income m, facing price P for private consumption and $\bar{P}_Q$ for the public good, attains

$$v(m) = \log m - (1-\gamma)\log P - \gamma \log \bar{P}_Q + K.$$

Equating $v(m_g)$ with u_g and solving for m_g yields the general formula $m_g = \varsigma_g^* \, (\bar{P}_Q / P_g)^\gamma$.

The remaining question is which reference price $\bar{P}_Q$ to use. If I use the couple's own effective price, $\bar{P}_Q = P\beta^{-e}$, then $m_f = \alpha^{1-\gamma} I$ and $m_m = (1-\alpha)^{1-\gamma} I$. Under this reference price, the index does not capture home production: for a wife who stays at home, it equals $\alpha^{1-\gamma} w^M$ regardless of her draw of β. I therefore use the market price, $\bar{P}_Q = P$, so that the index explicitly accounts for home production. In this case, $v(m) = \log(m/P) + K$, and setting $v(m_f) = u_f$ yields

$$m_f = \alpha^{1-\gamma}\beta^{e\gamma} I, \qquad m_m = (1-\alpha)^{1-\gamma}\beta^{e\gamma} I. \tag{B.4}$$

Rewriting the wife's index as $m_f = \alpha I \cdot \alpha^{-\gamma} \cdot \beta^{e\gamma}$ separates three forces. The first term, αI, is her GSR share. The second term, $\alpha^{-\gamma}$, corresponds to the Lindahl-price adjustment. The third, $\beta^{e\gamma}$, captures the value of home production for spouses.

Summing equation (B.4) across spouses, I can express the gains from the public nature of Q as

$$\frac{m_f + m_m}{\beta^{e\gamma} I} = G(\alpha, \gamma) \equiv \alpha^{1-\gamma} + (1-\alpha)^{1-\gamma} > 1 \quad \text{whenever } 0 < \gamma < 1.$$

Thus, the spouses would jointly require more income than the household possesses if each had to purchase the public good separately at its full market price.

Couples solve the mobility problem in equation (18) before β is realized. The relevant ex ante index is therefore the certain income $\bar{m}_g = \alpha_g^{1-\gamma} \, \mathbb{E}[\beta^{e\gamma} I(e)]$ that allows a person with

the same preferences to attain $\mathbb{E}[U_g]$.[61] With $\alpha_f \equiv \alpha$ and $\alpha_m \equiv 1-\alpha$,

$$\bar{m}_f = \alpha^{1-\gamma}\, B\, w^M, \qquad \bar{m}_m = (1-\alpha)^{1-\gamma}\, B\, w^M. \tag{B.5}$$

where B is the term defined in Section III. Summing equation (B.5) across spouses gives the ex ante counterpart of $G(\alpha,\gamma)$:

$$\bar{m}_f + \bar{m}_m = G(\alpha,\gamma)\, B\, w^M = G(\alpha,\gamma)\, w^M(1+r)\, \frac{\kappa/\gamma - \rho}{\kappa/\gamma - 1}.$$

Dividing by P gives the couple's money-metric indirect utility, $\bar{W}^c$, in Section III. Relative to household income when the wife works, $w^M(1+r)$, the spouses' combined index is therefore scaled up twice: once by $G(\alpha,\gamma)$, because the public good is shared, and once by the value of the option to stay at home.

5. MMWI for singles. The central assumption is that singles have the same preferences as spouses but cannot share the public good with another household member. A single of gender g with wage w^g therefore solves

$$\max \gamma \log Q^g + (1-\gamma)\log c^g \quad \text{s.t.} \quad P(c^g + Q^g) = w^g.$$

This problem implies $c^g = (1-\gamma)w^g/P$ and $Q^g = \gamma w^g/P$, with indirect utility $u_s^g = \log \frac{w^g}{P} + K$. The constant K does not affect choices and cancels out in every change in welfare. Thus, the model's $W^{s_g} = w^g/P$ is an exact representation of these preferences, and no quantitative result changes. What a single lacks is not the public good itself but its *publicness*: a single person must pay its full price. Accordingly, the MMWI for a single of gender g is

$$m^g = \frac{w^g}{P}.$$

6. Money-Metric-Weighted Aggregate Welfare. Starting from equation (35), I can express the per-person consumption-equivalent change in utility as

$$\bar{Z} = \frac{\sum N^{s_m} Z^{s_m} + \sum N^{s_f} Z^{s_f} + \sum N^c Z^c}{\sum N^{s_m} + \sum N^{s_f} + \sum N^c}.$$

The measure $\bar{Z}$ averages percentage changes defined relative to different bases: for a single, the base is w/P; for a spouse, it is a share of the household's full income, together with a shared public good. To express the per-person consumption-equivalent change in monetary units, I therefore construct a money-metric-weighted measure that aggregates across singles and couples.

[61] Household income is $I = w^M(1+r)$ when the wife works and $I = w^M$ when she stays at home, so $\beta^{e\gamma} I = w^M \max\{1+r, \beta^\gamma\}$ and $\mathbb{E}[\beta^{e\gamma} I] = B\, w^M$.

Because each spouse's expected utility is a constant multiple of the couple's expected utility, the consumption-equivalent change of each spouse is identical to that of the couple when expressed in hat notation. The money-metric-weighted measure is therefore

$$\bar{Z}^{\$} = \frac{\sum m_i \, Z_i}{\sum m_i} = \bar{Z} + \frac{\mathrm{Cov}(m_i, Z_i)}{\bar{m}}.$$

The covariance term captures whether the gains from the shock accrue to individuals with larger money metrics, such as spouses, or to those with smaller money metrics, such as singles.

C. Dynamic Hat Algebra

This appendix develops the Dynamic Hat Algebra used to solve and simulate the model. It expresses the equilibrium entirely in terms of changes over time and ratios across the with-shock and no-shock economies, so that I never have to recover the levels of the fundamentals. The appendix proceeds in two parts. I first state the two propositions that characterize the no-shock economy: one for the temporary equilibrium at a given point in time and another for the full sequential equilibrium, together with the algorithm used to solve them. I then derive the corresponding results for the with-shock economy, defined by an alternative path of sectoral productivities.

C.1 Solving the Model

Notation. Let $\dot{x}_{t+1} = x_{t+1}/x_t$ denote the proportional change in any variable x between t and $t+1$ in the no-shock economy. Let $\hat{x}_{t+1} = \dot{x}'_{t+1}/\dot{x}_{t+1}$, where $\dot{x}'_{t+1} = x'_{t+1}/x'_t$, denote the ratio of the changes in the with-shock economy to those in the no-shock economy. Primes denote levels in the with-shock economy throughout. For every feasible bundle $\boldsymbol{o}$, define $u^h_{\boldsymbol{o},t} = \exp(V^h_{\boldsymbol{o},t})$, and define regional expenditure as $E_{j,t} \equiv P_{j,t}Q_{j,t}$. Because local sectoral productivities $\{A_{j,n,t}\}$ are the only fundamentals that vary over time, a quantitative application begins with a path of productivity changes $\{\dot{A}_{j,n,t}\}$ and solves the equilibrium system below.

For couples, define the quality-adjusted value of destination $\boldsymbol{d}$ as

$$B_{\boldsymbol{d},t} = \mathcal{A}\left[(1+r_{\boldsymbol{d},t}) + \chi^{\kappa}\frac{\gamma}{\kappa-\gamma}\left(1+r_{\boldsymbol{d},t}\right)^{1-\kappa/\gamma}\right].$$

Because $\mathcal{A}$ is constant, it cancels from all proportional changes and cross-economy ratios of B. I retain it in the level expressions below only for completeness. The proportional change in B is

$$\dot{B}_{\boldsymbol{d},t+1} \equiv \frac{B_{\boldsymbol{d},t+1}}{B_{\boldsymbol{d},t}} = \frac{(1+r_{\boldsymbol{d},t+1}) + \chi^{\kappa}\frac{\gamma}{\kappa-\gamma}\left(1+r_{\boldsymbol{d},t+1}\right)^{1-\kappa/\gamma}}{(1+r_{\boldsymbol{d},t}) + \chi^{\kappa}\frac{\gamma}{\kappa-\gamma}\left(1+r_{\boldsymbol{d},t}\right)^{1-\kappa/\gamma}}. \tag{C.1}$$

Proposition C.1. *Given the temporary-equilibrium allocation at t, the temporary equilibrium at $t+1$ can be computed without knowing the levels of the fundamentals. It is characterized by the following system of equations:*

$$\begin{aligned}
\dot{w}^M_{j,n,t+1} &= \dot{p}_{j,n,t+1}\dot{A}^{(\sigma-1)/\sigma}_{j,n,t+1}\dot{M}^{-1/\sigma}_{j,n,t+1}\dot{Y}^{1/\sigma}_{j,n,t+1},\\
\dot{w}^F_{j,n,t+1} &= \dot{p}_{j,n,t+1}\dot{A}^{(\sigma-1)/\sigma}_{j,n,t+1}\dot{F}^{-1/\sigma}_{j,n,t+1}\dot{Y}^{1/\sigma}_{j,n,t+1},\\
\dot{Y}_{j,n,t+1} &= \dot{A}_{j,n,t+1}\left(s_{j,n,m,t}\dot{M}^{(\sigma-1)/\sigma}_{j,n,t+1} + s_{j,n,f,t}\dot{F}^{(\sigma-1)/\sigma}_{j,n,t+1}\right)^{\sigma/(\sigma-1)},
\end{aligned}$$

$$\dot{P}_{j,n,t+1} = \left(\sum_{i=1}^{L} \sum_{k=1}^{K} \pi_{ik,jn,t} \left(\dot{p}_{i,k,t+1} \right)^{1-\eta_n} \right)^{1/(1-\eta_n)},$$

$$\pi_{ik,jn,t+1} = \pi_{ik,jn,t} \left(\frac{\dot{p}_{i,k,t+1}}{\dot{P}_{j,n,t+1}} \right)^{1-\eta_n},$$

$$\dot{P}_{j,t+1} = \prod_{n=1}^{K} \dot{P}_{j,n,t+1}^{\iota_n}, \qquad \dot{E}_{j,t+1} = \dot{P}_{j,t+1} \dot{Q}_{j,t+1},$$

$$\dot{p}_{i,k,t+1} \dot{Y}_{i,k,t+1} = \sum_{j=1}^{L} \sum_{n=1}^{K} \varphi_{ik,jn,t} \frac{\pi_{ik,jn,t+1}}{\pi_{ik,jn,t}} \dot{E}_{j,t+1},$$

for all region-sector pairs (j,n)*, where* $s_{j,n,m,t} \equiv \frac{w^M_{j,n,t} M_{j,n,t}}{p_{j,n,t} Y_{j,n,t}}$ *and* $s_{j,n,f,t} \equiv \frac{w^F_{j,n,t} F_{j,n,t}}{p_{j,n,t} Y_{j,n,t}}$ *are the male and female shares of sectoral revenue, and* $\varphi_{ik,jn,t} \equiv \frac{\pi_{ik,jn,t}\, \iota_n P_{j,t} Q_{j,t}}{p_{i,k,t} Y_{i,k,t}}$ *is the share of origin-sector revenue from* (i,k) *sold to destination sector* (j,n)*. Moreover, for each destination bundle relevant to couples,* $\dot{r}_{\boldsymbol{d},t+1} = \dot{w}^F_{j,n_f,t+1} / \dot{w}^M_{j,n_m,t+1}$*,* $r_{\boldsymbol{d},t+1} = r_{\boldsymbol{d},t}\, \dot{r}_{\boldsymbol{d},t+1}$*,* $\rho_{\boldsymbol{d},t+1} = 1 - \chi^{\kappa} \left(1 + r_{\boldsymbol{d},t+1}\right)^{-\kappa/\gamma}$*, and* $\dot{B}_{\boldsymbol{d},t+1}$ *follows from equation* (C.1)*.*

Building on Proposition C.1, I next characterize the sequential equilibrium in time differences. Equation (20) implies that the ratio of within-location shares between $t+1$ and t is

$$\frac{\lambda^h_{\boldsymbol{od}|j,t+1}}{\lambda^h_{\boldsymbol{od}|j,t}} = \frac{\exp\left[\delta \left(V^h_{\boldsymbol{d},t+2} - V^h_{\boldsymbol{d},t+1}\right)\right]^{1/\nu}}{\Omega^h_{\boldsymbol{o}j,t+1} / \Omega^h_{\boldsymbol{o}j,t}}, \qquad \frac{\Omega^h_{\boldsymbol{o}j,t+1}}{\Omega^h_{\boldsymbol{o}j,t}} = \sum_{\boldsymbol{z} \in \mathcal{S}_h(j)} \lambda^h_{\boldsymbol{oz}|j,t} \left(\dot{u}^h_{\boldsymbol{z},t+2}\right)^{\delta/\nu},$$

where the second equality follows from the time invariance of $\mu_{\boldsymbol{oz}}$. Define the proportional change in the inclusive value of location j as

$$D^h_{\boldsymbol{o}j,t} \equiv \sum_{\boldsymbol{z} \in \mathcal{S}_h(j)} \lambda^h_{\boldsymbol{oz}|j,t} \left(\dot{u}^h_{\boldsymbol{z},t+2}\right)^{\delta/\nu}. \tag{C.2}$$

The location shares in equation (19) and the value function in equation (18) depend on continuation values only through the inclusive values. Their time differences can therefore be expressed entirely in terms of $D^h_{\boldsymbol{o}j,t}$.

Proposition C.2. *Given the economy's initial allocation, including the initial location shares* $\lambda^h_{\boldsymbol{o}j,0}$ *and within-location shares* $\lambda^h_{\boldsymbol{od}|j,0}$*, together with an anticipated convergent sequence of productivity changes, the sequential equilibrium in time differences can be solved without recovering the levels of the fundamentals. The solution satisfies the following system of*

nonlinear equations:

$$\lambda^h_{\boldsymbol{od}|j,t+1} = \frac{\lambda^h_{\boldsymbol{od}|j,t}\left(\dot{u}^h_{\boldsymbol{d},t+2}\right)^{\delta/\nu}}{\sum_{\boldsymbol{z}\in\mathcal{S}_h(j)}\lambda^h_{\boldsymbol{oz}|j,t}\left(\dot{u}^h_{\boldsymbol{z},t+2}\right)^{\delta/\nu}}, \qquad \forall h,$$

$$\lambda^h_{\boldsymbol{o}j,t+1} = \frac{\lambda^h_{\boldsymbol{o}j,t}\left(D^h_{\boldsymbol{o}j,t}\right)^{\nu/\theta}}{\sum_{\ell=1}^{L}\lambda^h_{\boldsymbol{o}\ell,t}\left(D^h_{\boldsymbol{o}\ell,t}\right)^{\nu/\theta}}, \qquad \forall h,$$

$$\dot{u}^h_{\boldsymbol{o},t+1} = \dot{W}^h_{\boldsymbol{o},t+1}\left(\sum_{\ell=1}^{L}\lambda^h_{\boldsymbol{o}\ell,t}\left(\sum_{\boldsymbol{z}\in\mathcal{S}_h(\ell)}\lambda^h_{\boldsymbol{oz}|\ell,t}\left(\dot{u}^h_{\boldsymbol{z},t+2}\right)^{\delta/\nu}\right)^{\nu/\theta}\right)^{\theta}, \qquad \forall h,$$

$$N^h_{\boldsymbol{d},t+1} = \sum_{\boldsymbol{o}\in\mathcal{S}_h}\lambda^h_{\boldsymbol{od}|j,t}\,\lambda^h_{\boldsymbol{o}j,t}\,N^h_{\boldsymbol{o},t}, \qquad \boldsymbol{d}\in\mathcal{S}_h(j), \quad \forall h,$$

$$M_{j,n,t+1} = N^{s_m}_{j,n,t+1} + \sum_{n_f=1}^{K} N^c_{j,n,n_f,t+1},$$

$$F_{j,n,t+1} = N^{s_f}_{j,n,t+1} + \sum_{n_m=0}^{K}\rho_{j,n_m,n,t+1}N^c_{j,n_m,n,t+1},$$

for all feasible bundles and all sectors at each t, where $D^h_{\boldsymbol{o}\ell,t}$ is defined in equation (C.2) and the sectoral prices, trade shares, and participation objects solve the temporary-equilibrium system in Proposition C.1.

Because the migration costs $\mu_{\boldsymbol{od}}$ are time-invariant, they cancel from all ratios and therefore need not be recovered in levels. The flow term in the couple's continuation value is the only one that contains an additional factor. In particular, because the couple's indirect utility is $W^c_{\boldsymbol{o},t} = B_{\boldsymbol{o},t}\,w^M_{i,k_m,t}/P_{i,t}$, its proportional change is $\dot{B}_{\boldsymbol{o},t+1}\dot{w}^M_{i,k_m,t+1}/\dot{P}_{i,t+1}$. Thus, the relevant term is the *ratio* defined in equation (C.1), rather than the level $B_{\boldsymbol{o},t+1}$.

Algorithm for Solving the Sequential Competitive Equilibrium. Given an initial equilibrium at $t=0$ and an anticipated convergent sequence of sectoral productivity changes $\left\{\dot{A}_{j,n,t}\right\}^{L,K,T}_{j=1,\,n=1,\,t=0}$, the algorithm is as follows:

1. Fix a sufficiently large terminal horizon T and guess a path for the changes in continuation values $\left\{\dot{u}^{h,(0)}_{\boldsymbol{o},t+1}\right\}^{T}_{t=0}$, subject to the terminal condition $\dot{u}^{h,(0)}_{\boldsymbol{o},T+1} = 1$ for every feasible bundle $\boldsymbol{o}$.

2. Decompose the initial migration matrix into location shares $\lambda^h_{\boldsymbol{o}j,0} = \sum_{\boldsymbol{d}\in\mathcal{S}_h(j)}\lambda^h_{\boldsymbol{od},0}$ and within-location shares $\lambda^h_{\boldsymbol{od}|j,0} = \lambda^h_{\boldsymbol{od},0}/\lambda^h_{\boldsymbol{o}j,0}$. Using Proposition C.2, update these objects forward in the following order: the within-location shares, $D^h_{\boldsymbol{o}j,t}$, and then the location shares.

3. Starting from the initial allocation, apply the law of motion to recover the entire path of household masses $\left\{N_{\boldsymbol{d},t}^{s_m,(0)}, N_{\boldsymbol{d},t}^{s_f,(0)}, N_{\boldsymbol{d},t}^{c,(0)}\right\}_{t=0}^{T}$, and aggregate these masses into sectoral male labor supplies $\left\{M_{j,n,t}^{(0)}\right\}_{t=0}^{T}$.

4. Solve the temporary equilibrium sequentially for each period $t \geq 0$:

   (a) Guess the participation rates $\{\rho_{\boldsymbol{d},t+1}^{(0)}\}$ and use them to construct the sectoral female labor supplies $\left\{F_{j,n,t+1}^{(0)}\right\}$.

   (b) Conditional on the labor aggregates and the path of fundamentals, solve the production block for $\left\{\dot{Y}_{j,n,t+1}^{(0)}, \dot{w}_{j,n,t+1}^{M,(0)}, \dot{w}_{j,n,t+1}^{F,(0)}\right\}$, the trade block for $\left\{\dot{p}_{j,n,t+1}^{(0)}, \dot{P}_{j,n,t+1}^{(0)}, \pi_{ik,jn,t+1}^{(0)}\right\}$, and the regional aggregation block for $\left\{\dot{P}_{j,t+1}^{(0)}, \dot{E}_{j,t+1}^{(0)}\right\}$.

   (c) Recover $\left\{\dot{r}_{\boldsymbol{d},t+1}^{(0)}\right\}$, construct the levels $r_{\boldsymbol{d},t+1}^{(0)} = r_{\boldsymbol{d},t}\dot{r}_{\boldsymbol{d},t+1}^{(0)}$, and update $\left\{\rho_{\boldsymbol{d},t+1}^{(1)}, \dot{B}_{\boldsymbol{d},t+1}^{(1)}\right\}$.

   (d) Iterate over steps (a) through (c) until the participation block converges.

5. Given the temporary-equilibrium path and the guessed continuation values, solve backward from T for the updated path $\left\{\dot{u}_{\boldsymbol{o},t+1}^{h,(1)}\right\}_{t=0}^{T}$ using the $\dot{u}^h$ equation in Proposition C.2. This backward solution requires both the location shares and the within-location shares.

6. Use the updated path as the new guess and repeat steps 2 through 5 until the sequential equilibrium converges.

C.2 Solving the With-Shock Equilibrium

The path of sectoral productivity fundamentals in the with-shock economy, $\{A'_{j,n,t}\}_{t\geq 1}$, defines the corresponding productivity hats $\{\hat{A}_{j,n,t}\}_{t\geq 1}$. Under CDP's timing, households do not anticipate the change at $t = 0$. Therefore, $\lambda'^{h}_{\boldsymbol{o}j,0} = \lambda^{h}_{\boldsymbol{o}j,0}$, $\lambda'^{h}_{\boldsymbol{od}|j,0} = \lambda^{h}_{\boldsymbol{od}|j,0}$, and $N'^{h}_{\boldsymbol{o},1} = N^{h}_{\boldsymbol{o},1}$.

Dividing each with-shock migration equation in Proposition C.2 by its no-shock counterpart, and using $\dot{\lambda}^{h}_{\boldsymbol{od}|j,t} = \lambda^{h}_{\boldsymbol{od}|j,t}/\lambda^{h}_{\boldsymbol{od}|j,t-1}$ and $\dot{\lambda}^{h}_{\boldsymbol{o}j,t} = \lambda^{h}_{\boldsymbol{o}j,t}/\lambda^{h}_{\boldsymbol{o}j,t-1}$, yields the following with-shock analogue of equation (C.2):

$$\hat{D}^{h}_{\boldsymbol{o}j,t} \equiv \sum_{\boldsymbol{z}\in\mathcal{S}_h(j)} \lambda'^{h}_{\boldsymbol{oz}|j,t-1}\, \dot{\lambda}^{h}_{\boldsymbol{oz}|j,t}\, \left(\hat{u}^{h}_{\boldsymbol{z},t+1}\right)^{\delta/\nu}. \tag{C.3}$$

Proposition C.3. *Given the no-shock economy and a convergent sequence of productivity changes in the with-shock economy relative to the no-shock path, the with-shock sequential*

equilibrium can be solved without recovering the fundamental levels. For $t \geq 2$, it is characterized by the following system of nonlinear equations:

$$\lambda'^{h}_{\boldsymbol{od}|j,t} = \frac{\lambda'^{h}_{\boldsymbol{od}|j,t-1}\, \dot{\lambda}^{h}_{\boldsymbol{od}|j,t}\, \left(\hat{u}^{h}_{\boldsymbol{d},t+1}\right)^{\delta/\nu}}{\hat{D}^{h}_{\boldsymbol{o}j,t}}, \qquad \forall h,$$

$$\lambda'^{h}_{\boldsymbol{o}j,t} = \frac{\lambda'^{h}_{\boldsymbol{o}j,t-1}\, \dot{\lambda}^{h}_{\boldsymbol{o}j,t}\, \left(\hat{D}^{h}_{\boldsymbol{o}j,t}\right)^{\nu/\theta}}{\sum_{\ell=1}^{L} \lambda'^{h}_{\boldsymbol{o}\ell,t-1}\, \dot{\lambda}^{h}_{\boldsymbol{o}\ell,t}\, \left(\hat{D}^{h}_{\boldsymbol{o}\ell,t}\right)^{\nu/\theta}}, \qquad \forall h,$$

$$\hat{u}^{h}_{\boldsymbol{o},t} = \hat{W}^{h}_{\boldsymbol{o},t} \left(\sum_{\ell=1}^{L} \lambda'^{h}_{\boldsymbol{o}\ell,t-1}\, \dot{\lambda}^{h}_{\boldsymbol{o}\ell,t} \left(\sum_{\boldsymbol{z}\in\mathcal{S}_h(\ell)} \lambda'^{h}_{\boldsymbol{oz}|\ell,t-1}\, \dot{\lambda}^{h}_{\boldsymbol{oz}|\ell,t}\, \left(\hat{u}^{h}_{\boldsymbol{z},t+1}\right)^{\delta/\nu} \right)^{\nu/\theta} \right)^{\theta}, \qquad \forall h,$$

$$N'^{h}_{\boldsymbol{d},t+1} = \sum_{\boldsymbol{o}\in\mathcal{S}_h} \lambda'^{h}_{\boldsymbol{od}|j,t}\, \lambda'^{h}_{\boldsymbol{o}j,t}\, N'^{h}_{\boldsymbol{o},t}, \qquad \boldsymbol{d}\in\mathcal{S}_h(j), \quad \forall h,$$

$$M'_{j,n,t+1} = N'^{s_m}_{j,n,t+1} + \sum_{n_f=1}^{K} N'^{c}_{j,n,n_f,t+1},$$

$$F'_{j,n,t+1} = N'^{s_f}_{j,n,t+1} + \sum_{n_m=0}^{K} \rho'_{j,n_m,n,t+1} N'^{c}_{j,n_m,n,t+1}.$$

In period $t = 1$, when households first learn the new path, the migration equations instead become

$$\lambda'^{h}_{\boldsymbol{od}|j,1} = \frac{\upsilon^{h}_{\boldsymbol{od}|j}\, \left(\hat{u}^{h}_{\boldsymbol{d},2}\right)^{\delta/\nu}}{\sum_{\boldsymbol{z}\in\mathcal{S}_h(j)} \upsilon^{h}_{\boldsymbol{oz}|j}\, \left(\hat{u}^{h}_{\boldsymbol{z},2}\right)^{\delta/\nu}},$$

$$\lambda'^{h}_{\boldsymbol{o}j,1} = \frac{\lambda^{h}_{\boldsymbol{o}j,1} \left(\sum_{\boldsymbol{z}\in\mathcal{S}_h(j)} \upsilon^{h}_{\boldsymbol{oz}|j}\, \left(\hat{u}^{h}_{\boldsymbol{z},2}\right)^{\delta/\nu}\right)^{\nu/\theta}}{\sum_{\ell=1}^{L} \lambda^{h}_{\boldsymbol{o}\ell,1} \left(\sum_{\boldsymbol{z}\in\mathcal{S}_h(\ell)} \upsilon^{h}_{\boldsymbol{oz}|\ell}\, \left(\hat{u}^{h}_{\boldsymbol{z},2}\right)^{\delta/\nu}\right)^{\nu/\theta}},$$

$$\hat{u}^{h}_{\boldsymbol{o},1} = \hat{W}^{h}_{\boldsymbol{o},1} \left(\sum_{\ell=1}^{L} \lambda^{h}_{\boldsymbol{o}\ell,1} \left(\sum_{\boldsymbol{z}\in\mathcal{S}_h(\ell)} \upsilon^{h}_{\boldsymbol{oz}|\ell}\, \left(\hat{u}^{h}_{\boldsymbol{z},2}\right)^{\delta/\nu} \right)^{\nu/\theta} \right)^{\theta}, \qquad \upsilon^{h}_{\boldsymbol{od}|j} \equiv \lambda^{h}_{\boldsymbol{od}|j,1} \left(\hat{u}^{h}_{\boldsymbol{d},1}\right)^{\delta/\nu}.$$

The with-shock temporary equilibrium is characterized by

$$\hat{w}^{M}_{j,n,t} = \hat{p}_{j,n,t} \hat{A}^{(\sigma-1)/\sigma}_{j,n,t} \hat{M}^{-1/\sigma}_{j,n,t} \hat{Y}^{1/\sigma}_{j,n,t},$$
$$\hat{w}^{F}_{j,n,t} = \hat{p}_{j,n,t} \hat{A}^{(\sigma-1)/\sigma}_{j,n,t} \hat{F}^{-1/\sigma}_{j,n,t} \hat{Y}^{1/\sigma}_{j,n,t},$$
$$\hat{Y}_{j,n,t} = \hat{A}_{j,n,t} \left(s_{j,n,m,t} \hat{M}^{(\sigma-1)/\sigma}_{j,n,t} + s_{j,n,f,t} \hat{F}^{(\sigma-1)/\sigma}_{j,n,t} \right)^{\sigma/(\sigma-1)},$$

$$\hat{P}_{j,n,t} = \left(\sum_{i=1}^{L} \sum_{k=1}^{K} \pi_{ik,jn,t} \left(\hat{p}_{i,k,t} \right)^{1-\eta_n} \right)^{1/(1-\eta_n)},$$

$$\pi'_{ik,jn,t} = \pi_{ik,jn,t} \left(\frac{\hat{p}_{i,k,t}}{\hat{P}_{j,n,t}} \right)^{1-\eta_n},$$

$$\hat{P}_{j,t} = \prod_{n=1}^{K} \hat{P}_{j,n,t}^{\iota_n}, \qquad \hat{E}_{j,t} = \hat{P}_{j,t} \hat{Q}_{j,t},$$

$$\hat{p}_{i,k,t} \hat{Y}_{i,k,t} = \sum_{j=1}^{L} \sum_{n=1}^{K} \varphi_{ik,jn,t} \frac{\pi'_{ik,jn,t}}{\pi_{ik,jn,t}} \hat{E}_{j,t},$$

$$\hat{r}_{\boldsymbol{d},t} = \frac{\hat{w}^F_{j,n_f,t}}{\hat{w}^M_{j,n_m,t}},$$

and the corresponding with-shock level objects satisfy

$$r'_{\boldsymbol{d},t} = r'_{\boldsymbol{d},t-1} \dot{r}_{\boldsymbol{d},t} \hat{r}_{\boldsymbol{d},t}, \qquad \rho'_{\boldsymbol{d},t} = 1 - \chi^{\kappa} \left(1 + r'_{\boldsymbol{d},t} \right)^{-\kappa/\gamma}, \qquad \hat{B}_{\boldsymbol{d},t} = \frac{B'_{\boldsymbol{d},t}}{B_{\boldsymbol{d},t}},$$

with $B'_{\boldsymbol{d},t}$ evaluated at $r'_{\boldsymbol{d},t}$. The no-shock revenue shares $s_{j,n,m,t}$, $s_{j,n,f,t}$ and revenue-allocation shares $\varphi_{ik,jn,t}$ are those of Proposition C.1, taken from the no-shock equilibrium.

Solution Algorithm for the With-Shock Equilibrium. Given the no-shock sequential equilibrium and a convergent sequence of changes in fundamentals, $\left\{ \hat{A}_{j,n,t} \right\}_{t \geq 1}$, that defines the with-shock economy, the with-shock equilibrium is computed as follows:

1. Because the temporary equilibrium and migration shares at $t = 0$ are unchanged, initialize all with-shock objects at their no-shock values.

2. Fix a sufficiently large horizon T and guess a path $\left\{ \hat{u}^{h,(0)}_{\boldsymbol{o},t} \right\}_{t=1}^{T+1}$, imposing $\hat{u}^{h,(0)}_{\boldsymbol{o},T+1} = 1$ for all feasible bundles $\boldsymbol{o}$ and household types h.

3. Use the guessed continuation values to compute the path of with-shock migration shares. At $t = 1$, construct $\upsilon^h_{\boldsymbol{od}|j}$ using the no-shock within-location shares and $\hat{u}^{h,(0)}_{\boldsymbol{d},1}$, and apply the period-1 equations in Proposition C.3. For $t \geq 2$, update, in sequence, the within-location shares, $\hat{D}^h_{\boldsymbol{o}j,t}$ from equation (C.3), and the location shares.

4. Starting from the same initial allocation as in the no-shock economy, use the with-shock shares to compute the implied path of with-shock household populations $\left\{ N'^{s_m,(0)}_{\boldsymbol{d},t}, N'^{s_f,(0)}_{\boldsymbol{d},t}, N'^{c,(0)}_{\boldsymbol{d},t} \right\}_{t=1}^{T+1}$.

5. Given the with-shock household masses, recover the path of with-shock sectoral male labor supply $\left\{ M'^{(0)}_{j,n,t} \right\}_{t=1}^{T+1}$.

6. Solve the with-shock temporary equilibrium period by period. For each $t \geq 0$, proceed as follows:

   (a) Guess the with-shock participation rates $\left\{\rho'^{(0)}_{\boldsymbol{d},t+1}\right\}$.

   (b) Use these participation rates to compute with-shock female labor supply $\left\{F'^{(0)}_{j,n,t+1}\right\}$.

   (c) Construct $\left\{\hat{M}^{(0)}_{j,n,t+1}\right\}$ and $\left\{\hat{F}^{(0)}_{j,n,t+1}\right\}$. Given these objects and the changes in fundamentals, recover $\left\{\hat{Y}^{(0)}_{j,n,t+1}\right\}$ from the production block. Next, solve the goods-market-clearing block for $\left\{\hat{p}^{(0)}_{j,n,t+1}, \hat{P}^{(0)}_{j,n,t+1}, \hat{P}^{(0)}_{j,t+1}, \pi'^{(0)}_{ik,jn,t+1}\right\}$ and recover $\left\{\hat{w}^{M,(0)}_{j,n,t+1}, \hat{w}^{F,(0)}_{j,n,t+1}\right\}$ from the firms' first-order conditions. Finally, recover $\left\{\hat{r}^{(0)}_{\boldsymbol{d},t+1}\right\}$.

   (d) Recover $\left\{r'^{(0)}_{\boldsymbol{d},t+1}\right\}$ and update $\left\{\rho'^{(1)}_{\boldsymbol{d},t+1}, B'^{(1)}_{\boldsymbol{d},t+1}\right\}$. If the updated $\rho'^{(1)}_{\boldsymbol{d},t+1}$ differs from its initial guess, return to step (b) and iterate until the participation block converges.

   (e) Repeat steps (a) through (d) sequentially for each period t.

7. Given the with-shock temporary-equilibrium path and the guessed continuation values, solve backward from T for the updated path $\left\{\hat{u}^{h,(1)}_{\boldsymbol{o},t}\right\}_{t=1}^{T+1}$ using the $\hat{u}^h$ equations in Proposition C.3. At $t = 1$, $\hat{u}^h_{\boldsymbol{o},1}$ appears on both sides through θ, so solve this equation as a fixed point.

8. Use the updated path as the new guess and repeat steps 3 through 7 until the continuation-value path converges.

9. Compute welfare using equation (34), with $\hat{\lambda}^h_{\boldsymbol{o}i,s}$ and $\hat{\lambda}^h_{\boldsymbol{oo}|i,s}$ constructed from the with-shock and no-shock location and within-location staying shares.

D. Quantification of the Model

This appendix details three elements of the model's quantification. First, I explain how I assign each non-employed married woman to her potential sector. Then, I describe how I build the migration matrices by household type. Finally, I describe the calibration of the couple's maximization problem.

D.1 Assignment of Non-Employed Wives to Their Potential Sector

The model assigns every wife to a productive sector, but the ACS does not indicate which sector a non-employed married woman would work in. In the ACS, which I use to construct the initial distribution of couples across states and sector pairs, $\{N^c_{\boldsymbol{o},0}\}$, a married woman who is out of the labor force is recorded as non-employed. Thus, I do not observe the counterfactual sector in which she would have worked if the couple had decided that she would participate in the market. In the model, couples choose the wife's productive sector first and her participation second, so she is never assigned to the non-employment sector itself. I therefore need to distribute the non-employed wives across productive sectors.

If I were to distribute them across productive sectors in the same proportions as employed wives, I would ignore a selection problem. The sectors in which married women are more likely to be employed are also the ones in which they are less likely to be non-employed. Following the distribution of employed wives would therefore assign a larger share of non-employed wives to the sectors that are least likely to be their potential sector.

To recover the potential sector, I return to the model. I drop the time index and write the labor supply of married women in state j and sector n as

$$F^c_{j,n} = \rho_{j,n} N^c_{j,n}.$$

The mass of non-employed wives in the same cell, which I denote by Non-$F^c_{j,n}$, is

$$\text{Non-}F^c_{j,n} = (1 - \rho_{j,n}) N^c_{j,n}.$$

The ratio of non-employed to employed wives is therefore

$$\frac{\text{Non-}F^c_{j,n}}{F^c_{j,n}} = \frac{(1 - \rho_{j,n})}{\rho_{j,n}},$$

which implies that the mass of non-employed wives in that cell is

$$\text{Non-}F^c_{j,n} \;=\; \underbrace{F^c_{j,n}}_{\text{observed}} \times \underbrace{\frac{1-\rho_{j,n}}{\rho_{j,n}}}_{\text{the model's odds of non-employment}} .$$

Dividing by $\rho_{j,n}$ therefore corrects the selection problem and assigns each non-employed wife to her potential productive sector.

These cell-level masses are in shares rather than headcounts, so nothing in the model's odds forces them to sum to the number of non-employed married women actually observed in state j. I therefore normalize them within each state, rescaling the $\{\text{Non-}F^c_{j,n}\}_n$ proportionally so that they add up to the observed state total.

D.2 Building the Migration Matrix

Since I solve the model with Dynamic Hat Algebra, the time-invariant mobility costs $\mu_{\boldsymbol{od}}$ cancel from the recursion that governs the choice probabilities (Proposition C.2). The recursion requires only two initial objects for each household type: the location shares $\lambda^h_{\boldsymbol{o}j,0}$ of equation (19) and the within-location shares $\lambda^h_{\boldsymbol{od}|j,0}$ of equation (20).

Following CDP's Appendix E, I build the initial migration matrix as the product of an interstate probability from the ACS and a sectoral transition probability from the Current Population Survey (CPS), where an interstate mover's sectoral transition is imputed from the destination state's sector matrix:

$$\lambda^h_{(i,k)(j,n),0} = \underbrace{f^h_{ij}}_{\text{ACS}} \times \underbrace{P^h_j(n \mid k)}_{\text{CPS}} .$$

Summing over the destination sectors of location j gives $\lambda^h_{\boldsymbol{o}j,0} = f^h_{ij}$, and dividing by that sum gives $\lambda^h_{\boldsymbol{od}|j,0} = P^h_j(n \mid k)$.[62]

I build the matrices in three steps: interstate flows from the ACS, sectoral transitions from the CPS, and a mapping of both onto the model's full grid. First, I derive interstate flows from the ACS migration question for 2001–2007, which records the state of residence one year earlier. The resulting state-to-state matrix is therefore annual, whereas the model period is quarterly. Following CDP's Appendix E, I assume that interstate moves are evenly distributed throughout the year and that a household makes at most one such move annually, so the quarterly matrix retains one-fourth of the moves and imputes the remaining three-

[62] Where an origin puts no mass on a location, the within-location share is undefined. I set it to zero rather than to a uniform distribution, since the two factors only ever enter as a product, and a uniform distribution would place mass on destinations that the data indicate are unreachable once the location share moves away from zero.

fourths as non-moves:

$$f^q_{i\to j} = \tfrac{1}{4} f_{i\to j} \quad (j \neq i), \qquad f^q_{i\to i} = \tfrac{1}{4} f_{i\to i} + \tfrac{3}{4}.$$

I apply this adjustment to every quarter from 2000 to 2007. The adjustment is quantitatively important: combining the *annual* interstate probability with the *quarterly* sector matrix without it would inflate mobility by roughly a factor of six, implying a quarterly own-state retention rate of 0.959 for single men against 0.994 in CDP. The error would fall entirely on the location margin, which is calibrated to the population's response to the shock.

Next, I estimate sectoral transitions from the CPS on fewer sectors than the model's 22 sectors plus non-employment, because disaggregating by household type would otherwise leave too few observations per cell. I therefore pool the 12 manufacturing sub-sectors into a single manufacturing sector and the 8 service sub-sectors into a single service sector, leaving an aggregated five-sector structure: non-employment, manufacturing, retail, construction, and services. I estimate transitions on this aggregation separately for single men, single women, and couples, with couples moving on the joint space of both spouses' groups.

Combining the first two steps gives a transition matrix across five sectors and 50 states, by household type, which I then map onto the model's grid of 22 productive sectors. Let $\tilde{\boldsymbol{o}}$ denote the origin bundle in the five-sector grid and $\tilde{\boldsymbol{d}}$ denote the corresponding destination bundle. The production side considers 22 productive sectors across 50 states. Applying the same sectoral aggregation to the production side would lose the dispersion across manufacturing sectors in the China Trade Shock. I map the matrices onto the full grid by factorizing each transition probability into a group-level move and a within-group destination. For an individual of gender $g \in \{m, f\}$ in a household of type h,

$$\underbrace{P_{h,g}(\boldsymbol{d} \mid \boldsymbol{o})}_{\text{model's grid}} = \underbrace{P_h(\tilde{\boldsymbol{d}} \mid \tilde{\boldsymbol{o}})}_{\text{ACS + CPS, by household type}} \times \underbrace{P_g(\boldsymbol{d} \mid \tilde{\boldsymbol{d}},\ \boldsymbol{o})}_{\text{within-group composition, by gender}},$$

so that household type governs *which* broad sector, and which state, a household moves to, and the within-group composition governs *where within* that sector it lands. Because this factorizes a single conditional distribution, the rows of the resulting matrix sum to one by construction. The within-group composition only redistributes mass among sectors of the same group *in the same state*, so it leaves the location shares unchanged and acts only on the within-location shares.

To build the within-group composition, I split the CPS sample by sex and compute, within manufacturing and within services, the share of men and of women going to each sub-sector. This step matters: men and women sort differently within both groups, with men concentrated in heavy and durable manufacturing and women in lighter, non-durable

goods. Without it, the model would miss one of the channels through which the shock affected men and women differently.

D.3 Calibration of the Couple's Maximization Problem

This subsection documents the calibration of the couple's maximization problem in equation (22), with $\beta \sim \text{Pareto}(\chi, \kappa)$. Four parameters govern it: the wife's weight in the household objective, α; the public-good expenditure share, γ; and the lower bound and shape of the home-production shock, χ and κ. Because I follow Chiappori and Mazzocco (2017)'s method to solve the couple's problem, I identify the four parameters *sequentially*, each from its own source and each mapped onto a distinct piece of variation.

1) α and γ from Expenditure Data

The optimal allocations for private and public consumption in equation (23) are Cobb-Douglas shares, so both parameters map exactly onto expenditure shares. The parameter α maps onto the share of female private consumption in total private consumption, and γ maps onto the share of public expenditure in total expenditure:

$$\alpha = \frac{c_f^*}{c_f^* + c_m^*}, \qquad \gamma = \frac{q^*}{q^* + c_f^* + c_m^*}.$$

Data. I use the Consumer Expenditure Interview Survey (CEX), 2000–2007, to construct three inputs.[63] First, the *assignable private goods* c_m and c_f are adult apparel and footwear.[64] Second, the *public good* q is the market input to home production: because $Q = \beta^e q$, the object I need is what the household combines with the wife's home time. It comprises shelter, fuels and utilities, food at home, cable and internet services, do-it-yourself repair materials, and household operations such as domestic services, lawn care, and appliance repair. Household operations are the *market substitutes* for home time, so omitting them would measure q selectively on precisely the margin the model is about. Third, I construct *total expenditure* Y as consumption only.[65]

The sample consists of household-year pairs with complete twelve-month expenditure records. Couples are intact married couples with both spouses present, aged 25–65; singles are one-person consumer units in the same age band. I drop households containing a 16- or 17-year-old, because the CEX books apparel for ages 16 and over under the adult men's and

[63] The complete list of expenditure codes entering each of the three objects is available upon request.

[64] Children's, boys', girls', and infants' apparel are excluded, since the model has no children.

[65] The CEX expenditure file also carries asset stocks, debt balances, loans, principal repayments, and asset-sale receipts. All of these are excluded, as are transfers out of the household and gifts.

women's codes, so a teenager's clothing would otherwise be read as a spouse's assignable private consumption. The couples and singles samples are disjoint by construction.

Estimating α. I estimate the wife's weight in the household objective, α, from apparel, the only assignable good in the CEX, correcting for the fact that women spend more on apparel than men. Apparel is only a *subset* of private consumption, so the raw apparel ratio, 0.60 in this sample, is not α. Taken at face value, the raw ratio would imply that the wife has the larger Pareto weight. To recover α, I follow Browning et al. (2013): I estimate an Engel curve for apparel among *singles* of each gender and recover the resource shares ϱ_f and ϱ_m such that, evaluated at each spouse's own resources $\varrho_g Y$, these Engel curves reproduce the aggregate apparel spending observed in couples. Using both ϱ_f and ϱ_m, I recover $\alpha = \varrho_f/(\varrho_f + \varrho_m)$. The correction is substantial: it moves α from 0.60 to 0.47, which *reverses* which spouse has the larger bargaining power.

Voena (2015) estimates a value of 0.3, which is one-third smaller than the one I estimate. The two numbers do not align because α captures different objects in the two papers. In Voena (2015)'s world, α prices the wife's outside option given changes in divorce laws across US states, while in my model, α reflects her observed consumption. In addition, she estimates α using data from the PSID between 1980 and 1990, years when female labor force participation was lower than in the 2000s, and women's bargaining power is expected to rise as they enter the labor force.

Estimating γ. I measure the public-good expenditure share, γ, directly from expenditure data rather than as the residual implied by the resource shares. One alternative would be to set γ equal to the residual $1 - \varrho_f - \varrho_m$. The residual is 0.13, implying that the couple spends roughly \$5,400 a year on public goods, while couples are *observed* spending \$22,300 a year on shelter, utilities, and food alone. The residual is this small because the apparel Engel curves are essentially flat ($R^2 = 0.004$). Browning et al. (2013)'s method allows me to recover the *ratio*, but it carries almost no information about the *level* of resources, and the residual absorbs the error. I therefore measure γ directly as the share of total spending devoted to market inputs for home production, including shelter, utilities, food at home, and market substitutes for home time, such as domestic services and appliance repair. This yields $\gamma = 0.4887$.

Del Boca and Flinn (1995) estimate γ to be 0.204. The two values differ because 0.204 is an estimate based on divorced parents, whose only joint consumption concerns their children; they cannot enjoy economies of scale in housing, utilities, or transportation, among others. For couples who live together, joint consumption is therefore expected to account for a

significant share of total consumption.

Standard errors. The CEX is a stratified, multistage, clustered sample, and the BLS does not publish the design variables needed for an analytic variance. Thus, following the BLS recommendation, I re-run the *entire* estimator on each of the 44 replicate weight vectors and take the spread across replicates as the sampling variance. The resulting standard errors are 0.021 for α and 0.006 for γ.

2) κ/γ from the China Trade Shock in the PSID

As shown in Section IV, the participation rule satisfies

$$\log\big(1-\rho(r)\big) = \underbrace{\kappa \log \chi}_{\text{intercept}} - \underbrace{(\kappa/\gamma)}_{\text{slope}} \log(1+r). \tag{D.1}$$

This equation is a structural relation, not a regression. Cross-sectional variation in the relative wage r is not exogenous: r is an equilibrium object. In addition, I observe r only when the wife participates in the market. I therefore use the empirical design in Section II as the identification strategy for the participation block.

In particular, I identify κ/γ from the experiment in which the *husband's* industry is exposed. The wife is not directly displaced, so the shock affects her participation decision only through r, which is exactly the exclusion restriction required by equation (D.1). Because the equation is linear, the parameter is a Wald ratio:

$$\frac{\kappa}{\gamma} = -\frac{\Delta \log(1-\rho)}{\Delta \log(1+r)}. \tag{D.2}$$

To construct the numerator of equation (D.2), I use the estimates in Panel B of Table 3. Wives of exposed husbands raise labor-force participation by 4.2 p.p. from a baseline of 0.78, so ρ increases from 0.78 to 0.822. Then, the numerator is:

$$\Delta \log(1-\rho) = \log(1-0.822) - \log(1-0.78) = -0.212.$$

To construct the denominator, I estimate the static version of equation (5) on log(Hourly Wage) for each spouse. The sample consists of individuals who have at least one observation with positive wages before and during the shock. Table D.1 shows that when the husband is exposed to the shock, his hourly wage decreases by 16.4 log points while her wage decreases by 4.2 log points. Since the husband's wage falls by more than his wife's, the relative wage she faces rises. Using the estimates in Table D.1, r increases by 12.2 log points, which implies an increase of 13%. The r implied by the baseline mean is 0.88.

However, this number includes only employed individuals, while the numerator considers all individuals in the sample. Then, the correct value for r_0 is 0.607. Thus, the denominator of equation (D.2) is

$$\Delta \log(1+r) = \log(1+0.686) - \log(1+0.607) = 0.0477.$$

Substituting the numerator and the denominator into equation (D.2),

$$\frac{\kappa}{\gamma} = -\frac{-0.212}{+0.048} = 4.44.$$

With $\gamma = 0.4887$, I obtain $\kappa = \gamma \cdot \frac{\kappa}{\gamma} = 2.17$, which satisfies the integrability restriction $\kappa > \gamma$.

Table D.1: Husband's Exposure to the China Trade Shock and log(Hourly Wage)

	(1) Husband	(2) Wife
Exposed Husband=1	-0.078*	0.150***
	(0.042)	(0.052)
(Exposed Husband=1)x(Post=1)	-0.164*	-0.042
	(0.086)	(0.070)
Observations	12,023	9,979
R^2	0.35	0.31
Baseline Mean	3.31	2.91

Notes. Robust standard errors are two-way clustered at the husband's industry and individual levels. Given that I restrict the sample to individuals who have at least one observation with positive wages before and during the shock, the number of observations in this table does not match the number of observations in Fact 3. * for p<0.10, ** for p<0.05, *** for p<0.01.

3) χ as a Level Restriction

The participation rule in equation (D.1) is log-linear in $1-\rho$: its *slope* in $\log(1+r)$ identifies κ/γ, and its *intercept* identifies χ. With κ/γ estimated above and $\kappa = \gamma \cdot (\kappa/\gamma)$, χ is the single remaining free parameter, and I set it so that the model reproduces the observed *level* of married women's participation in 2000Q1, $\rho_0 = 0.621$. Married women's participation is therefore a *targeted* moment and is not reported as an untargeted success of the model.

I calibrate χ only on couples in which the husband is employed in a productive sector, because the model misses the participation of wives of non-employed husbands. In the data, wives of non-employed husbands work *less*, whereas the model predicts that they work *more*. Since ρ is monotone in r, and a non-employed husband's income is the benefit b_j, the ratio $r = w_f/b_j$ is mechanically the largest anywhere in the model, inducing wives to work. The data instead reflect strong positive assortative matching in the marriage market (Browning et al., 2014), which the model abstracts from. I denote the bundles in which the husband is employed by $\bar{\boldsymbol{o}}$.

Let $m_{\bar{\boldsymbol{o}},0}$ be the mass of couples in bundle $\bar{\boldsymbol{o}}$ at $t = 0$ and $r_{\bar{\boldsymbol{o}},0} = w^F_{\bar{\boldsymbol{o}},0}/w^M_{\bar{\boldsymbol{o}},0}$ its wage ratio. I then set χ such that the mass-weighted average participation rate the model generates equals the one in the data:

$$\sum_{\bar{\boldsymbol{o}}} m_{\bar{\boldsymbol{o}},0}\, \rho(r_{\bar{\boldsymbol{o}},0}; \chi) \;=\; \rho_0. \tag{D.3}$$

Since ρ is strictly decreasing in χ, equation (D.3) has a unique root, and it yields $\chi = 1.92$. This satisfies the support restriction $\chi \geq 1$ and leaves the participation margin interior in every cell.

E. Additional Results in the Quantitative Exercises

E.1 The Same Trade Shock in Every Economy

The household block changes who bears the China Trade Shock, but not the size of the shock where trade operates. In Table E.1, I compare CDP with the baseline economy, the all-singles economy, the all-singles economy without migration, and the three economies in which I shut down migration, partially shut down participation, or both. Manufacturing jobs per capita fall by between 0.197 and 0.206 p.p. in every economy I study, within 4.2% of the baseline. Welfare in the rest of the world does not move either: the mean gain of the other 36 regions is between 0.156% and 0.158% in every economy, and their gains correlate with those in CDP at 0.997. The sectors that drive the shock do not change either. I compute each sector's contribution to the change in employment at $t = 199$, as in Figures 2 and 5 of Caliendo et al. (2019). Across manufacturing industries, the contributions to the manufacturing employment decline in the baseline and the all-singles economy have a rank correlation of 0.94, and across non-manufacturing sectors, the contributions to the non-manufacturing employment gain have a rank correlation of 0.98. In both economies, computers, electronics, and furniture lead the decline, and other services and health lead the gain.

Table E.1: Same Shock in Every Economy

	CDP	Baseline	All-singles	All-singles, no migration	No migration	No participation	No migration & no participation
Manufacturing, $t = 199$							
Manuf. jobs per capita (p.p.)	−0.228	−0.206	−0.202	−0.200	−0.202	−0.200	−0.197
% change vs Baseline	+11.0	—	−1.6	−2.9	−1.7	−2.6	−4.2
Rest of the world (36 regions)							
Mean welfare gain (%)	0.158	0.156	0.158	0.157	0.156	0.156	0.156
Corr. with CDP	—	0.997	0.997	0.997	0.997	0.997	0.997

Notes. Responses at $t = 199$. Manufacturing jobs per capita is manufacturing employment divided by population, expressed in p.p.; % change vs. baseline is $100 \times (X/\text{baseline} - 1)$. CDP is the economy of Caliendo et al. (2019). Rest of the world: consumption-equivalent welfare gains of the 36 regions (35 countries and the rest of the world; China excluded), computed as in Caliendo et al. (2019) and summed over $t = 2, \ldots, 199$. The economies are defined in Table 7.

E.2 Effects of Shutting Down Migration

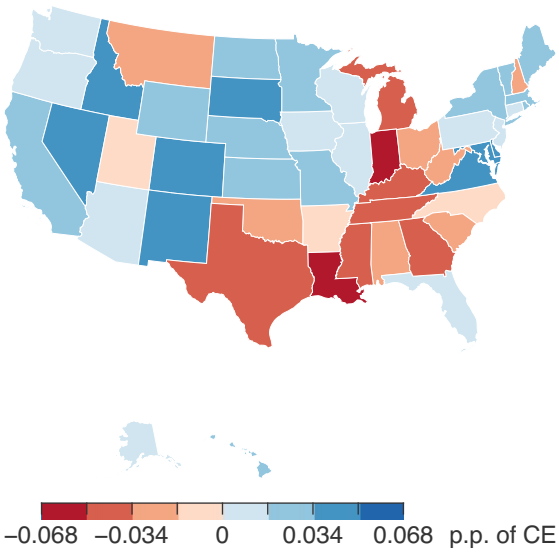
Shutting down migration barely moves aggregate welfare (+1.0%; Table 9), even though the employment response is 12% smaller. Moving is how people reach jobs, not how they

capture gains, and since sector switching is still allowed, a displaced worker still has their primary adjustment margin. This near-null, however, hides large spatial effects: welfare becomes 61% more spatially unequal (the coefficient of variation of state welfare gains rises from 0.28 to 0.45; Panel B of Table E.4), and the correlation between a state's welfare change in this exercise and its population response in the baseline is +0.98. Migration redistributes gains across states: the losers are the exposed states that were losing population, led by Indiana, Louisiana, and Kentucky; the winners are the destination states that were gaining population, which no longer have their gains diluted by arrivals (Figure E.1). Couples make this redistribution larger: without migration, the welfare gain of the people living in states losing people falls by 0.037 p.p. in the baseline economy and by 0.026 p.p. in the all-singles economy (Panel B of Table E.4).

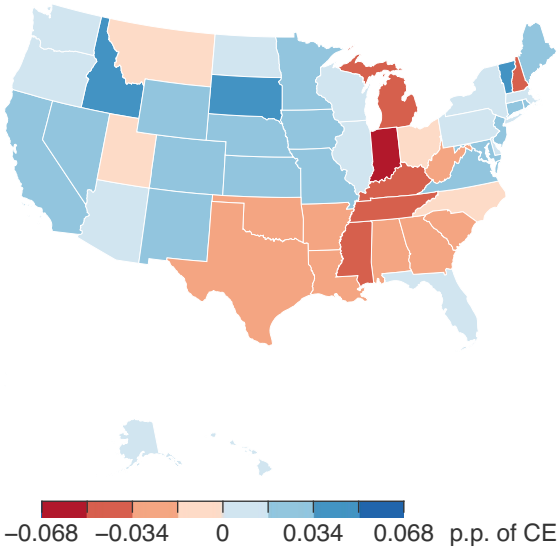
Unexposed couples benefit from shutting down migration (their gain rises by 2%; Table E.4) because they do not have to bear the costs of mobility decisions made by exposed households. When the exposed relocate, they disproportionately move to areas where the unexposed already live; their arrival increases local labor supply and pushes local wages down. With migration shut down, nobody arrives, and the unexposed keep a tighter local market.

Figure E.1: Change in State Welfare from Shutting Down Migration

(a) Baseline: no migration minus baseline



(b) All-singles: no migration minus all-singles



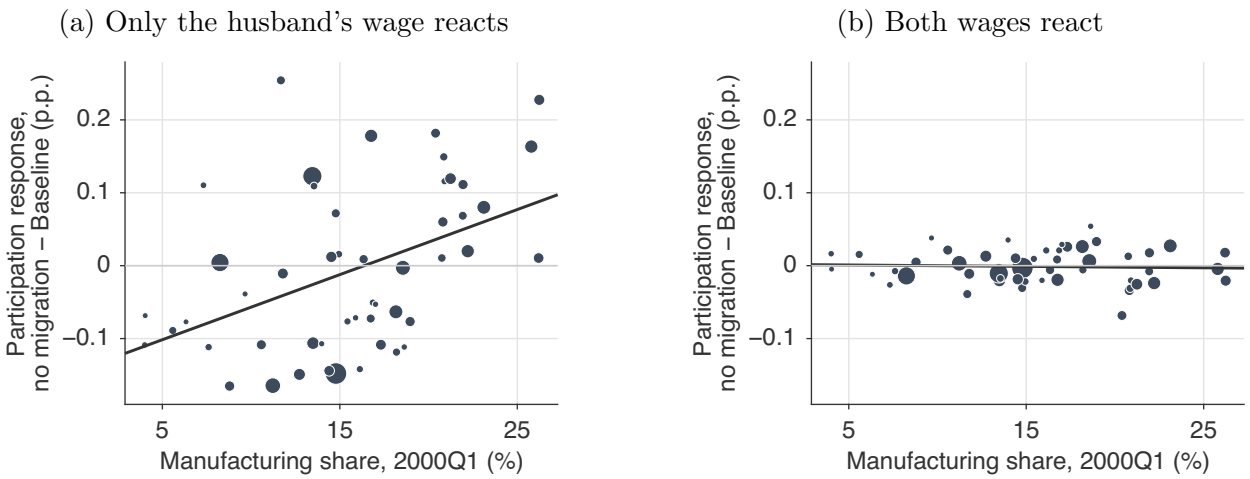
Notes. Change in state money-metric welfare, in p.p., for the people living in the state in 2000Q1, weighting every economy by the baseline money metrics. One common scale.

Participation does not compensate for the lack of spatial mobility: the participation

response of married women is 3.4% smaller in the counterfactual economy without migration (Panel C of Table E.3). Although this result may seem counterintuitive, it is driven by how couples decide on her participation. Given that ρ is homogeneous of degree zero in wages, it depends only on $r = w_f/w_m$ and not on the level, and this counterfactual moves both spouses' wages by nearly the same amount, leaving r intact to the third decimal. The participation response falls the least for couples in which the husband is exposed (-2.8%, -3.0%) and the most for unexposed couples (-3.5%). In states that lose people, shutting down migration reduces the shock's effect on men's employment rate from 0.10 to 0.03 p.p., whereas the effect on women's employment falls by a smaller amount, from 0.15 to 0.12 p.p.

Wives do not step in when migration is shut down, because general equilibrium undoes the partial-equilibrium incentive. Figure E.2 plots, state by state, the change in married women's participation response when migration is shut down. If only the husband's wage reacted, wives' participation would rise most where manufacturing was concentrated (correlation $+0.45$, panel (a)). Once both spouses' wages react, this relationship disappears (-0.05, panel (b)), and the change follows the population response to migration in the state ($+0.69$): participation falls most in the states people would have left, where both spouses' wages fall together.

Figure E.2: Why Wives Do Not Step In When Migration Is Shut Down

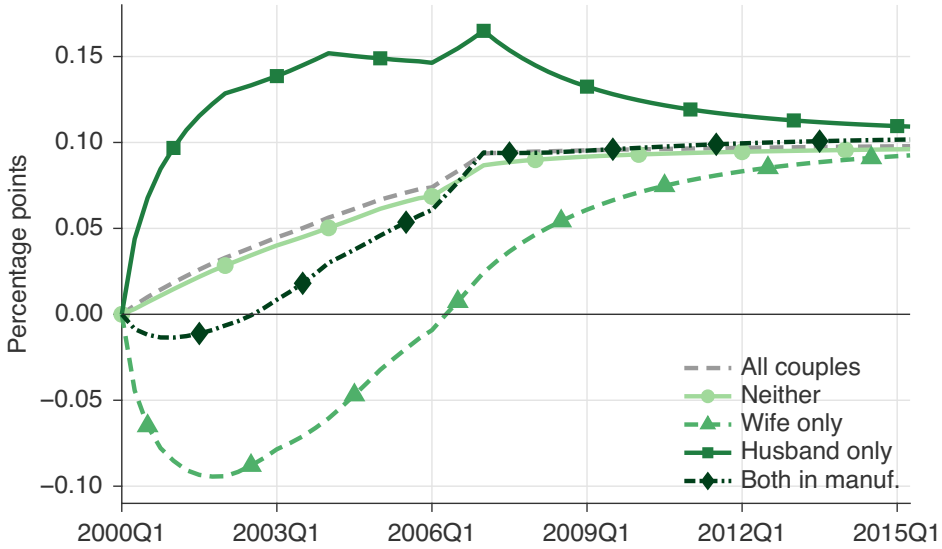


Notes. Each dot is a state, sized by its number of couples. The vertical axis is the change in married women's participation response at $t = 199$, in p.p., when migration is shut down, for the state's own married women. In panel (a), it is a partial-equilibrium calculation in which only the husband's wage takes its value from the economy without migration. The horizontal axis is the state's manufacturing share of employment in 2000Q1.

E.3 Participation as a Mitigation Mechanism

Married women's participation response differs by the couple's exposure in the short run, as discussed in Section V.2. Figure E.3 plots it over the first 62 quarters; its long-run values are in Table 6.

Figure E.3: Married Women's Short-Run Participation Response by Couples' Exposure



Notes. Baseline economy. Change in married women's quarterly participation rate (with-shock minus no-shock economy), in p.p., over the first 62 quarters (2000Q1–2015Q2), for the couples that started in each cell at $t = 1$. Couples are tagged by both spouses' sectors at $t = 1$; a non-working wife is tagged by her potential sector

Employment and migration when participation is partially shut down. Partially shutting down participation has the largest employment effect of the two margins (Table E.3): while the manufacturing decline is only 3% smaller in the counterfactual, the employment response is 24% smaller. If wives cannot respond to the shock by increasing their participation, the overall employment effect is lower. In addition, through general equilibrium effects, the employment response is lower for married men (Panel B of Table E.3).

When participation is partially shut down, couples with an exposed husband move slightly more: the couples' migration response rises from 0.0141 to 0.0146 p.p. when only the husband is exposed and from 0.0078 to 0.0080 p.p. when both are, whereas all other couples move less (Panel D of Table E.3). Moving does not make up for the loss: shutting down migration costs couples in which only the husband is exposed 3.4% of their gain, whether or not wives' participation is partially shut down (Panel B of Table 10). Conversely, wives do not work more when migration is shut down: their participation response is 3.4% smaller without migration, and an additional 2.7% smaller when participation is also partially shut down.

For couples with an exposed husband, the two margins are therefore asymmetric substitutes: they move more when wives cannot respond, but wives do not work more when they cannot move.

Valuing wives' time at home. In the baseline economy, a non-working wife's time is valued as home production, through $B_{\boldsymbol{o},t}$ in the couple's payoff

$$\frac{w^{M}_{i,k_m,t}}{P_{i,t}} \cdot B_{\boldsymbol{o},t}$$

To measure how much of the gains from the added-worker effect this valuation absorbs, I recompute couples' welfare with every decision held fixed and two alternative values of $B_{\boldsymbol{o},t}$. Under *market income only*, a non-working wife's time is worth nothing,

$$B^{\text{Market}}_{\boldsymbol{o},t} = \mathcal{A}\,(1 + \rho_{\boldsymbol{o},t}\, r_{\boldsymbol{o},t}).$$

Under *valued like singles*, a non-working wife receives the non-employment benefit, as a non-employed single does,

$$B^{\text{Singles}}_{\boldsymbol{o},t} = \mathcal{A}\left[1 + \rho_{\boldsymbol{o},t}\, r_{\boldsymbol{o},t} + (1 - \rho_{\boldsymbol{o},t})\,\frac{b_i}{w^{M}_{i,k_m,t}}\right],$$

with b_i held fixed in real terms.

Valuing wives' time at home like singles' raises the baseline economy's aggregate gain, such that it would be 8.5% higher than the all-singles economy (Table 9). The valuation also reorders couples: couples in which only the wife is exposed gain 0.022% rather than 0.161%, and those in which only the husband is exposed gain 0.320% rather than 0.140%. When the wife's wage falls, the couple falls back on home production; when the husband's falls, the couple gives up home production as she enters the market.

The two margins are weak complements. Table E.2 compares the welfare change from shutting down both margins with the sum of shutting down each one alone. For every group, shutting down both costs slightly less than the sum of the two, by between 0.0 and 0.4 percentage points; participation is worth 4.6% of the baseline gain with migration and 4.4% without it.

Partially shutting down participation lowers welfare in almost every state, with or without migration. Figure E.4 maps the change in state welfare from partially shutting down participation, with migration (no participation minus baseline) and without it (no migration & no participation minus no migration), on a common scale. With migration, all 50 states lose; without migration, 47 do, and Louisiana, South Carolina, and Texas gain slightly. The

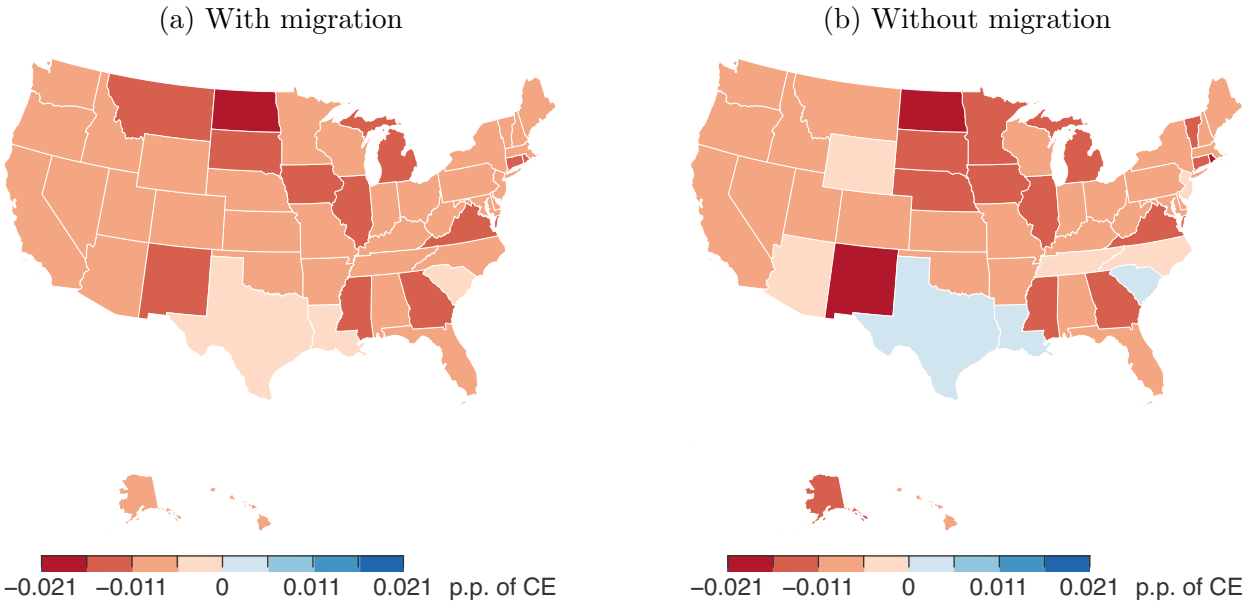
two maps correlate at 0.98 across states.

Table E.2: Welfare Effects of Shutting Down Migration and Participation

	Baseline (% CE)	No migration (%)	No participation (%)	No migration & no participation (%)	Sum of the two (%)	Interaction (p.p.)
Aggregate	0.175	+1.0	−4.6	−3.4	−3.7	+0.2
Single men	0.157	+2.9	−2.4	+0.7	+0.6	+0.1
Single women	0.234	+0.9	+8.8	+9.7	+9.7	+0.0
Couples	0.170	+0.7	−7.2	−6.2	−6.5	+0.3
Neither	0.180	+1.6	−6.9	−5.0	−5.2	+0.3
Wife only	0.161	+0.7	−5.0	−4.0	−4.3	+0.3
Husband only	0.140	−3.4	−10.2	−13.2	−13.6	+0.3
Both in manuf.	0.125	−2.7	−7.6	−9.9	−10.3	+0.4

Notes. The first column reports the baseline economy's money-metric consumption-equivalent gain (%); the next three columns report 100 × (economy/baseline − 1), weighting every economy by the baseline money metric at 2000Q1. The sum of the two is the sum of the no migration and no participation columns, and the interaction is the no migration & no participation column minus that sum, in points of the percentage change.

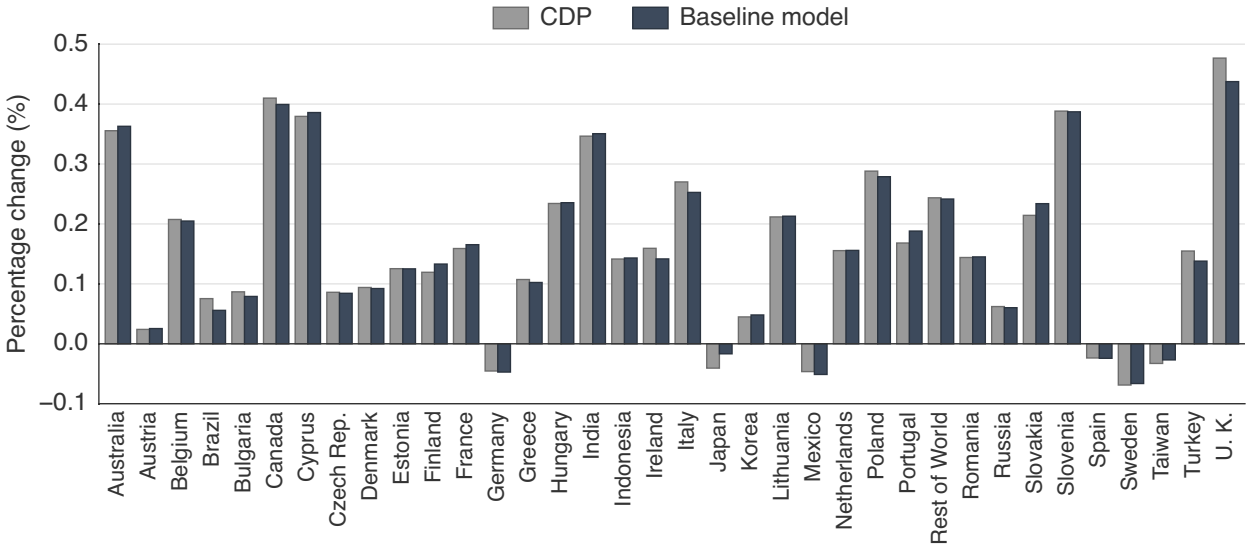
Figure E.4: Change in State Welfare from Partially Shutting Down Participation



Notes. Change in state money-metric welfare, in p.p., for the people living in the state in 2000Q1, weighting every economy by the baseline money metrics. Panel (a) is no participation minus baseline; panel (b) is no migration & no participation minus no migration. One common scale.

E.4 Additional Figures

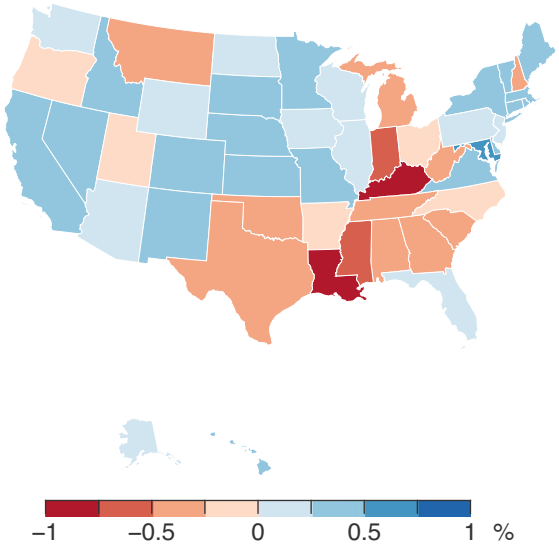
Figure E.5: Comparison of Welfare Effects in the Rest of the World



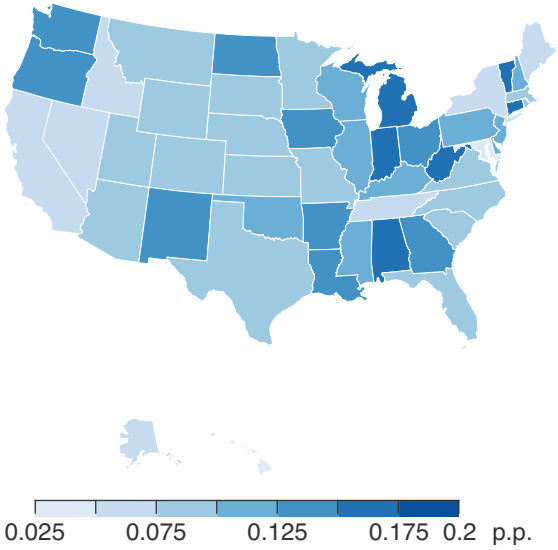
Notes. Consumption-equivalent welfare gains of the 36 regions (35 countries and the rest of the world; China excluded), calculated in both models as in Caliendo et al. (2019) and summed over $t = 2, \ldots, 199$.

Figure E.6: The China Trade Shock across States

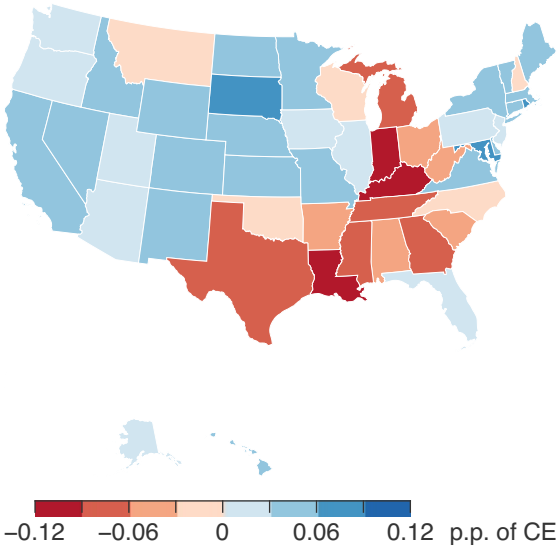
(a) Population change (%)



(b) Married women's participation (p.p.)



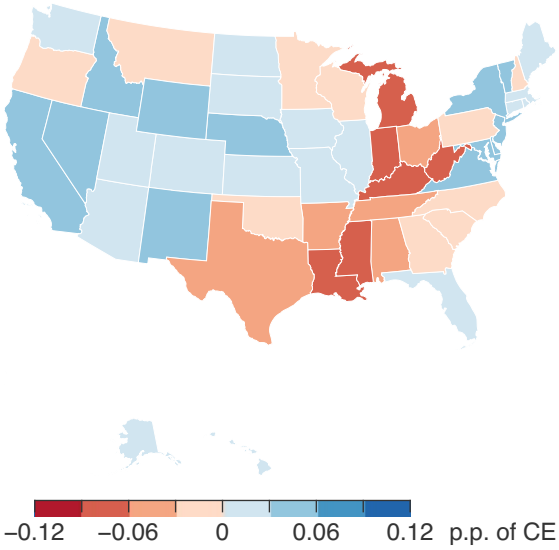
(c) State welfare minus national (p.p.)



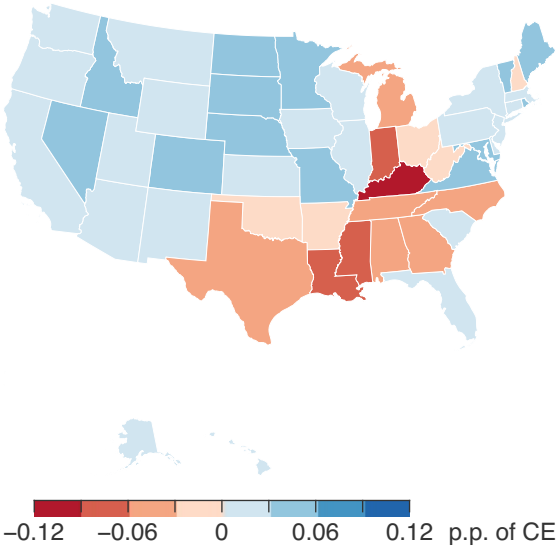
Notes. Panels (a) to (c) are long-run responses ($t = 199$): the change in the state's population as a percentage of its no-shock level, which sums to zero across states because the national population is fixed; the change in married women's participation rate, in p.p., averaged over the couples living in the state; and the state's money-metric welfare gain minus the national gain (0.175), in p.p., for the people living in the state in 2000Q1.

Figure E.7: Distribution of Welfare Effects by Household Type

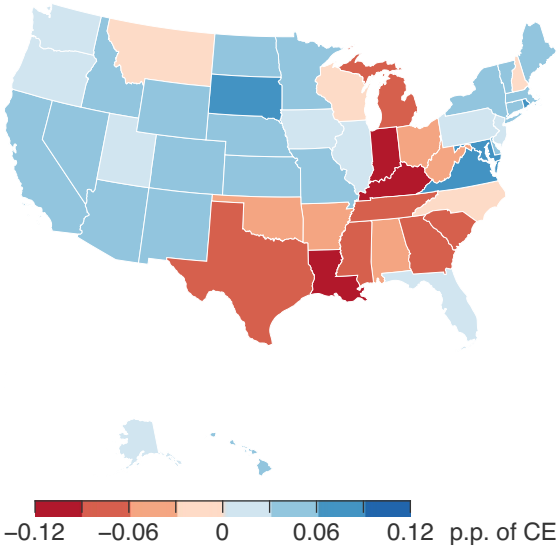
(a) Single men



(b) Single women

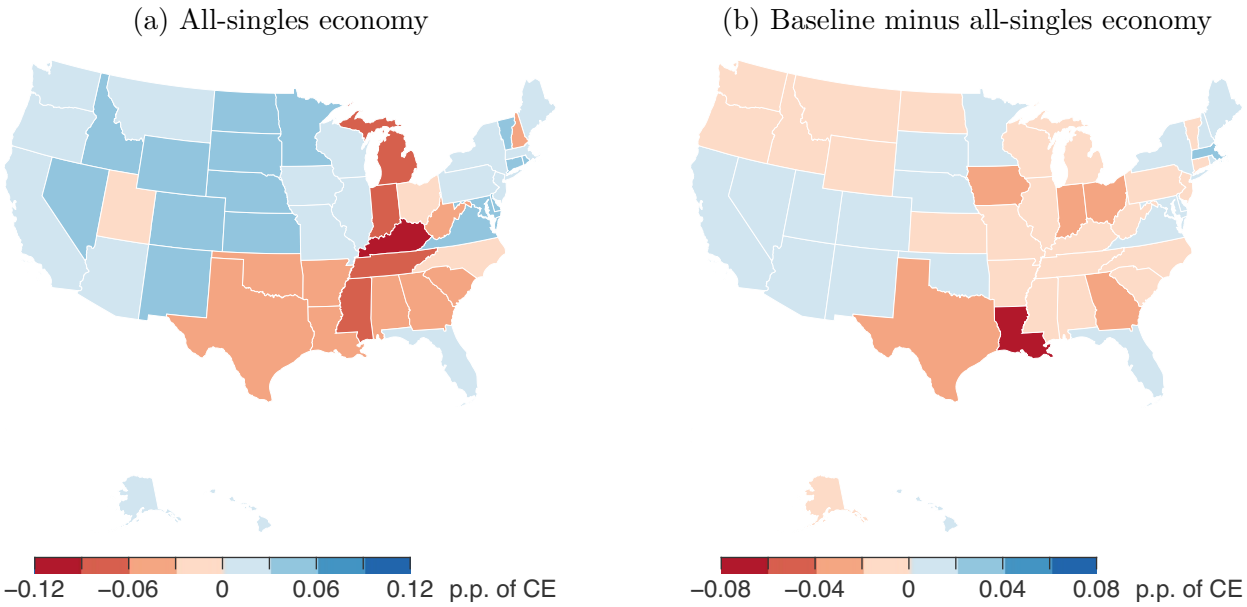


(c) Couples



Notes. State welfare minus the group's own national welfare, money-metric consumption-equivalent, in p.p., for the people living in the state in 2000Q1. The population-weighted coefficient of variation of state welfare gains is 0.23 for single men, 0.16 for single women, and 0.31 for couples.

Figure E.8: Distribution of Welfare Effects in the All-Singles Economy



Notes. Money-metric consumption-equivalent welfare gains, in p.p., for the same people located at their 2000Q1 state, weighting both economies by the baseline money metrics. Panel (a) is state welfare minus the national gain of the all-singles economy (0.178), on the same scale as Figure E.6c. Panel (b) is the baseline economy's state welfare minus the all-singles economy's.

E.5 Additional Tables

Table E.3: Responses to the China Trade Shock across Economies: All Groups

	Baseline	All-singles	All-singles, no migration	No migration	No participation	No migration & no participation
Panel A. Manufacturing						
Manuf. share of employment (p.p.)	−0.293	−0.291	−0.286	−0.287	−0.285	−0.279
Panel B. Employment rate (p.p.)						
Aggregate	0.112	0.132	0.119	0.099	0.085	0.073
Men	0.099	0.092	0.074	0.079	0.090	0.072
Women	0.124	0.170	0.161	0.117	0.080	0.075
Single men	0.106	0.092	0.075	0.082	0.104	0.081
Single women	0.158	0.170	0.161	0.143	0.171	0.157
Married men	0.096	0.093	0.074	0.079	0.086	0.069
Married women	0.111	0.170	0.161	0.107	0.043	0.041
Panel C. Married women's participation (p.p.)						
All couples	0.101			0.097	0.041	0.040
Neither	0.100			0.096	0.041	0.040
Wife only	0.101			0.097	0.041	0.040
Husband only	0.105			0.102	0.042	0.042
Both in manuf.	0.103			0.100	0.042	0.041
Panel D. Migration						
Migration response of the exposed (p.p.)	+0.0166	+0.0194	—	—	+0.0170	—
Share living in a different state because of the shock (%)						
Aggregate	0.150	0.130	—	—	0.148	—
Single men	0.151	0.150	—	—	0.150	—
Single women	0.127	0.116	—	—	0.123	—
Married men	0.159	0.151	—	—	0.158	—
Married women	0.159	0.116	—	—	0.158	—
Couples' migration response (p.p.)						
All couples	−0.0066			—	−0.0071	—
Neither	−0.0109			—	−0.0116	—
Wife only	−0.0062			—	−0.0066	—
Husband only	+0.0141			—	+0.0146	—
Both in manuf.	+0.0078			—	+0.0080	—
Panel E. Wages and labor supply						
Female real-wage response (%)	0.405	0.362	0.365	0.405	0.442	0.442
Male real-wage response (%)	0.236	0.258	0.265	0.243	0.230	0.237
Relative wage r response (%)	0.169	0.104	0.100	0.162	0.212	0.205
Female labor-supply elasticity	0.452	0.682	0.644	0.426	0.276	0.258

Notes. Responses at $t = 199$, in levels. Panel A is in p.p. of employment and Panel B in p.p. of population. Panel C is the change in married women's quarterly participation rate, in p.p., by the couple's exposure at $t = 1$. In Panel D, the migration response of the exposed is the difference between the shares of manufacturing and non-manufacturing workers who have changed state at least once by $t = 199$, in p.p., counting each person once. Panel E reports employment-weighted log changes in real wages (×100); the female labor-supply elasticity is the log female employment response over the log female real-wage response.

Table E.4: Welfare Effects of the China Trade Shock across Economies: All Groups

	Baseline economy		Counterfactual economies				
	Home production	Wives' time valued as singles'	All-singles	All-singles, no migration	No migration	No participation	No migration & no participation
Panel A. Welfare gain (% CE)							
Aggregate	0.175	0.194	0.178	0.180	0.176	0.167	0.169
Single men	0.157	0.157	0.166	0.170	0.162	0.153	0.158
Single women	0.234	0.234	0.194	0.195	0.236	0.255	0.257
Couples	0.170	0.193	0.178	0.180	0.171	0.157	0.159
Married men	0.170	0.193	0.164	0.167	0.171	0.157	0.159
Married women	0.170	0.193	0.193	0.194	0.171	0.157	0.159
Couples, by exposure							
Neither	0.180	0.180	0.184	0.186	0.182	0.167	0.171
Wife only	0.161	0.022	0.174	0.175	0.162	0.153	0.155
Husband only	0.140	0.320	0.161	0.159	0.135	0.126	0.121
Both in manuf.	0.125	0.136	0.155	0.153	0.122	0.115	0.113
Panel B. Spatial							
CV of state welfare	0.278		0.204	0.327	0.448	0.289	0.462

Notes. Money-metric consumption-equivalent gains (%), summed over $t = 2, \ldots, 199$. Each row represents the same people across all economies, and each economy weights each person by their baseline money metric in 2000Q1. In the economies with couples, married men and women carry the couple's welfare; in the all-singles economies, they are the same people as singles, and couples' rows average the two former spouses. The first two columns are the baseline economy: the first values a non-working wife's time as home production, as in the model, and the second values it as a single, at the non-employment benefit b_j. In Panel B, CV is the population-weighted coefficient of variation of state welfare gains. Weighting the all-singles economy by its own wages instead, its aggregate gain is 0.177.